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Parabolic Anderson Problem and Intermittency

René A. Carmona
S. A. Molchanov



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Abstract

We consider the stochastic partial differential equation

$$\frac{\partial u}{\partial t} = \kappa \Delta u + \xi_t(x)u.$$

The potential $\xi_t(x)$ is assumed to be a mean zero homogeneous Gaussian field. We pay special attention to the white noise case. In order to minimize the technical difficulties we consider only the case the discrete Laplacian Δ on the lattice $\mathbb{Z}^d$. We prove existence and uniqueness (for almost every realization of the random potential) for nonnegative initial conditions. These results are proved by means of the Feynman-Kac representation of the minimal solutions. Infinite dimensional Ito and Stratonovich equations are needed to study the white noise case. We then prove that the solutions have moments of all orders. In the case of a white noise potential we derive a family of closed equations for these moments.

We then prove the existence of the moment Lyapunov exponents and we study their dependence upon the diffusion constant κ . As a consequence, we show that there is full intermittency of the solution when the dimension d is not greater than 2 while the same intermittency only holds for large values of the diffusion constant in higher dimensions. The fundamental equation can be viewed as a parabolic Anderson model and this phase transition is natural from the point of view of localization theory.

Finally, the last chapter is devoted to the study of the almost sure Lyapunov exponent. We prove its existence and we derive their asymptotic behavior for small κ .

Key words and phrases: Random Parabolic Equation, Nonstationary Anderson Problem, Large Time Asymptotics, Intermittency.

Chapter I

INTRODUCTION

The subject of the present monograph is the investigation of the asymptotic properties of the solutions of the parabolic partial differential equation:

$$\frac{\partial u}{\partial t} = \kappa \Delta u + \xi_t(x)u \quad (\text{I.1})$$

where $\xi_t(x)$ is a random potential. This potential may be time dependent and we shall not rule out the possibility that it is a Schwartz distribution instead of a function. This equation plays an important role in chemical kinetics. But it can also be regarded as a particular case of more realistic equations. We have in mind, for example, the equation of the magnetic field in a random flow. This equation reads:

$$\frac{\partial \vec{H}}{\partial t} = \kappa \Delta \vec{H} + (\vec{v}(t, x) \nabla) \vec{H} - (\vec{H}, \nabla) \vec{v}. \quad (\text{I.2})$$

Equation (I.2) differs from (I.1) in two respects. First it is a vector equation instead of a scalar one. Second it has a first order term. Nevertheless, one expects that the results of the qualitative analysis will be the same. There is also some analogy between these equations and the heat equation which describes the time evolution of the temperature field in the system atmosphere-ocean. We now give a simple derivation of equation (I.1). It is presented as an example of a noninteracting particle system.

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Let us consider an ensemble of particles and let us denote by x_i 's the initial positions of these particles. Let us assume that $x_i \in \mathbb{R}^d$. We are mostly interested in the case $d = 3$ but it will be interesting to consider the effects of the dimension d on the results of our mathematical investigations. We choose to represent configurations of particles by point measures. Our initial configuration is of the form:

$$\eta_0(x) = \sum \delta_{x_i}(x),$$

where we use the notation δ_x for the unit mass at the point x . We now describe the dynamics of the system. We assume that each particle has a Brownian motion with diffusion coefficient κ and that the various particles move independently of each other. We also assume that, at each time t any particle at the location x splits into two identical particles (which start independent Brownian motions from the point x) with rate $\xi_t^+(x)dt$ and that the same particle dies (i.e. disappears) with rate $\xi_t^-(x)dt$. The configuration of particles at time t is then denoted by:

$$\eta_t(x) = \sum \delta_{x(t)}(x).$$

A first approach to the time evolution of the ensemble of particles is to consider the occupation measures. For each domain D one sets:

$$\eta_t(D) = \#\{i; x_i(t) \in D\}$$

for the number of particles in D at time t . For each domain D and each time t , $\eta_t(\cdot)$ is a random measure and it is natural to investigate the properties of its average:

$$\mu_t^{(1)}(D) = \langle \eta_t(D) \rangle$$

It is easy to show that this mean value, when regarded as a function of the domain D for t fixed, is in fact a measure which has a density. We denote this density by $m_1(t, x)$. In other words we have:

$$\mu_t^{(1)}(D) = \int_D m_1(t, x) dx.$$

This density has the interpretation of the *mean density of particles at time t at the point x* . It is possible to prove that this function satisfies the partial differential equation (I.1) in the sense that:

$$\frac{\partial m_1(t, x)}{\partial t} = \kappa \Delta_x m_1(t, x) + \xi_t(x) m_1(t, x). \quad (\text{I.3})$$

provided we set:

$$\xi_t(x) = \xi_t^+(x) - \xi_t^-(x).$$

Adding an initial condition to this equation requires some assumption on the initial distribution of the particles. It has many applications despite its simplicity. Let us assume for example that the particles at time $t = 0$ are distributed throughout the whole entire space according to a Poisson point process with intensity λ . Then it is clear that the initial mean density $m_1(0, x)$ is a constant independent of x because of the homogeneity of the point process. This constant is equal to λ . So in this particular case the initial condition for the equation (I.3) is:

$$m_1(0, x) \equiv \lambda.$$

The model which we described above is the simplest one for noninteracting particles. For example it can be used as a mathematical model for the diffusion of plankton in the upper layer of the ocean. In this case, if one denotes by $u(t, x)$ the concentration of plankton at time t at the location x where this location is measured in the moving frame following the stream of the large scale motion of the ocean currents, then equation (I.1) can be shown to be appropriate for this function $u(t, x)$. The choice of a moving coordinate system implies that we do not consider large scale (random) currents in this equation. The short time and local correlation parts of the ocean currents and the molecular Brownian motion are brought into the model via the diffusion part of the partial differential operator given by $\kappa \Delta$. The evolution of a given plankton colony depends strongly upon the temperature and the salinity of the ocean, upon the sun irradiation and upon most of the local characteristics of the ocean. Consequently, it is reasonable to assume that the plankton concentration satisfies an equation of the type (I.1) with a random potential $\xi_t(x)$ homogeneous in the space variable x . Moreover, the time correlation of this potential process is very short and it seems possible to assume that this correlation is actually a *delta* function or in other words that it is a white noise in time. This assumption leads to potential fields with singularities depending on the dimension. Some of these problems (existence of solutions, moment equations, intermittency, ...) are considered in [8] while the large time behavior of the solutions is studied in [5] for time independent random potentials.

The above discussion concerns the *continuous case* where the space variable x runs through the continuous space $\mathbb{R}^d$. This generality creates some difficult technical problems the solutions of which obscure the simple nature of some qualitative phenomena. The present memoir is devoted to the analysis of this problem in the case of the discrete lattice $\mathbb{Z}^d$ instead of the continuous case $\mathbb{R}^d$. We shall thus consider

equation (I.1) for $t \geq 0$ and $x \in \mathbb{Z}^d$ the operator Δ being the usual discrete analog of the Laplacian. It is defined by:

$$[\Delta f](x) = \sum_{|x-y|=1} [f(y) - f(x)] \quad (\text{I.4})$$

We shall see that all the qualitative properties which are expected on physical grounds are actually present in this approximate mathematical model. We shall not attempt to discretize the time variable t . The random potential becomes a family $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ of random variables which we assume to form a mean zero *homogeneous* ergodic process. The homogeneity assumption says that, for each finite set $\{(t_1, x_1), \dots, (t_n, x_n)\}$ and for each $t \geq 0$ and $x \in \mathbb{Z}^d$ the random vectors $(\xi_{t_1}(x_1), \dots, \xi_{t_n}(x_n))$ and $(\xi_{t_1+t}(x+x_1), \dots, \xi_{t_n+t}(x+x_n))$ have the same distribution. It is important to notice that we call homogeneous what is very often called *stationary*. The reason for our choice is to avoid misunderstandings when the word stationary is used with the meaning it has in the theory of dynamical systems and differential equations. Stationarity has to do with time independence in all cases but we shall restrict its use to mean that the coefficients of an equation are independent of time. The zero mean assumption is not restrictive. Indeed, this mean has to be constant because of the homogeneity assumption and if this constant was not zero, the solution of the problem (I.1) would be obtained by multiplying the solution of this (similar) problem with the centered potential process by the exponential of t times the mean. This would modify some of the results below but only in a trivial way. We shall not recall the definition of ergodicity. The most important tool in the study of a homogeneous process (also called a field) is its covariance. Because of the homogeneity and the mean zero assumption the covariance of the potential $\xi_t(x)$ is given by the formula:

$$\Gamma(t, x) = \langle \xi_s(y) \xi_{s+t}(y+x) \rangle$$

where we use the notation $\langle \cdot \rangle$ to denote the expectation over the probability space on which the random field $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is defined. The above right hand side does not depend on s or y because of the homogeneity assumption. From the physical point of view there are two *correlation scales*. They can be given a precise mathematical meaning when the correlation function Γ decays fast enough. If this correlation function is exponentially decaying in the time and the space variables one defines the *time correlation length* τ and the *space correlation length* L as the inverses of the corresponding rates of exponential decay. For the sake of definiteness, one

defines L by the formula:

$$L = \inf\{L'; |\Gamma(0, \ell)| \leq \frac{1}{2} \Gamma(0, 0) \quad |\ell| \geq L'\}$$

and τ as the number:

$$\tau = \inf\{\tau'; |\Gamma(s, 0)| \leq \frac{1}{2} \Gamma(0, 0) \quad s \geq \tau'\}.$$

These correlation lengths can be understood in the following way. If one considers the processes $\{\xi_t(Lx); t \geq 0\}$ for fixed $x \in \mathbb{Z}^d$, then the definition of L makes their correlation very weak and we can assume, at least at the heuristic level, that they are essentially independent. For the sake of the present discussion we shall assume that $L = 1$. This does not change qualitatively the phenomena which we expect and it will facilitate the discussion.

Since we are working with the lattice approximation, the diffusion is governed by the finite difference operator (III.12). Since we keep the time as a real variable without discretization, we now have two different time scales.

- τ the time correlation length introduced above.
- τ_1 the average time it takes for a particle to exit a *space correlation cell*. τ_1 is consequently of the order of L^2/κ . Also, τ_1 can be viewed as the mean time between two successive jumps of the continuous time random walk governed by the discrete Laplacian (III.12) since we assume that $L = 1$.

The relative values of these two time scales play a crucial role in the characteristics of the system. Let us consider the various possibilities.

- when τ is much smaller than τ_1 , i.e. when $\tau/\tau_1 \ll 1$, the time correlation is very much like a delta function at the origin and the field $\xi_t(x)$ can be approximate by an independent family (parametrized by $x \in \mathbb{Z}^d$) of white noise processes $\{\xi_t(x); t \geq 0\}$. In this case, the equation (I.1) can be rewritten as a stochastic integral equation driven by an infinite dimensional Wiener process. It can then be investigate as a discrete analog of a stochastic partial differential equation.
- when τ is much larger than τ_1 , i.e. when $\tau/\tau_1 \gg 1$, one says that one is in the *stationary adiabatic regime*. Indeed, the characteristics of the potential $\xi_t(x)$ vary very slowly and one expects that one can use the results of the stationary case (i.e. time independent case) as a good approximation. We shall pay a special attention to this case in the present paper.

- finally in the intermediate case where τ and τ_1 are of the same order is difficult to study the transition between the two regimes. We shall not discuss this case here.

The present paper is devoted to the mathematical study of the stochastic parabolic problem:

$$\begin{cases} \frac{\partial u}{\partial t} = \kappa \Delta u + \xi_t(x)u \\ u(0, x) \equiv 1. \end{cases} \quad (I.5)$$

It can be viewed as a parabolic Anderson problem with a possibly time dependent potential. We describe below three different problems in which this Anderson parabolic problem comes up naturally.

Branching Process in Random Environment

The fundamental equation (I.5) can be viewed as an infinite system of coupled stochastic (difference) equations. We spend some time on the interpretation of this equation in terms of branching processes and we discuss in details its connection with reaction-diffusion equations in general and more precisely the so-called Kolmogorov-Petrovskii-Piskunov equation in particular.

As explained before we restrict ourselves to the particular case of a system of noninteracting Brownian particles on the lattice $\mathbb{Z}^d$. The dynamics of the system of particles is given by two functions $\xi^+(t, x)$ and $\xi^-(t, x)$. They represent respectively the rates at which a particle being at $x \in \mathbb{Z}^d$ at time t splits into two identical particles or dies. One also needs to know the diffusion constant κ which determines the holding time at each point of the lattice: a particle at time t at $x \in \mathbb{Z}^d$ jumps during the time interval $[t, t + dt)$ to any of its $2d$ neighboring point x' (i.e. such that $|x' - x| = 1$) with probability κdt . We denote by $n(t, x)$ the number of particles being at the site time x of the lattice $\mathbb{Z}^d$ at time t . We compute its moment generating function. For $s < t$, $x, y \in \mathbb{Z}^d$ and $\alpha > 0$ we set:

$$M_\alpha(s, x, t, y) = \mathbb{E}\{e^{-\alpha n(s, x)} \mid n(t, \cdot) = \delta_y(\cdot)\}. \quad (I.6)$$

It is then easy to derive, for (t, y) fixed the *backward Kolmogorov's equation*:

$$0 = \frac{\partial M_\alpha}{\partial s} + \kappa \Delta_x M_\alpha + \xi^+(s, x)M_\alpha^2 - (\xi^+(s, x) + \xi^-(s, x))M_\alpha + \xi^-(s, x) \quad (I.7)$$

which is satisfied for $s \leq t$ and which is to be complemented with the *terminal condition*:

$$M_\alpha(t, x, t, y) = e^{-\alpha} \delta_y(x) + (1 - \delta_y(x)).$$

Unfortunately this equation may not have a unique solution. In any case, if we denote by $m_1(s, x, t, y)$ the first moment of $n(t, x)$ under the same condition, i.e.

$$m_1(s, x, t, y) = \mathbb{E}\{n(s, x) \mid n(t, \cdot) = \delta_y(\cdot)\}$$

then, using the fact that:

$$m_1(s, x, t, y) = -\frac{\partial M_\alpha}{\partial \alpha} \Big|_{\alpha=0}$$

one gets the following equation for m_1 :

$$0 = \frac{\partial m_1}{\partial s} + \kappa \Delta_x m_1 + [\xi^+(s, x) - \xi^-(s, x)]m_1 \quad (I.8)$$

with the terminal condition:

$$m_1(t, x, t, y) = \delta_y(x).$$

One gets similar equations for the higher moments by taking more derivatives. For example the equation for the second moment:

$$m_2(s, x, t, y) = \mathbb{E}\{n(t, y)^2 \mid n(s, x) = \delta_y(x)\}$$

is obtained by taking one extra derivative. It reads:

$$0 = \frac{\partial m_2}{\partial s} + \kappa \Delta_x m_2 + (\xi^+(s, x) - \xi^-(s, x))m_2 - \xi^+(s, x)m_1^2 \quad (I.9)$$

with the same initial-boundary condition:

$$m_2(t, x, t, y) = \delta_y(x).$$

The first part of the right hand side is the same as for m_1 . The only difference comes the term $\xi^+(s, x)m_1^2(s, x, t, y)$. Note that this last term can be viewed as a source term if the equation for the first moment has already been solved. As we already explained, similar equations can be derived for the higher moments:

$$m_p(s, x, t, y) = (-1)^p \frac{\partial^p M_\alpha}{\partial \alpha^p} \Big|_{\alpha=0}.$$

These equations are not very easy to deal with because of the *boundary condition* which assume the knowledge of the number of particles at the terminal time t . Moreover, this condition states that there is a single particle at a specific point. In particular, this is far from the desirable condition:

$$n(s, x) \equiv 1.$$

This is the main reason for an investigation of the more complicated *forward equation* which we derive now. In order to do so we need to consider the number of particles $n(s, x)$ as a function $n(s)$ of $x \in \mathbb{Z}^d$. If $\alpha = \alpha(x)$ is a function with compact support (i.e. $\alpha(x) = 0$ except for finitely many x) then the duality:

$$\langle \alpha, n(s, \cdot) \rangle = \sum_x \alpha(x) n(s, x)$$

makes sense and the moment generating function M_α can be defined as a Laplace transform on a functional space by the formula:

$$M_\alpha(s, t) = \mathbb{E}\{e^{-\langle \alpha, n(t) \rangle} | n(s)\}. \quad (\text{I.10})$$

We use the same arguments as before to derive the Kolmogorov's equation. But this time we give details because the results seem to be less standard. We consider the three possible transitions for the number of particles in an infinitesimal time interval $[t, t + dt)$.

- $n(t, y) \longrightarrow n(t, y) + 1$ with probability

$$\xi^+(t, y) n(t, y) dt + \sum_{|y' - y| = 1} \kappa n(t, y') dt + O(dt^2)$$

- $n(t, y) \longrightarrow n(t, y) - 1$ with probability

$$\xi^-(t, y) n(t, y) dt + 2d\kappa n(t, y) dt + O(dt^2)$$

- $n(t, y) \longrightarrow n(t, y)$ with probability

$$1 - \xi^+(t, y) n(t, y) dt - \sum_{|y' - y| = 1} \kappa n(t, y') dt - \xi^-(t, y) n(t, y) dt - 2d\kappa n(t, y) dt + O(dt^2)$$

Next we notice that, up to terms of order $O(dt^2)$, the $n(t + dt, y) - n(t, y)$ may be regarded as independent for different sites y . Consequently, one easily checks that:

$$M_\alpha(s, t + dt) - M_\alpha(s, t) = -\mathbb{E}\{e^{-\langle \alpha, n(t) \rangle} \sum_y (e^{-\alpha(y)} - 1) [\xi^+(t, y) n(t, y) + \sum_{|y' - y| = 1} \kappa n(t, y')] + \sum_y (e^{\alpha(y)} - 1) \xi^-(t, y) n(t, y) + 2d\kappa n(t, y)\} dt + O(dt^2).$$

If one notices that:

$$\frac{\partial M_\alpha}{\partial \alpha(y)} = -\mathbb{E}\{e^{-\langle \alpha, n(t) \rangle} n(t, y)\}$$

one immediately gets the (formal) equation for M_α :

$$\begin{aligned} \frac{\partial M_\alpha}{\partial t} = & \sum_y (e^{-\alpha(y)} - 1) [\xi^+(t, y) \frac{\partial M_\alpha}{\partial \alpha(y)} \\ & + \sum_{|y' - y| = 1} \kappa \frac{\partial M_\alpha}{\partial \alpha(y')}] + \sum_y (e^{\alpha(y)} - 1) (\xi^-(t, y) + 2d\kappa) \frac{\partial M_\alpha}{\partial \alpha(y)} \end{aligned} \quad (\text{I.11})$$

with the initial condition $M_\alpha(s, s) = e^{\alpha(x)}$ if one wants to consider the initial condition $n(s, y) = \delta_x(y)$, or:

$$M_\alpha(s, s) = \exp[\sum_x \alpha(x)]$$

if one wants to consider the initial condition $n(s, y) \equiv 1$. In any case, equation (I.11) makes it possible to derive equations for the moments and the correlation coefficients of the field $n(t, y)$. Let us consider for example the case $s = 0$ and $n(0, x) \equiv 1$. Then we have:

$$m_1(t, y) = \langle n(t, y) \rangle = -\frac{\partial M_\alpha}{\partial \alpha(y)} \Big|_{\alpha=0}$$

and:

$$m_2(t, x, y) = \langle n(t, x) n(t, y) \rangle = \frac{\partial^2 M_\alpha}{\partial \alpha(x) \partial \alpha(y)} \Big|_{\alpha=0}$$

and consequently, we have first:

$$\frac{\partial m_1}{\partial t} = \kappa \Delta m_1 + [\xi^+(t, y) - \xi^-(t, y)] m_1 \quad (I.12)$$

with initial condition $m_1(0, y) \equiv 1$. Notice that equation (I.12) is exactly the fundamental equation (I.1) the investigation of which is the main topic of the present paper. Taking one more derivative we get:

$$\begin{aligned} \frac{\partial m_2}{\partial t} = & \kappa (\Delta_x + \Delta_y) m_2 + [\xi^+(t, x) + \xi^+(t, y) - \xi^-(t, x) - \xi^-(t, y)] m_2 \\ & + \delta_x(y) \left[(\xi^+(t, x) + \xi^-(t, x) + 2\kappa d) m_1(t, x) + \kappa \sum_{|x'-x|=1} m_1(t, x') \right] \end{aligned} \quad (I.13)$$

with initial condition $m_1(0, x, y) \equiv 1$. This equation is remarkably different from the equation we obtained for the second moment from the backward Kolmogorov equation.

Most of the results derived in the present paper concern the situation where the field $\xi(t, y) = \xi^+(t, y) - \xi^-(t, y)$ is some sort of white noise in time for each fixed y . This assumption is justified because it appears naturally as a limiting case of Gaussian fields with short time correlations. In this case, the equation for m_1 can be given a rigorous meaning: indeed, it can be rewritten as a stochastic integral equation of the Ito's type in an infinite dimensional space. Unfortunately, the equations for $m_2(t, y)$ and $m_2(t, x, y)$ do not make sense in this limit because we have $\xi^\pm(t, y) \equiv \infty$ and $\xi^+(t, y) + \xi^-(t, y) \equiv \infty$ identically. Nevertheless, there are very interesting cases for which the equations for all the moments of $n(t, y)$ make sense. This is the case for example when $\xi^\pm(t, y)$ is the time derivative of a Lévy process (i.e. a process with stationary independent increments) with sample paths of bounded variations, the simplest possible case being obviously a difference of two Poisson processes. These models are considered in [1].

Hydrodynamics and Burger's Equation

Burger's equation is a fundamental equation of hydrodynamics and astrophysics. It describes self gravitating media where, attraction takes place instead of the repulsion between the *liquid particles* (as it is the case for ordinary liquids such as water). It has the form:

$$\frac{\partial \vec{v}(t, x)}{\partial t} + (\vec{v}(t, x) \cdot \nabla) \vec{v}(t, x) = \kappa \Delta \vec{v}(t, x) + f(t, x) \quad (I.14)$$

$$\begin{aligned} \text{curl} \vec{v}(t, x) &\equiv 0 \\ \vec{v}(0, x) &= \vec{v}_0(x), \quad x \in \mathbb{R}. \end{aligned} \quad (I.15)$$

Here $\vec{v}(t, x)$ is the velocity field of the fluid, $f(t, x)$ is an external force, $\kappa > 0$ is the viscosity, $t \geq 0$ is the time and the space variable $x \in \mathbb{R}^d$ varies in $d = 1, 2$ or 3 dimensions. The pressure is absent because the nonlinear term $(\vec{v}(t, x) \cdot \nabla) \vec{v}(t, x)$ is a gradient. Burger's equation has the same form as the Navier-Stokes equation. The difference concerns the class of velocity fields. Navier-Stokes equation is concerned with incompressible velocity fields satisfying $\text{div} \vec{v}(t, x) \equiv 0$ while Burger's equation deals with potential fields. Because of this restriction, the nonlinear equation (I.14) can be reduced to a linear equation via the substitution:

$$\vec{v}(t, x) = -2\kappa \nabla \log \varphi(t, x).$$

Indeed, equation (I.14) becomes the following (linear) parabolic equation:

$$\frac{\partial \varphi(t, x)}{\partial t} = \kappa \Delta \varphi(t, x) + F(t, x) \varphi(t, x) \quad (I.16)$$

with an appropriate initial condition. The new potential $F(t, x)$ satisfies:

$$\nabla F(t, x) = f(t, x).$$

and the initial condition becomes:

$$-2\kappa \nabla \varphi_0(x) = \vec{v}_0(x).$$

Consequently, when the original external force is random, equation (I.16) can be viewed as a particular case of the so-called Anderson parabolic problem. Notice that the potential F can be time-dependent. The solution of Burger's equation can then be written in the form:

$$\vec{v}(t, x) = -2\kappa \frac{\nabla \varphi(t, x)}{\varphi(t, x)}$$

and the analysis of the asymptotic behavior of $\vec{v}(t, x)$ as $t \rightarrow \infty$ reduces to the asymptotic analysis of $\varphi(t, x)$ and its gradient.

Advection-Convection Equation for the Temperature Field

Let us consider the following parabolic equation for the passive scalar field $T(t, x)$:

$$\frac{\partial T(t, x)}{\partial t} + \tilde{v}(t, x) \cdot \nabla T(t, x) = \kappa \Delta T(t, x) \quad (\text{I.17})$$

$$T(0, x) = T_0(x), \quad x \in \mathbb{R}. \quad (\text{I.18})$$

where the initial condition $T_0(x)$ is assumed to be compactly supported. This equation plays an important role in the analysis of turbulent diffusion. It is used for example in the analysis of the temperature field of incompressible fluids. See for example [16]. In [3], Avellaneda and Majda gave, a complete analysis of the following particular case. The dimension d is equal to 2 and the 2-dimensional velocity field of the form:

$$\tilde{v}(x, y) = \begin{bmatrix} 0 \\ v(x) \end{bmatrix}$$

where we use now the notation $(x, y) \in \mathbb{R}^2$ for the space variable. In this case, the equation (I.17) becomes:

$$\frac{\partial T(t, x, y)}{\partial t} + v(x) \frac{\partial T(t, x, y)}{\partial y} = \kappa \Delta T(t, x, y).$$

The authors also assume that the function $v(x)$ is a realization of a stationary mean zero Gaussian process with a spectrum of the Kolmogorov type. It is not difficult to show that the solution $T(t, x, y)$ satisfies:

$$|T(t, x, y)| \leq c_1 e^{-c_2(|x|^2 + |y|^2)}$$

for some positive constants c_1 and $c_2 = c_2(t)$. One can thus consider the two-sided Laplace transform in y , namely the function $\hat{T}_\lambda(t, x)$ defined by:

$$\hat{T}_\lambda(t, x) = \int_{-\infty}^{+\infty} T(t, x, y) e^{\lambda y} dy.$$

Then, the parabolic equation for $T(t, x, y)$ becomes the following equation:

$$\frac{\partial \hat{T}_\lambda(t, x)}{\partial t} = \kappa \frac{\partial^2 \hat{T}_\lambda(t, x)}{\partial x^2} + (\lambda^2 - \lambda v(x)) \hat{T}_\lambda(t, x) \quad (\text{I.19})$$

with the initial condition:

$$\hat{T}_\lambda(0, x) = \int_{-\infty}^{+\infty} T(0, x, y) e^{\lambda y} dy.$$

Again the transformed equation is a parabolic Anderson problem with a random potential. The homogenization for equation (I.17) can be reformulated in terms of this new equation as the asymptotic analysis of the solution of (I.19) for $t \rightarrow \infty$ and $\lambda \rightarrow 0$.

Intermittency

We shall show that a solution of equation (I.1) exists and is unique under very general conditions. The solution $u(t, x)$ is a homogeneous ergodic field in x for each fixed value of t but is not homogeneous in t . In fact, the field and its moments have non trivial exponential behaviors as $t \rightarrow \infty$. This behavior can be interpreted in terms of the notion of *intermittency*. The latter refers to unusually large fluctuations of the field. In the language of particles this means that the distribution of the particles following the dynamics given by (I.17) is highly nonuniform. For large values of the time t there exist spots where the concentration is very high, the distances between these spots being very large, and most of the *mass* of our ensemble of particles being concentrated in these spots. The mathematical notion of intermittency is defined as follows. Let us assume that the p -th moment of the solution exists, call it $m_p(t)$, and that the limit:

$$\gamma_p = \lim_{t \rightarrow \infty} \frac{1}{t} \log m_p(t)$$

exists. This limit is called the p -th moment Lyapunov exponent of the solution. It depends on the parameters $\kappa, \tau, \dots$ of the problem but we shall not consider this dependence in the present discussion. We say that the family of homogeneous fields $\{u(t, x); x \in \mathbb{Z}^d\}$ is intermittent when $t \rightarrow \infty$ if one has:

$$\gamma_1 < \frac{\gamma_2}{2} < \dots < \frac{\gamma_p}{p} < \dots$$

It is not difficult to check that one always has:

$$\gamma_1 \leq \frac{\gamma_2}{2} \leq \dots \leq \frac{\gamma_p}{p} \leq \dots$$

Moreover, we shall also prove that, if one of the above inequality is a strict inequality, then all the subsequent inequalities are also strict. This proves that intermittency is in fact proven when the first inequality is shown to be strict. Since it may not be completely obvious that the above mathematical definition actually captures the physical phenomenon which we described earlier, we spend some time to shed some light on the implications of our mathematical definition. Most of the ideas are borrowed from [26].

There is another parametrization of equation (I.1). It corresponds to another physical regime. Let us do the substitution $s = \kappa t$ and let us set $v(s, x) = u(s/\kappa, x)$. Then the function $v(s, x)$ satisfies the parabolic equation:

$$\frac{\partial v}{\partial s} = \Delta v + \frac{1}{\sqrt{\kappa}} \xi_1(s, x) v(s, x)$$

where the new stochastic field $\xi_1(s, x)$ is defined by:

$$\xi_1(s, x) = \frac{1}{\sqrt{\kappa}} \xi\left(\frac{s}{\kappa}, x\right).$$

If we consider that the random field ξ_1 is given, the new problem can be rewritten in the form:

$$\begin{cases} \frac{\partial u}{\partial t} &= \Delta u + \sqrt{\sigma} \xi_1(t, x) u(t, x) \\ u(0, \cdot) &= 1. \end{cases} \quad (\text{I.20})$$

Notice that the covariance of the new mean zero Gaussian random field is given by:

$$\langle \xi_1(s, x) \xi_1(t, y) \rangle = \frac{1}{\kappa} \Gamma_0\left(\frac{s-t}{\kappa}\right) \Gamma_1(x-y)$$

This form is particularly important because of the dimensionless character of the constant κ . We shall have to study both problem (I.5) and problem (I.20). They are both of physical importance because they correspond to two different regimes.

Summary of the Results

We close this introduction with a short summary of the contents of the present work. Chapter II is devoted to the proofs of existence and uniqueness of the solution of (I.1). This part is very much in the spirit of [10] where the stationary case was studied. It is important to recall that the word stationary is given the meaning it has in the theory of dynamical systems: the coefficients are *time-independent*. We first assume that the time correlation of the potential field $\xi_t(x)$ is given by a function. The equation can be studied path by path and the existence and uniqueness problems are approached via the Feynman-Kac representation of minimal solutions. We then consider the case where the time correlation of the potential field is a delta function at the origin. We reformulate the equation as an Ito stochastic integral equation driven by an infinite dimensional Wiener process. We also consider the case of the corresponding Stratonovich integral equation. These equations are easily solved in the appropriate weighted Hilbert spaces. Next, we give a diffusion approximation result

which shows that the problem of the delta-time correlation appears as a natural limit of more realistic physical situations. We close this chapter with a proof of the existence of moments of all orders for the solutions of equation (I.1). We also derive a set of closed equations for the moments in the case of a delta-time correlation. They read:

$$\frac{\partial m_p(t, x_1, \dots, x_p)}{\partial t} = \kappa [\Delta_{x_1} + \dots + \Delta_{x_p} + V(x_1, \dots, x_p)] m_p(t, x_1, \dots, x_p) \quad (\text{I.21})$$

where the potential function $V(x_1, \dots, x_p)$ is given by:

$$V_p(x_1, \dots, x_p) = \sum_{1 \leq i < j \leq p} \Gamma_1(x_i - x_j) \quad (\text{I.22})$$

and where Γ_1 is the spatial covariance of the random field $\xi_t(x)$. It is interesting to notice that these equations are simply heat equations for a multiparticle Schrödinger operator.

Chapter III is devoted to the proof of the existence of the moment Lyapunov exponents $\gamma_p(\kappa)$ and to their quantitative analysis. The existence is proved in great generality. In the white noise case, these exponents are characterized as the spectral radii of the above multiparticle Schrödinger operators. Some classical arguments from spectral theory and simple perturbation arguments give the properties of $\gamma_p(\kappa)$ as functions of the order p of the moment under consideration, of the diffusion constant κ and of the dimension d . We also give detailed asymptotic formulas for $\gamma_p(\kappa)$ both as $\kappa \rightarrow 0$ and $\kappa \rightarrow \infty$. In any case, we prove that full intermittency holds for all values of κ in dimensions $d = 1$ and $d = 2$ while a similar result depends on the values of κ in higher dimensions. Indeed, we prove that $\gamma_p(\kappa)$ vanishes for large values of κ if $d \geq 3$. We view this result as a phase transition in the spirit of the theory of localization for disordered Anderson models. Finally, we consider the general case of a non-delta correlated potential field $\xi_t(x)$. We mostly study the Lyapunov exponents as functions of the coupling constant σ . We prove similar asymptotic estimates of $\gamma_p(\sigma)$ both for σ large and for σ small. Finally we investigate the intermittency of the solution field as before.

Chapter IV is devoted to the proof of the existence and the investigation of the behavior of the almost sure Lyapunov exponents $\tilde{\gamma}(\kappa)$. The existence of the almost sure exponents is obtained (as usual) from the subadditive ergodic theorem. We were unable to prove the existence for general bounded initial conditions $u_0(x)$. We have to restrict ourselves to initial conditions with bounded supports. We show that the value

of the exponent does not depend on the initial condition and we determine the asymptotic behavior of $\tilde{\gamma}(\kappa)$ as $\kappa \rightarrow 0$. The proofs of the lower bound and the proof of the upper bound are given in separate sections. These estimates are the main technical results of the whole analysis. The proofs are rather involved. They are more difficult than the *soft analysis* proofs of the estimations on the moment Lyapunov exponents. But they are of crucial importance because they shed light on the phenomenon of intermittency and the presence of very high picks separated by large distances.

We summarize in a short and last chapter the analogies of the results proved in this monograph for the time dependent random parabolic problem and the results and/or predictions of the Anderson's theory of (time independent) random Schrödinger operators.

Chapter II

EXISTENCE AND UNIQUENESS PROBLEMS

This section is devoted to the discussion of the existence of solutions of the equation (I.1). We show existence and uniqueness of a solution under various conditions on the random potential $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$. For a large class of potentials ξ the equation (I.1) has a mathematical meaning for each realization of the random field $\xi_t(x)$ and it can be solved by first fixing the *sample path* of ξ and then treating the problem as a deterministic one. For this reason we first consider the case of a deterministic potential. We introduce a specific set of notations and we review the classical results which we use. We then consider the more general random case.

II.1 The Deterministic Problem

Some of the results concerning existence and uniqueness of a solution for the random parabolic equation (I.1) are obtained for each fixed $\omega \in \Omega$ from results on deterministic equations. This is especially the case when the random potential $\xi_t(x)$ is a bona fide function of t for each x and ω . The limiting case of the white noise in time will have to be treated by different methods.

II.1.1 The Feynman-Kac Representation

It may seem paradoxical, but we choose to use probabilistic techniques to study the deterministic version of (I.1). To this effect, we introduce a specific set of probabilistic notations. It deals with the continuous time nearest neighbor random walk on the lattice $\mathbb{Z}^d$. This process can also be regarded as the lattice version of the process of Brownian motion because it is a Markov process with independent increments and because its infinitesimal generator is the discrete Laplacian Δ defined in (III.12).

We let $\mathbf{W} = \mathbf{W}^{(d)}$ be the space of right continuous functions with left limits from $\mathbb{R}_+$ into $\mathbb{Z}^d$. For each $t \geq 0$, we denote by X_t , the coordinate function $\mathbf{W} \ni w \rightarrow X_t(w) = w(t) \in \mathbb{Z}^d$. We use the notation $\mathcal{W}_t = \mathcal{W}_t^{(d)}$ for the smallest σ -field of subsets of $\mathbf{W}$ for which all the X_s 's for $0 \leq s \leq t$ are measurable. The simpler notation $\mathcal{W}$ will be used in the case $t = \infty$. For each $s \geq 0$ and $x \in \mathbb{Z}^d$ we denote by $\mathbb{P}_x$ the probability measure on $(\mathbf{W}, \mathcal{W})$ of the *Brownian motion* or *continuous time (nearest neighbor) random walk* on $\mathbb{Z}^d$ starting from x at time 0. The corresponding expectation will be denoted by $\mathbb{E}_x$. Remember that under $\mathbb{P}_x$, the process $\{X_t; t \geq 0\}$ is at x at time 0, it stays there for an exponential holding time, and then jumps at one of the $2d$ neighbors of x in $\mathbb{Z}^d$ with equal probabilities. It stays there an exponential holding time independent of everything else, and jumps again to one of the $2d$ neighbors with equal probabilities, ...

We did not talk about the value α of the parameter of the exponential distribution of the time spent at each site of the lattice. A convenient choice will be $\alpha = 2\kappa d$.

For each $t \geq 0$ and $w \in \mathbf{W}$ we denote by $N(t, w)$ the number of jumps of the path w before time t . $\{N(t); t \geq 0\}$ is a Poisson process with parameter $\alpha = 2\kappa n$ independent of the sites of $\mathbb{Z}^n$ visited by the path.

The operator $H_0 = \kappa\Delta$ is self-adjoint and bounded. The operator semigroup generated by H_0 is given by:

$$[e^{tH_0} f](x) = \mathbb{E}_x\{f(X_t)\}$$

and, for each function $u_0(x)$ on the lattice $\mathbb{Z}^d$, the function $u(t, x) = \mathbb{E}_x\{u_0(X_t)\}$ satisfies:

$$\frac{\partial u}{\partial t} = H_0 u \quad (\text{II.1})$$

and the initial condition $u(0, x) = u_0(x)$. We shall prove a more general fact below.

We now investigate the parabolic equation (I.1) with a deterministic potential $\xi_t(x)$. So, let $\xi_t(x)$ be a real valued function defined on $\mathbb{R}_+ \times \mathbb{Z}^n$. The classical Feynman-Kac perturbation theory of parabolic equations says that the function $u(t, x)$ defined by:

$$u(t, x) = \mathbb{E}_x\{u_0(X_t)e^{\int_0^t \xi_{t-s}(X_s)ds}\}, \quad (\text{II.2})$$

is the right candidate for a solution. Here, the initial condition u_0 is any nonnegative function on the lattice $\mathbb{Z}^n$.

We are about to show that this is indeed the case, but first, we need to rewrite the integral in the exponent in a different way. We shall use a special set of notations which are justified by the following consideration. One would like to solve equation (I.1) when the potential $\xi_t(x)$ is not a function but merely the derivative (with respect to t) in the sense of distributions of a function. This is the typical case of a random potential of the white noise type.

II.1.2 Special Notations

For each function ξ_t of the real variable t we denote by $\tilde{\xi}_t$ the antiderivative which vanishes at 0, i.e.

$$\tilde{\xi}_t = \int_0^t \xi_s ds. \quad (\text{II.3})$$

We also consider a function $\zeta_t(x)$ on $\mathbb{R}_+ \times \mathbb{Z}^n$ and a piecewise constant function X_t in $\mathbb{Z}^n$. One can think for example of a sample path of the random walk. Let us denote by $0 = T_0 < T_1 < T_2 < \dots$ the successive times of jump and by $X_0, X_1, X_2, \dots$ the successive sites visited by the random walk. In order to avoid the technical problems created by the time reversal we shall not make any assumptions on the values of the function X_s at the times of jumps, i.e. for $s = T_j$. In other words we only assume that:

$$X_j = X_t, \quad t \in (T_j, T_{j+1}), \quad j = 0, 1, \dots$$

Under these conditions we set:

$$\int_0^t d\zeta_s(X_s) = \sum_{j \geq 0} \zeta_{T_{j+1} \wedge t}(X_j) - \zeta_{T_j \wedge t}(X_j). \quad (\text{II.4})$$

In particular, with this notation we have:

$$\int_0^t \xi_s(X_s) ds = \int_0^t d\tilde{\xi}_s(X_s) \quad (\text{II.5})$$

and also:

$$\begin{aligned} \int_0^t \xi_{t-s}(X_s) ds &= \sum_{j \geq 0} \int_{t \wedge T_j}^{t \wedge T_{j+1}} \xi_{t-s}(X_j) ds \\ &= \sum_{j \geq 0} \tilde{\xi}_{t-t \wedge T_j}(X_j) - \tilde{\xi}_{t-t \wedge T_{j+1}}(X_j) ds \\ &= - \int_0^t d\tilde{\xi}_{t-s}(X_s). \end{aligned}$$

There are some inconvenience with the notation (II.4). In particular, even though we obviously have:

$$\int_0^t \xi_{t-s}(X_s) ds = \int_0^t \xi_s(X_{t-s}) ds$$

we only have:

$$\int_0^t d\zeta_{t-s}(X_s) ds = - \int_0^t d\zeta_s(X_{t-s}) ds.$$

II.1.3 Existence Problems

We now come back to the problem of the existence of a solution of equation (I.1). We assume that we are given a function $\zeta_t(x)$ which is continuous in t for each fixed $x \in \mathbb{Z}^n$ and a nonnegative function $u_0(x)$. $\zeta_t(x)$ is the antiderivative (with respect to the time variable t) of the potential function $\xi_t(x)$ appearing in equation (I.1) and $u_0(x)$ is the initial condition. We consider the function $u(t, x)$ given by the Feynman-Kac formula:

$$u(t, x) = \mathbb{E}_x \{ u_0(X_t) e^{\int_0^t d\zeta_s(X_{t-s})} \}, \quad (\text{II.6})$$

and we assume that the expectation defining $u(t, x)$ is finite for all $t \geq 0$ and $x \in \mathbb{Z}^d$. Notice that the quantity $\int_0^t d\zeta_s(X_{t-s})$ is an additive functional in the sense that:

$$\begin{aligned} \int_0^{t+h} d\zeta_s(X_{t+h-s}) &= \int_0^t d\zeta_s(X_{t+h-s}) + \int_t^{t+h} d\zeta_s(X_{t+h-s}) \\ &= \left(\int_0^t d\zeta_s(X_{t-s}) \right) \circ \theta_h + \int_t^{t+h} d\zeta_s(X_{t+h-s}) \end{aligned}$$

where θ_h denotes as usual the time shift. The Markov property gives:

$$u(t+h, x) = \mathbb{E}_x \{ u(t, X_h) e^{\int_t^{t+h} d\zeta_s(X_{t+h-s})} \}.$$

and consequently:

$$\begin{aligned} u(t+h, x) &= \mathbb{E}_x \{ u(t, X_h) e^{\int_t^{t+h} d\zeta_s(X_{t+h-s})}; h < T_1 \} \\ &\quad + \mathbb{E}_x \{ u(t, X_h) e^{\int_t^{t+h} d\zeta_s(X_{t+h-s})}; h \geq T_1 \} \\ &= u(t, x) e^{\zeta_{t+h}(x) - \zeta_t(x)} e^{-2\kappa n h} \\ &\quad + \mathbb{E}_x \{ e^{\zeta_{t+h-T_1}(x) - \zeta_{t+h}(x)} \frac{1}{2n} \sum_{|y-x|=1} u(t+h-T_1, y); h \geq T_1 \} \end{aligned}$$

and:

$$\begin{aligned} u(t+h, x) - u(t, x) &= u(t, x) e^{\zeta_{t+h}(x) - \zeta_t(x)} [e^{-2\kappa n h} - 1] \\ &\quad + u(t, x) e^{-\zeta_t(x)} [e^{\zeta_{t+h}(x) - \zeta_t(x)} - e^{\zeta_t(x)}] \\ &\quad + \mathbb{E}_x \{ e^{\zeta_{t+h-T_1}(x) - \zeta_{t+h}(x)} \frac{1}{2n} \sum_{|y-x|=1} u(t+h-T_1, y); h \geq T_1 \} \end{aligned} \quad (\text{II.7})$$

Let us consider the problem of the continuity of the function $u(t, x)$. In order to do so we assume that $t \geq 0$ and $x \in \mathbb{Z}^n$ are fixed, and we let the time variable varies while the space variable is kept fixed. In other words we consider, for t and x fixed, the limit $h \rightarrow 0$ in the equation (II.7) above. It is clear that, because of our continuity assumption on $\zeta_t(x)$ the first two terms of the right hand side go to 0. The convergence toward 0 of the third term is equivalent to the convergence toward 0 of:

$$\mathbb{E}_x \{ u(t+h-T_1, y); h \geq T_1 \} = \int_0^h u(t+s) e^{-\kappa s} ds$$

which is clear by our finiteness assumption.

Remark:

When the initial condition $u_0(x)$ is nonnegative it is interesting to notice that the function $u(t, x)$ defined by formula (II.6) is most of the time unbounded, even when $u_0(x)$ is bounded. Indeed, the obvious lower bound:

$$u(t, x) \geq \mathbb{E}_x\{u_0(X_t)e^{\int_0^t d\zeta_s(X_{t-s})}; N(t) = 0\} \quad (\text{II.8})$$

$$= u_0(x)e^{\zeta_t(x)-\zeta_0(x)}\mathbb{P}_0\{N(t) = 0\} \quad (\text{II.9})$$

$$= u_0(x)e^{\zeta_t(x)-\zeta_0(x)}e^{-2d\kappa t} \quad (\text{II.10})$$

is obtained by restricting the expectation to the paths which do not jump before time t , i.e. the paths which leave x after time t , and this lower bound shows that, for fixed $t > 0$, the function $u(t, x)$ can be unbounded even when $u_0(x)$ is bounded, as long as the increment of ζ are unbounded. We shall come back to this elementary remark in several occasions.

We now consider the problem of the differentiability with respect to t of the function $u(t, x)$. In order to do so we divide both sides of the equation (II.7) by h . Obviously the first term goes to $-2\kappa nu(t, x)$ when $h \rightarrow 0$. The first factor in the expectation converges to 1 and is uniformly bounded because $T_1 \leq h$. Since $u(t, x)$ is continuous in t for each fixed x , the expectation is easily seen to converge (after division by h) to $\kappa\Delta u(t, x) + 2\kappa nu(t, x)$ because $h^{-1}\mathbb{P}\{h \geq T_1\}$ converges to $\alpha = 2\kappa n$. The problem reduces to the convergence of the term:

$$\frac{e^{-\zeta_t(x)}e^{\zeta_{t+h}(x)} - e^{\zeta_t(x)}}{h}. \quad (\text{II.11})$$

Notice that, in the particular case of a differentiable function one has:

$$e^{\zeta_{t+h}(x)} - e^{\zeta_t(x)} = h \frac{\partial \zeta_t(x)}{\partial t} e^{\zeta_t(x)} + o(h)$$

which shows that (II.11) converges to $\partial \zeta_t(x)/\partial t$. We summarize these facts in a Proposition.

Proposition II.1.1 *Let us assume that the function $\zeta_t(x)$ is continuous in t for fixed x and that the right hand side of formula (II.6) is finite for every choice of t and x . Then the limit:*

$$\lim_{h \searrow 0} \frac{u(t+h, x) - u(t, x)}{h}$$

exists if and only if the limit:

$$\xi_t(x) = \lim_{h \searrow 0} e^{-\zeta_t(x)} \frac{e^{\zeta_{t+h}(x)} - e^{\zeta_t(x)}}{h}$$

exists and in this case the function u is solution of the parabolic equation:

$$\frac{\partial u}{\partial t} = \kappa\Delta u + \xi u.$$

Obviously,

$$\xi_t(x) = \frac{\partial \zeta_t(x)}{\partial t}$$

whenever ζ is differentiable in t with a locally bounded partial derivative.

The following corollary will play a crucial role in the sequel.

Corollary II.1.2 *Let us assume that the function $\xi_t(x)$ is measurable in t for fixed x , that $u_0(x) \geq 0$ for all $x \in \mathbb{Z}^d$ and that the right hand side of formula (II.6) is finite for every choice of t and x . Then the function u defined by formula (II.6) is the minimal solution of the parabolic equation:*

$$\frac{\partial u}{\partial t} = \kappa\Delta u + \xi u.$$

Proof:

The fact that u is a solution is an immediate consequence of the above proposition. In proving the second claim, we give a *stochastic analysis* justification of the Feynman-Kac formula. This will explain why this formula appears naturally in the study of equation (I.1) with a potential $\xi_t(x)$ given by a function.

Let us assume that the function $v(t, x)$ is continuously differentiable in t for each fixed $x \in \mathbb{Z}^n$. Let us assume that $t \geq 0$ and $x \in \mathbb{Z}^d$ are fixed and that $X_0 = x$. Then, for every $s \in (0, t)$ ordinary calculus gives:

$$v(t-s, X_s) = v(t, x) - \int_{0+}^s \frac{\partial v}{\partial t}(t-\alpha, X_\alpha) d\alpha + \sum_{0 < \alpha \leq s} [v(t-\alpha, X_\alpha) - v(t-\alpha, X_{\alpha-})].$$

It is natural to *compensate* the sum of jumps in the right hand side by adding and subtracting its compensator. Consequently, if we denote by ν the Lévy measure of $\{X_t; t \geq 0\}$, we get:

$$\begin{aligned} v(t-s, X_s) &= v(t, x) - \int_{0+}^s \frac{\partial v}{\partial t}(t-\alpha, X_\alpha) d\alpha \\ &\quad + \int_{0+}^s \int [v(t-\alpha, X_{\alpha-} + y) - v(t-\alpha, X_{\alpha-})] d\alpha \nu(dy) + M_s \\ &= v(t, x) + \int_{0+}^s \left[-\frac{\partial v}{\partial t}(t-\alpha, X_\alpha) + \kappa \Delta v(t-\alpha, X_{\alpha-}) \right] d\alpha + M_s \quad (\text{II.12}) \end{aligned}$$

where M_s is a local martingale. Let us now set:

$$v(t-s, X_s) = V_s + M_s \quad \text{and} \quad A_s = \int_0^s \xi_{t-\alpha}(X_{\alpha-}) d\alpha.$$

Using formula (II.12) and integration by parts (for semimartingales) one gets:

$$\begin{aligned} v(t-s, X_s) e^{A_s} &= v(t, x) + \int_0^s v(t-\alpha, X_\alpha) d(e^{A_\alpha}) + \int_0^s e^{A_\alpha} d\alpha v(t-\alpha, X_\alpha) \\ &\quad + [v(t-s, X_s), e^{A_s}] \\ &= v(t, x) + \int_0^s v(t-\alpha, X_\alpha) \xi_{t-\alpha}(X_{\alpha-}) e^{A_\alpha} d\alpha + \int_0^s e^{A_\alpha} dV_\alpha \\ &\quad + \int_0^s e^{A_\alpha} dM_\alpha. \quad (\text{II.13}) \end{aligned}$$

Notice that $\tilde{M}_s = \int_0^s e^{A_\alpha} dM_\alpha$ is another local martingale. Consequently, if we now assume that $v(t, x)$ is a solution of our fundamental equation (I.1), then we have $v(t-\alpha, X_{\alpha-}) \xi_{t-\alpha}(X_{\alpha-}) + \partial_t V_\alpha = 0$ and formula (II.13) can be rewritten as:

$$v(t-s, X_s) e^{A_s} = v(t, x) + \tilde{M}_s \quad (\text{II.14})$$

which is a local martingale in $s \in (0, t)$. Consequently, for a sequence $\{T_n; n \geq 1\}$ of stopping times increasing to t we have:

$$v(t, x) = \mathbb{E}_x \{ v(t-T_n, X_{T_n}) e^{\int_0^{T_n} \xi_{t-\alpha}(X_\alpha) d\alpha} \}$$

and, if we also assume that v is a nonnegative solution, we can use Fatou's lemma in the limit $n \rightarrow \infty$ and get:

$$v(t, x) \geq \mathbb{E}_x \{ v(0, X_t) e^{\int_0^t \xi_{t-\alpha}(X_\alpha) d\alpha} \}.$$

This completes the proof. ■

Remark: It was proven in [10] that if the potential function $\xi_t(x)$ is stationary (recall that this means that the function $\xi_t(x)$ is actually independent of the time variable t) and if it is not percolating from below (in a sense given in [10]) then the right hand side of formula (II.6) is actually finite for every t and x . Consequently, if one can prove uniqueness of the solution of equation (I.1), the whole *existence-uniqueness* problem reduces to proving that the expectation in formula (II.6) makes sense for all $t \geq 0$ and $x \in \mathbb{Z}^n$. This solves the existence problem in terms of properties of the potential function ξ (at least when the latter is time independent) instead of the sufficient condition involving the finiteness of the expectation defining $u(t, x)$ in formula (II.6).

We now consider in detail the problem of the uniqueness of the solution of equation (I.1).

II.1.4 Uniqueness

It is possible to prove uniqueness of the solution of the parabolic equation (I.1) in classes of functions whose spatial behavior at infinity is at most exponential. In this respect it is convenient to introduce weighted Hilbert spaces $\mathcal{H}_w = \ell_w^2(\mathbb{Z}^d)$ and to check that the solution $u(t, x)$ given by the Feynman-Kac representation (II.6) does belong to some of these spaces. We postpone the definition of these spaces for we shall not use them now.

But because of our special interest in nonnegative initial conditions $u_0(x)$ we shall first consider the uniqueness problem in the class of nonnegative solutions $u(t, x)$. We shall assume that the potential function $\xi_t(x)$ satisfies the following growth condition: for each $T > 0$ there exists a positive constant c_T for which:

$$|\xi_t(x)| \leq c_T \sqrt{1 + \log_+ |x|}, \quad x \in \mathbb{Z}^d, \quad 0 \leq t \leq T. \quad (\text{II.15})$$

We shall see later that, under a mild condition on the covariance function, this condition is almost surely satisfied by the sample realizations of typical homogeneous Gaussian fields.

The uniqueness is proven by first considering the problem in finite domains. In this case the problem can easily be solved for the solution has a nice representation in

term of its boundary values and a Poisson-type kernel. The general case can then be obtained by a limiting procedure in which the finite domains extend to the whole lattice $\mathbb{Z}^d$. For each $r > 0$ we consider the cube Λ_r defined by:

$$\Lambda_r = \{x \in \mathbb{Z}^d; |x| \leq r\}$$

and we denote by τ_r the first exit time of the random walk, i.e.

$$\tau_r = \inf\{t > 0; X_t \in \partial\Lambda_r\}$$

where we used the notation:

$$\partial\Lambda_r = \{x \in \mathbb{Z}^d; |x| = r\}$$

for the inner boundary of the set Λ_r . The Markov property gives:

$$\begin{aligned} u(t, x) &= \mathbb{E}_x\{u_0(X_t)e^{\int_0^t \xi_s(X_{t-s})ds}; t < \tau_r\} \\ &\quad \mathbb{E}_x\{u(t - \tau_r, X_{\tau_r})e^{\int_{t-\tau_r}^t \xi_s(X_{t-s})ds}; \tau_r \leq t\} \end{aligned} \quad (\text{II.16})$$

Let $v(t, x)$ be a solution of (I.1) which satisfies the same initial condition as u , i.e. such that $v(0, x) = u_0(x)$ for all $x \in \mathbb{Z}^d$.

The limiting procedure is controlled by an estimate of the Poisson kernel. The latter is given by the joint distribution $\pi_x^r(y, dt)$ of the time and the location of exit from Λ_r , i.e.

$$\pi_x^r(y, dt) = \mathbb{P}_x\{X_{\tau_r} = y, \tau_r \in dt\}$$

and more precisely in its asymptotic behavior as $r \rightarrow \infty$. Using the independence of the times of jumps and the sites of the lattice visited by the random walk and using standard properties of Poisson processes one gets:

Lemma II.1.3 *The exist distribution $\pi_x^r(y, dt)$ has a density $\pi_x^r(y, t)$ given by the formula:*

$$\pi_x^r(y, t) = \sum_{k \geq r} \frac{(2d\kappa)^k}{(k-1)!} t^{k-1} (2d)^{-k} N_r(x, y) \quad (\text{II.17})$$

where $N_r(x, y)$ denotes the numbers of paths γ of nearest neighbors sites such that $\gamma(0) = x$, $|\gamma(1)| < r, \dots, |\gamma(k-1)| < r$ and $\gamma(k) = y$.

This density can be estimated in the following way:

Lemma II.1.4 *As $r \rightarrow \infty$ one has:*

$$\pi_x^r(y, t) \sim \frac{t^{r-1} (e\kappa)^r r e^{-r \log r - \sum_{1 \leq j \leq d} |y_j| \log(|y_j|/r)}}{(2\pi)^{d/2} \sqrt{(1 \vee |y_1|) \cdots (1 \vee |y_d|)}}$$

uniformly in t restricted to a bounded interval for $|y| = r$.

The proof of this estimate is based on the fact that the main contribution to the infinite sum of (II.17) comes from the first term corresponding to $k = r$ and repeated use of Stirling's formula. We omit the details.

Proposition II.1.5 *Uniqueness of nonnegative solutions of equation (I.1) holds whenever the potential function $\xi_t(x)$ satisfies condition (II.15).*

One has to prove that, if $u(t, x)$ is a nonnegative solution satisfying $u(0, x) \equiv 0$ then $u(t, x)$ is identically zero. The proof uses the result of Corollary II.1.2 and the above estimate on the exit density $\pi_x^r(y, t)$. We do not give the details because the procedure is quite standard.

II.2 The Random Case

II.2.1 Setting of the Problem

We assume that $\{\zeta_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is a mean zero Gaussian field defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We shall use the notation $\langle \cdot \rangle$ for the expectation of a random variable over this probability space, i.e.:

$$\langle \Phi \rangle = \int_{\Omega} \Phi(\omega) d\mathbb{P}(\omega).$$

The distribution of the Gaussian process ζ is determined by its covariance function. We shall assume that it is the tensor product of the covariance of a Gaussian process on the half line $[0, \infty)$ by the covariance of a Gaussian field on $\mathbb{Z}^d$. In other words, we shall assume that its covariance is of the form:

$$\Gamma((s, x), (t, y)) = \Gamma_0(s, t) \Gamma_1(x, y). \quad (\text{II.18})$$

We shall always assume that the spatial covariance Γ_1 is homogeneous, i.e.

$$\Gamma_1(x, y) = \Gamma_1(x - y).$$

We shall say that ζ is a **spatial white noise**, or that it is a white noise in space if:

$$\Gamma_1(x - y) = \delta_0(x - y).$$

When the Gaussian process ζ is the random potential $\xi_t(x)$ appearing in equation (I.1), we shall also assume that the **time covariance** Γ_0 is homogeneous in the sense that it is a function of the difference $s - t$, i.e.

$$\Gamma_0(s, t) = \Gamma_0(s - t)$$

and we shall say that ξ is a **time white noise** if:

$$\Gamma_0(s, t) = \delta_0(s - t).$$

It is well known that a rigorous mathematical definition of such a white noise process requires some extra work. Our way out will be to consider ξ as the time derivative of a (one parameter) Wiener process with values in an infinite dimensional space. This interpretation is very natural for the study of the stochastic equation (I.1). Indeed, the latter can then be interpreted as an Ito equation in infinite dimensions. In any case, we shall consider the time antiderivative:

$$\zeta_t(x) = \int_0^t \xi_s(x) ds$$

because it always makes sense.

We shall say that ξ is a **space-time white noise** if it is a white noise both in time and in space. In the sequel, the only case of interest for which the random potential $\xi_t(x)$ does not satisfy the assumption of time stationarity is not satisfied is the piecewise constant approximation of white noise for which:

$$\Gamma_0^{(\epsilon)}(s, t) = \delta_0([s/\epsilon] - [t/\epsilon])$$

where we use the notation $[\cdot]$ for the integer part of a nonnegative real number and where ϵ is a fixed positive number. The time evolution of such a process is given by the following simple description: it is equal to a Gaussian field $\{\xi_0(x); x \in \mathbb{Z}^d\}$ with covariance Γ_1 as long as $0 \leq t < \epsilon$, then it is equal to a Gaussian field $\{\xi_1(x); x \in \mathbb{Z}^d\}$ which is independent of the previous one and has the same distribution as long as $\epsilon \leq t < 2\epsilon$, then it is equal to a Gaussian field $\{\xi_2(x); x \in \mathbb{Z}^d\}$ which is independent of the previous ones and has the same distribution as long as $2\epsilon \leq t < 3\epsilon, \dots$ It

is clear that such a process gives a good approximation of a time white noise when the parameter ϵ is small. This fact is physically obvious. We give a mathematical justification below. Even though we shall refrain from checking all the necessary details, it is not difficult to see that all of the results which we state and prove in the homogeneous case remain valid in the case of this piecewise constant approximation of white noise.

II.2.2 Continuity of the Feynman-Kac Formula

Throughout this section we assume that $\{\zeta_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is a mean zero Gaussian field with a covariance function given by a tensor product. As we explained above, ζ should be interpreted as the antiderivative $\tilde{\xi}$ of the random potential ξ . As such it should not be expected to be homogeneous in time. We assume that its covariance $\tilde{\Gamma}((s, x), (t, y))$ is of the form:

$$\langle \zeta_s(x) \zeta_t(y) \rangle = \tilde{\Gamma}((s, x), (t, y)) = \tilde{\Gamma}_0(s, t) \Gamma_1(x - y)$$

where the time covariance $\tilde{\Gamma}_0$ is determined by a measure μ on the first quadrant $[0, \infty) \times [0, \infty)$ of the (s, t) -plane in the sense that:

$$\tilde{\Gamma}_0(s, t) = \mu([0, s] \times [0, t]).$$

Notice that the measure μ is necessarily symmetric with respect to the diagonal of the first quadrant. In the particular case $\zeta_t(x) = \tilde{\xi}_t(x)$ where ζ is the (time) antiderivative of a homogeneous field $\xi_t(x)$ with covariance $\Gamma = \Gamma_0 \otimes \Gamma_1$ then one has:

$$\begin{aligned} \langle \zeta_s(x) \zeta_t(y) \rangle &= \int_0^s \int_0^t \langle \xi_\alpha(x) \xi_\beta(y) \rangle d\alpha d\beta \\ &= \int_0^s \int_0^t \Gamma_0(\alpha - \beta) d\alpha d\beta \Gamma_1(x - y) \end{aligned}$$

so that:

$$\tilde{\Gamma}_0(s, t) = \int_0^s \int_0^t \Gamma_0(\alpha - \beta) d\alpha d\beta.$$

Notice that, in this particular case, one has $\mu(ds, dt) = \Gamma_0(s - t) ds dt$ and consequently, μ is invariant under the shifts parallel to the first diagonal in the sense that:

$$\mu(A) = \mu((t, t) + A), \quad t \geq 0, \quad A \in \mathcal{B}_{[0, \infty) \times [0, \infty)} \quad (\text{II.19})$$

This shift invariance is a reasonable assumption which can be used to generalize the results beyond the case of the time antiderivative of a time homogeneous field. Note also that, still in this particular case, one has:

$$|\mu|([0, t] \times [0, t]) \leq t \int |\Gamma_0(\tau)| d\tau.$$

Recall that the covariance function Γ_1 is nonnegative definite and consequently, it is bounded and:

$$\Gamma_1(0) = \sup\{\Gamma_1(x); x \in \mathbb{Z}^d\}. \quad (II.20)$$

We first check that, for every $t > 0$ and every $x \in \mathbb{Z}^d$ the Feynman-Kac expectation:

$$\mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t-s})} \right\}$$

makes sense for almost every realization of the random field ζ . The following simple calculation will be of crucial importance.

$$\begin{aligned} \langle \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t-s})} \right\} \rangle &= \mathbb{E}_x \left\{ \langle e^{\int_0^t d\zeta_s(X_{t-s})} \rangle \right\} \\ &= \mathbb{E}_x \left\{ e^{\frac{1}{2} \langle (\int_0^t d\zeta_s(X_{t-s}))^2 \rangle} \right\}. \end{aligned}$$

The expectation in the exponent can be computed explicitly because, for each fixed sample path of the continuous time random walk, the random variable $\int_0^t d\zeta_s(X_{t-s})$ is Gaussian with mean zero. Moreover:

$$\begin{aligned} \langle \int_a^b d\zeta_s(X_{t-s}) \int_c^d d\zeta_s(X_{t-s}) \rangle &= \sum_{h,k \geq 0} \langle [\zeta_{(t-T_h) \wedge b}(X_h) - \zeta_{(t-T_{h+1}) \vee a}(X_h)] \\ &\quad [\zeta_{(t-T_k) \wedge d}(X_k) - \zeta_{(t-T_{k+1}) \vee c}(X_k)] \rangle \\ &= \sum_{h,k \geq 0} \Gamma_1(X_h - X_k) \mu([(t-T_{h+1}) \wedge b, (t-T_h) \vee a] \\ &\quad \times [(t-T_{k+1}) \wedge d, (t-T_k) \vee c]) \\ &= \int_a^b \int_c^d \Gamma_1(X_{t-t_1} - X_{t-t_2}) \mu(dt_1, dt_2). \end{aligned} \quad (II.21)$$

which gives:

$$\left\langle \left(\int_0^t d\zeta_s(X_{t-s}) \right)^2 \right\rangle = \int_0^t \int_0^t \Gamma_1(X_{t-t_1} - X_{t-t_2}) d\mu(dt_1, dt_2). \quad (II.22)$$

The final result is stated as a lemma for future reference.

Lemma II.2.1 *Let us assume that the covariance Γ of the mean zero Gaussian field $\{\zeta_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is a tensor product $\Gamma = \Gamma_0 \otimes \Gamma_1$ where Γ_0 is given by a symmetric measure μ on the first quadrant $[0, \infty) \times [0, \infty)$. Then, for every $t > 0$, $x \in \mathbb{Z}^d$ and $p > 0$ one has:*

$$\langle \mathbb{E}_x \left\{ e^{p \int_0^t d\zeta_s(X_{t-s})} \right\} \rangle \leq e^{p^2 \Gamma_1(0) |\mu|([0, t] \times [0, t]) / 2}. \quad (II.23)$$

The first consequence of this apriori estimate is the existence, for each fixed $(t, x) \in [0, \infty) \times \mathbb{Z}^d$ of a random variable:

$$u(t, x) = \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t-s})} \right\}.$$

on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which the random field ζ is defined. We now show that the family of random variables so defined has a continuous version. This will be a consequence of the crucial estimate which follows:

Lemma II.2.2 *On the top of the assumptions of the previous lemma we assume that, there is a positive constant $\epsilon > 0$ such that, for every $T > 0$ there exists a constant $c_T > 0$ such that:*

$$\mu([t, t+h] \times [0, T]) \leq c_T h^{(1+\epsilon)/2}, \quad t, h \geq 0, \quad t+h \leq T. \quad (II.24)$$

Then, for each fixed $\kappa > 0$ and $T > 0$ there exists a constant $c = c(\kappa, T)$ which depends on Γ_0 only through the constant c_T and for which:

$$|u(t+h, x) - u(t, x)|^4 \leq c \Gamma_1(0)^2 h^2 \quad (II.25)$$

for all $x \in \mathbb{Z}^d$, $t \geq 0$ and $h \geq 0$ such that $t+h \leq T$.

Proof:

We first write:

$$u(t+h, x) - u(t, x) = \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t+h-s})} [e^{\int_t^{t+h} d\zeta_s(X_{t+h-s})} - 1] \right\} \\ + \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t-s})} [e^{\int_0^t d\zeta_s(X_{t+h-s}) - \int_0^t d\zeta_s(X_{t-s})} - 1] \right\}.$$

Consequently we have:

$$\langle |u(t+h, x) - u(t, x)|^4 \rangle \leq 8 \left\langle \left| \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t+h-s})} [e^{\int_t^{t+h} d\zeta_s(X_{t+h-s})} - 1] \right\} \right|^4 \right\rangle \\ + 8 \left\langle \left| \mathbb{E}_x \left\{ e^{\int_0^t d\zeta_s(X_{t-s})} [e^{\int_0^t d\zeta_s(X_{t+h-s}) - \int_0^t d\zeta_s(X_{t-s})} - 1] \right\} \right|^4 \right\rangle \\ = 8 \langle |\mathbb{E}_x \{ e^{A_1} (e^{B_1} - 1) \}|^4 \rangle + 8 \langle |\mathbb{E}_x \{ e^{A_2} (e^{B_2} - 1) \}|^4 \rangle,$$

where we set:

$$A_1 = \int_0^t d\zeta_s(X_{t+h-s}) \quad \text{and} \quad B_1 = \int_t^{t+h} d\zeta_s(X_{t+h-s}),$$

and:

$$A_2 = \int_0^t d\zeta_s(X_{t-s}) \quad \text{and} \quad B_2 = \int_0^t d\zeta_s(X_{t+h-s}) - \int_0^t d\zeta_s(X_{t-s}).$$

Each term is controlled by the estimate:

$$\langle |\mathbb{E}_x \{ e^{A_j} (e^{B_j} - 1) \}|^4 \rangle \leq \mathbb{E}_x \{ \langle e^{12A_j} \rangle \}^{1/3} \mathbb{E} \{ \langle (e^B - 1)^6 \rangle \}^{2/3}. \quad (\text{II.26})$$

which follows easily from Hölder's inequality. Indeed, Lemma II.2.1 above can be used to control the first factors of the right hand side of (II.26). In both cases (i.e. for $j = 1, 2$) one gets:

$$\mathbb{E}_x \{ \langle e^{12A_j} \rangle \}^{1/3} \leq e^{24\Gamma_1(0)\mu([0,T] \times [0,T])} \quad (\text{II.27})$$

for all $t \geq 0$ and $h \geq 0$ as long as $t+h \leq T$. The second factor of the right hand side of (II.26) is controlled by the estimate:

$$\langle (e^B - 1)^6 \rangle = e^{18\langle B^2 \rangle} - 6e^{25\langle B^2 \rangle/2} + 15e^{8\langle B^2 \rangle} \\ - 20e^{9\langle B^2 \rangle/2} + 15e^{2\langle B^2 \rangle} - 6e^{\langle B^2 \rangle/2} + 1 \\ = c_0 \langle B^2 \rangle^3 \quad (\text{II.28})$$

which holds for some $c_0 > 0$ and any mean zero Gaussian random variable B as long as $\langle B^2 \rangle$ is bounded. In the first case, the argument proving formula (II.21) gives:

$$\langle B_1^2 \rangle \leq \Gamma_1(0) \mu([t, t+h] \times [t, t+h]),$$

which gives:

$$\mathbb{E}_x \{ \langle (e^{B_1} - 1)^6 \rangle \}^{2/3} \leq c_0 \Gamma_1(0)^2 \mu([t, t+h] \times [t, t+h])^2 \\ \leq c_0 c_T \Gamma_1(0)^2 h^{1+\epsilon}. \quad (\text{II.29})$$

The second case is treated as follows. Using (II.21) and (II.22) we get:

$$\langle B_2^2 \rangle = \langle \left(\int_0^t d\zeta_s(X_{t+h-s}) \right)^2 \rangle + \langle \left(\int_0^t d\zeta_s(X_{t-s}) \right)^2 \rangle \\ - 2 \langle \left(\int_0^t d\zeta_s(X_{t+h-s}) \right) \left(\int_0^t d\zeta_s(X_{t-s}) \right) \rangle \\ = \int_0^t \int_0^t [\Gamma_1(X_{t+h-s_1} - X_{t+h-s_2}) - 2\Gamma_1(X_{t+h-s_1} - X_{t-s_2}) \\ + \Gamma_1(X_{t-s_1} - X_{t-s_2})] \mu(ds_1, ds_2) \\ \leq 2\Gamma_1(0) \int_0^t \int_0^t (1 \wedge |X_{t+h-s_1} - X_{t-s_1}| + 1 \wedge |X_{t+h-s_2} - X_{t-s_2}|) \mu(ds_1, ds_2)$$

where we used the obvious bound:

$$|\Gamma_1(x) - \Gamma_1(y)| \leq 2\Gamma_1(0)(1 \wedge |x - y|).$$

Since the paths X_s change only by jumps of size 1 we have:

$$\langle B_2^2 \rangle \leq 4\Gamma_1(0) c_T N(t+h) h^{(1+\epsilon)/2}.$$

This gives:

$$\mathbb{E}_x \{ \langle (e^{B_2} - 1)^6 \rangle \}^{2/3} \leq 16\Gamma_1(0)^2 c_T^2 \mathbb{E} \{ N(T)^3 \}^{2/3} h^{1+\epsilon}. \quad (\text{II.30})$$

Putting together (II.27), (II.29) and (II.30) gives the desired result. ■

The first important consequence of the estimate (II.25) is that the stochastic field $\{u(t, x); t \geq 0, x \in \mathbb{Z}^d\}$ has a version with continuous sample paths. Henceforth we shall assume that we are working with this specific version.

We now come back to the lower bound (II.8) which was used to prove that the Feynman-Kac formula could give an unbounded solution in the deterministic case. We derive a similar lower bound with a slightly different argument. Jensen's inequality gives:

$$\begin{aligned} u(t, x) &\geq e^{\mathbb{E}_x \left\{ \int_0^t d\zeta_s(X_{t-s}) \right\}} \\ &= e^{\eta_t(x)} \end{aligned}$$

where:

$$\eta_t(x) = \mathbb{E}_x \left\{ \int_0^t d\zeta_s(X_{t-s}) \right\}$$

is a mean zero Gaussian field with covariance:

$$\langle \eta_s(x) \eta_t(y) \rangle = \int_0^s \int_0^t \mathbb{E}_0 \otimes \mathbb{E}_0 \{ \Gamma_1(x - y + X_{t-t_1}^{(1)} - X_{t-t_2}^{(2)}) \} d\mu(dt_1, dt_2)$$

where $X^{(1)}$ and $X^{(2)}$ are independent copies of the continuous time random walk on the lattice $\mathbb{Z}^d$. It is then easy to see that, under mild conditions on the measure $\mu(dt_1, dt_2)$ giving the time covariance and the spatial covariance Γ_1 , that the Gaussian field $\eta_t(x)$ satisfies:

$$\mathbb{P} \left\{ \liminf_{|x| \rightarrow \infty} \frac{|\eta_t(x)|}{\sqrt{1 + \log_+ |x|}} > 0 \right\} = 1.$$

See for example Lemma II.2.4 below for a similar result. This implies that the solution $u(t, x)$ given by the Feynman-Kac formula is unbounded if the initial condition is constant over the lattice.

The estimate (II.25) was used in conjunction with Kolmogorov's criterion to give the existence of a continuous version of the Feynman-Kac formula. It can also be used to give the following tightness result.

Lemma II.2.3 *Let us consider a family of mean zero Gaussian random fields $\{\zeta_t^{(\epsilon)}(x); t \geq 0, x \in \mathbb{Z}^d\}$ parametrized by $\epsilon > 0$ such that, for each $\epsilon > 0$ the covariance $\Gamma^{(\epsilon)}$ of $\zeta_t^{(\epsilon)}(x)$ is given by a tensor product $\Gamma^{(\epsilon)} = \Gamma_0^{(\epsilon)} \otimes \Gamma_1^{(\epsilon)}$ where $\Gamma_0^{(\epsilon)}$ is given by a symmetric measure $\mu^{(\epsilon)}$ on the first quadrant $[0, \infty) \times [0, \infty)$. We assume that:*

$$\sup_{\epsilon > 0} \Gamma_1^{(\epsilon)}(0) < \infty$$

and that assumption (II.24) is satisfied for each $\epsilon > 0$ with a constant c_T independent of ϵ . Then the family of distributions of the corresponding family $\{u^{(\epsilon)}(t, x); t \geq 0, x \in \mathbb{Z}^d\}$ of solution random fields is tight.

We shall use this fact in several instances and in particular in the case when:

$$\Gamma_1^{(\epsilon)}(x) = \Gamma_1(x)$$

for a fixed (bounded) spatial covariance Γ_1 and when the dependence upon ϵ is obtained through a scaling of the time variable t .

II.2.3 The Case of a Homogeneous Potential Field

As before we assume that $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is a mean zero homogeneous Gaussian field with a covariance function $\Gamma = \Gamma_0 \otimes \Gamma_1$ given by a tensor product and such that Γ_0 is integrable. We shall use the same notation as in the previous section.

We showed that the random function $u(t, x)$ defined by the Feynman-Kac formula (II.6) is a solution of the random parabolic equation (I.1) for $\mathbb{P}$ -almost all $\omega \in \Omega$. Indeed, we proved that $u(t, x)$ so defined is almost surely a continuous function of t and consequently, Proposition II.1.1 gives the desired result. This solves the problem of the existence of a solution. The uniqueness problem is solved by proving that the random potential $\xi_t(x)$ satisfies almost surely the condition (II.15) under which we proved uniqueness of the solution. We proceed to the proof of such a property of the random potential. The proof of the following estimate is standard in the theory of homogeneous Gaussian fields and the result we prove is presumably known. We include it for the sake of completeness.

Lemma II.2.4 *If the homogeneous (mean zero) Gaussian field $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ satisfies the above condition and the time covariance Γ_0 is Hölder continuous at 0 in the sense that there exists a $\delta > 0$ such that for every $T > 0$ one has:*

$$|\Gamma_0(t) - \Gamma_0(0)| \leq c_T t^\delta, \quad 0 \leq t \leq T,$$

then for every $T > 0$ we have:

$$\mathbb{P}\left\{\sup_{0 \leq t \leq T, x \in \mathbb{Z}^d} \frac{|\xi_t(x)|}{\sqrt{1 + \log_+ |x|}} < \infty\right\} = 1 \quad (\text{II.31})$$

Note that the assumption on the covariance implies the existence of a version of $\xi_t(x)$ which has continuous sample paths. We shall not need this fact in this section though.

Proof:

We fix $T > 0$ and for each integer $n \geq 1$ and $x \in \mathbb{Z}^d$ we define the event $A_{n,x}$ by:

$$A_{n,x} = \left\{ \sup_{k=0, \dots, 2^n-1} |\xi_{(k+1)T2^{-n}}(x) - \xi_{kT2^{-n}}(x)| \geq q^n \sqrt{2\Gamma_1(0)} \rho \sqrt{1 + \log_+ |x|} \right\}$$

where the constants $0 < q < 1$ and $\rho > 0$ will be chosen later. If ξ denotes any standard $N(0, 1)$ random variable we have:

$$\begin{aligned} \mathbb{P}\{A_{n,x}\} &\leq 2^n \mathbb{P}\left\{|\xi| \geq \frac{q^n \rho \sqrt{1 + \log_+ |x|}}{\sqrt{\Gamma_0(0) - \Gamma_0(T2^{-n})}}\right\} \\ &\leq 2^n \mathbb{P}\left\{|\xi| \geq T^{-\delta} 2^{n\delta} q^n \rho \sqrt{1 + \log_+ |x|}\right\} \\ &\leq c' \exp[n \log 2 - (2^\delta q)^2 n \rho^2 (1 + \log_+ |x|)/2] \end{aligned}$$

and consequently, as long as $2^\delta q > 1$, one has:

$$\sum_{n \geq 1} \mathbb{P}\{A_{n,x}\} \leq c' e^{-\rho^2(1 + \log_+ |x|)/2}$$

and:

$$\sum_{n \geq 1, x \in \mathbb{Z}^d} \mathbb{P}\{A_{n,x}\} < \infty$$

provided ρ is large enough. Here (and in the following) c' stands for a positive constant the value of which may change from line to line. The first Borel-Cantelli lemma implies that at most finitely many of the events $A_{n,x}$ can occur with probability 1. Now, if $0 \leq t \leq T$ we set:

$$\frac{t}{T} = \frac{\epsilon_1}{2} + \frac{\epsilon_2}{2^2} + \frac{\epsilon_3}{2^3} + \dots$$

for a sequence $\epsilon_1, \epsilon_2, \epsilon_3, \dots$ of numbers in $\{0, 1\}$ and we write:

$$\xi_t(x) = \xi_0(x) + [\xi_{\epsilon_1 T/2}(x) - \xi_0(x)] + [\xi_{\epsilon_2 T/2^2}(x) - \xi_{\epsilon_1 T/2}(x)] + [\xi_{\epsilon_3 T/2^3}(x) - \dots]$$

Consequently one has:

$$\begin{aligned} |\xi_t(x)| &\leq |\xi_0(x)| + \rho \sqrt{1 + \log_+ |x|} \sum_{1 \leq n < \infty} q^n \\ &\leq |\xi_0(x)| + c_1 \sqrt{1 + \log_+ |x|} \end{aligned}$$

for some random constant $c = c(\omega)$. One can also apply the first Borel-Cantelli lemma to the $N(0, 1)$ random variables $\xi_0(x)$ and show the existence almost sure of another random constant c_1 for which:

$$|\xi_0(x)| + c_1 \sqrt{1 + \log_+ |x|}$$

for all $x \in \mathbb{Z}^d$. Putting together these two estimates completes the proof. ■

Putting everything together we obtain:

Proposition II.2.5 *Let us assume that the homogeneous (mean zero) Gaussian field $\{\xi_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ has a covariance $\Gamma = \Gamma_0 \otimes \Gamma_1$ of the tensor product type and the time covariance Γ_0 is integrable and Hölder continuous at the origin (in the sense given in the above lemma). Then the random parabolic equation (I.1) has $\mathbb{P}$ -almost surely a unique nonnegative solution $u(t, x)$ and this solution is given by the Feynman-Kac formula (II.6).*

II.2.4 The Case of a White Noise Potential

We now consider the case of a potential process which is a white noise in the time variable. The interpretation of the stochastic equation (I.1) relies on the interpretation of $\xi_t(x) dt$ as $d\zeta_t(x)$ where $\{\zeta_t(x); t \geq 0\}$ is a Wiener process for each x . Indeed, if one rewrites (I.1) as a system of integral equations one gets:

$$u(t, x) = 1 + \int_0^t \kappa[\Delta u(s, \cdot)](x) ds + \int_0^t u(s, x) \xi_s(x) ds$$

or equivalently:

$$u(t, x) = 1 + \int_0^t \kappa[\Delta u(s, \cdot)](x) ds + \int_0^t u(s, x) d\left(\int_0^s \xi_{s'}(x) ds'\right)$$

which can be replaced by the equation:

$$u(t, x) = 1 + \int_0^t \kappa [\Delta u(s, \cdot)](x) ds + \int_0^t u(s, x) \circ d\zeta_s(x). \quad (\text{II.32})$$

We shall see in the next section that this stochastic integral equation has to be understood in the sense of Stratonovich and this is the reason why we used the standard notation $\circ$ in the right most stochastic integral. We shall solve this equation as an evolution equation in the weighted Hilbert space $\mathcal{H}_w = \ell_w^2(\mathbb{Z}^d)$. Given a positive function $w(x)$ on the lattice $\mathbb{Z}^d$, this space is defined as the set of complex functions $u(x)$ satisfying:

$$\|u\|_w = \sqrt{\sum_{x \in \mathbb{Z}^d} w(x) |u(x)|^2} < \infty.$$

This space is a separable Hilbert space for the inner product derived from the polarization of the formula giving its norm. We shall assume that the weight w is such that there exist positive constants c_1 and c_2 for which:

$$c_1 \leq \frac{w(x)}{w(x')} \leq c_2 \quad (\text{II.33})$$

whenever $|x - x'| \leq 1$. Because of this condition, the discrete Laplacian Δ has a bounded extension to the weighted space $\mathcal{H}_w$. It is possible to rewrite the equation (II.32) as an equation in the Ito's sense. See formula (II.36) below. The new equation has exactly the same form. The only change is in the drift term: a linear operator has to be added to the drift $\kappa\Delta$. Consequently, in order to treat both the Stratonovich and the Ito's cases simultaneously, we show existence and uniqueness of the solution of the following equation:

$$u(t, x) = 1 + \int_0^t [Bu(s, \cdot)](x) ds + \int_0^t u(s, x) \circ d\zeta_s(x). \quad (\text{II.34})$$

in the Ito sense by considering a drift term $[Bu(s, \cdot)](x)$ where B is a bounded operator on the Hilbert space $\mathcal{H}_w$. Notice that for any (possibly random) function $v(t, x)$ one has:

$$\begin{aligned} \left\langle \sup_{0 \leq s \leq t} \left\| \int_0^s v(\alpha, \cdot) d\zeta_\alpha(\cdot) \right\|_w^2 \right\rangle &\leq \sum_x w(x) \left\langle \sup_{0 \leq s \leq t} \left(\int_0^s v(\alpha, x) d\zeta_\alpha(x) \right)^2 \right\rangle \\ &= \Gamma_1(0) \int_0^t \langle \|v(s, \cdot)\|_w^2 \rangle ds. \end{aligned}$$

Consequently, if we begin with $u^{(0)} \in \mathcal{H}_w$ and if we set:

$$u^{(n+1)}(t) = u^{(n)}(t) + \int_0^t Bu^{(n)}(s) ds + \int_0^t u^{(n)}(s) d\zeta_s$$

then we get a sequence of random functions with values in the space $\mathcal{H}_w$ and the classical proof can be used in the present infinite dimensional setting to show existence and uniqueness of a solution of equation (II.32) whether it is understood in the Ito or Stratonovich sense.

The above argument is standard. This proof is so short that we reproduced it instead of referring to one of the numerous papers or texts on infinite dimensional stochastic differential equations. Indeed, giving the proof seemed shorter than any attempt we made at checking that the assumptions of a specific theorem were satisfied.

Finally we note that, in order for the constant function $u_0(x) \equiv 1$ to be used as an initial condition for the stochastic equation (II.32) we shall assume that:

$$\sum_x w(x) < \infty. \quad (\text{II.35})$$

We now consider the case of equation (II.32) understood in the Stratonovich sense. This equation can be rewritten in the Ito sense in the form:

$$u(t, x) = 1 + \int_0^t [Bu(s, \cdot)](x) ds + \int_0^t u(s, x) d\zeta_s(x). \quad (\text{II.36})$$

with:

$$B = \kappa\Delta + \Gamma_1$$

where Γ_1 denotes the convolution operator associated to the spatial part of the covariance. If we assume that this covariance is integrable, i.e.

$$\sum_x \Gamma_1(x) < \infty \quad (\text{II.37})$$

then the drift operator B is bounded and the above result applies. We summarize the results of this section in the following:

Theorem II.2.6 *Let us assume that the weight function $w(x)$ satisfies the conditions (II.33) and (II.35). Then the Ito stochastic differential equation (II.36) where the*

stochastic integral is understood in the Ito's sense, has a unique solution, say $u^{(I)}(t, x)$, in the weighted Hilbert space $\mathcal{H}_w$. If moreover the spatial covariance Γ_1 is integrable (i.e. if condition (II.37) is satisfied), then the Stratonovich equation (II.32) also has a unique solution, say $u^{(S)}(t, x)$, in $\mathcal{H}_w$. In this case we have:

$$u^{(I)}(t, x) = e^{-t\Gamma_1(0)/2} u^{(S)}(t, x) \quad (\text{II.38})$$

II.2.5 A Diffusion Limit Approximation Result

The purpose of the present section is to justify the study of the white noise case (presented in the previous section) as a limiting object of more realistic models. We consider a family of Gaussian fields $\{\zeta_t^{(\epsilon)}(x); t \geq 0, x \in \mathbb{Z}^d\}$ parametrized by a parameter $\epsilon > 0$ and we consider the limit $\epsilon \rightarrow 0$ of the corresponding family of solutions of equation (I.1). We shall try to write the results in this general setting, but the typical example we have in mind corresponds to the case:

$$\zeta_t^{(\epsilon)}(x) = \int_0^t \xi_s^{(\epsilon)}(x) ds \quad (\text{II.39})$$

where $\xi_t^{(\epsilon)}(x)$ is a Gaussian field satisfying:

$$\xi_t^{(\epsilon)}(x) = \frac{1}{\epsilon} \xi_{t/\epsilon^2}(x), \quad t \geq 0, \quad x \in \mathbb{Z}^d$$

for some fixed mean zero Gaussian field $\xi_t(x)$. As usual we shall assume that its covariance is the tensor product of a homogeneous time covariance and a homogeneous spatial covariance. The asymptotic regime of the *small parameter* ϵ tending to 0 is usually called "diffusion approximation".

We set the notations and we state the assumptions corresponding to the more general framework in which we want to work. The need for this greater generality comes from a couple of models which are almost of the type described above and which we shall need to investigate in the sequel.

We shall consider various equations of the type (I.1) or (II.32). We shall always restrict ourselves to nonnegative solutions so that uniqueness of the solution holds. Consequently, we consider that a given bounded nonnegative function $u_0(x)$ has been chosen once for all to be used as initial condition for these equations. The cases of interest to us correspond to $u_0(x) \equiv 1$ and to $u_0(x) = \delta_z(x)$ for a fixed site $z \in \mathbb{Z}^d$.

We assume that, for each $\epsilon > 0$, the covariance $\Gamma^{(\epsilon)}$ of $\zeta_t^{(\epsilon)}(x)$ is given by a tensor product $\tilde{\Gamma}^{(\epsilon)} = \tilde{\Gamma}_0^{(\epsilon)} \otimes \Gamma_1^{(\epsilon)}$ where $\tilde{\Gamma}_0^{(\epsilon)}$ is given by a symmetric measure $\mu^{(\epsilon)}$ on the first quadrant $[0, \infty) \times [0, \infty)$ in the sense that:

$$\tilde{\Gamma}_0^{(\epsilon)}(s, t) = \mu^{(\epsilon)}([0, s] \times [0, t]), \quad s, t \geq 0.$$

We also assume that:

$$\sup_{\epsilon > 0} \Gamma_1^{(\epsilon)}(0) < \infty$$

and that assumption (II.24) is satisfied for each $\epsilon > 0$ with a constant c_T independent of ϵ .

For each $\epsilon > 0$ we assume that $\zeta_t^{(\epsilon)}(x)$ is differentiable in t , i.e. that it is of the form (II.39) and we consider the solution $u^{(\epsilon)}(t, x)$ of the equation (I.1) with the random potential $\xi_t^{(\epsilon)}(x)$ defined by formula (II.39). We proved in Lemma II.2.3 that the family of distributions of the corresponding solution random fields $\{u^{(\epsilon)}(t, x); t \geq 0, x \in \mathbb{Z}^d\}$ was tight. Since the random field $\zeta_t^{(\epsilon)}(x)$ is differentiable in time, the solution $u^{(\epsilon)}(t, x)$ is given by the Feynman-Kac formula:

$$u^{(\epsilon)}(t, x) = \mathbb{E}\{u_0(X_t) e^{\int_0^t \xi_s^{(\epsilon)}(X_{t-s}) ds}\}. \quad (\text{II.40})$$

Let us fix an arbitrary sequence $\{\epsilon_n; n \geq 1\}$ of positive numbers converging to 0. Extracting a subsequence if necessary, we can assume without any loss of generality that both $\zeta_t^{(\epsilon)}(x)$ and $u^{(\epsilon)}(t, x)$ converge in distribution. Let us denote the limits by $\zeta_t(x)$ and $u(t, x)$ respectively. Using Skorohod representation theorem, we can assume without any loss of generality that the convergences are in fact almost sure. One may have to redefine all the stochastic processes and fields on a new probability space but this will not affect the rest of the proof. The goal of this section is to identify $u(t, x)$ as the solution of the Stratonovich equation (II.32). We prove:

Theorem II.2.7 *Let us assume (for the sake of simplicity) that $\Gamma_1^{(\epsilon)} = \Gamma_1$, that $u_0(x)$ is a nonnegative function on the lattice $\mathbb{Z}^d$, that assumption (II.24) is satisfied for each $\epsilon > 0$ with a constant c_T independent of ϵ , that $\zeta_t^{(\epsilon)}(x)$ is differentiable in t and that:*

$$\lim_{\epsilon \searrow 0} \tilde{\Gamma}_0^{(\epsilon)}(s, t) = s \wedge t$$

for all $s, t \geq 0$. For each $\epsilon > 0$ we denote by $u^{(\epsilon)}(t, x)$ the solution of the fundamental equation (I.1) with potential $\xi_t(x)$ equal to the time derivative of $\zeta_t^{(\epsilon)}(x)$ and initial condition $u_0(x)$. Then the field $u^{(\epsilon)}(t, x)$ converges in distribution as $\epsilon \searrow 0$ toward the solution of the Stratonovich equation (II.32).

Proof:

The discussion above shows that $u^{(\epsilon)}(t, x)$ is tight. It also shows that it actually converges because $\zeta_t^{(\epsilon)}(x)$ converges in distribution toward an infinite dimensional Brownian motion $\zeta_t(x)$ with covariance $(s \wedge t)\Gamma_1(x - y)$ and because the limit of any subsequence, say $u(t, x)$, is necessarily given by the Feynman-Kac formula. This fact is easily proven by using the almost sure convergence given by the Skorohod representation discussed above and the uniform integrability given by the estimate (II.25). It remains only to identify this $u(t, x)$ as the solution of the Stratonovich equation (II.32). The rest of the proof follows the lines of the finite dimensional case as treated in [22]. It is long and technical so we only give the main ideas. Let us introduce the Gaussian field $\zeta_t^{(n)}(x)$ which is continuous and piecewise linear in t and which satisfies $\zeta_t^{(n)}(x) = \zeta_t(x)$ when t is of the form $k2^{-n}$. Notice that its time derivative $\xi_t^{(n)}(x)$ is given by the formula:

$$\xi_t^{(n)}(x) = 2^n (\zeta_{t_n^+}(x) - \zeta_{t_n}(x))$$

if one uses the notations $t_n = [2^n t]/2^n$ and $t_n^+ = ([2^n t] + 1)/2^n$. We then consider the solution $u^{(n)}(t, x)$ of equation (I.1) for the potential $\xi_t^{(n)}(x)$. It is given by the classical Feynman-Kac formula, i.e.

$$u^{(n)}(t, x) = \mathbb{E}_x \{ u_0(X_t) e^{\int_0^t \xi_{t-s}^{(n)}(X_s) ds} \}.$$

For each fixed sample path X of the random walk, the integral in the exponential converges to the desired quantity. More precisely,

$$\lim_{\epsilon \searrow 0} \int_0^t \xi_{t-s}^{(n)}(X_s) ds = \int_0^t d\zeta_{t-s}(X_s) ds.$$

Moreover, one can compute the moments of the exponential of this integral in the usual way and conclude that the family is uniformly integrable. This implies that, for each fixed t and x , the random variable $u^{(n)}(t, x)$ converges, in all L^p -spaces to the random variable $u(t, x)$ given by the Feynman-Kac formula (II.6). A similar argument shows that the finite dimensional distributions also converge. It remains to prove that

the finite dimensional marginal distributions of $u(t, x)$ coincide with the corresponding marginal distributions of the solution of the Stratonovich equation (II.32). As above, we only argue the case of the single distribution of $u(t, x)$ for t and x fixed. The desired conclusion follows from the fact that:

$$M_t = u(t, x) - u_0(x) - \int_0^t [\Delta + \Gamma_1] u(s, \cdot)(x) ds$$

is a martingale and that:

$$[M, M]_t = \Gamma_1(0) \int_0^t |u(s, x)|^2 ds.$$

We are working with any filtration $\mathcal{F}_t$ which is admissible for the infinite dimensional Brownian motion ζ_t i.e. a filtration with respect to which all of the processes $\{\zeta_t(x); t \geq 0\}$ are Brownian motions. The proof is as follows. First one notices that:

$$\begin{aligned} \langle u(t+h, x) - u(t, x) - \int_t^{t+h} \Delta u(s, x) ds \mid \mathcal{F}_t \rangle \\ = \lim_{n \rightarrow \infty} \langle u^{(n)}(t+h, x) - u^{(n)}(t, x) - \int_t^{t+h} \Delta u^{(n)}(s, x) ds \mid \mathcal{F}_t \rangle \\ = \lim_{n \rightarrow \infty} \langle \int_t^{t+h} \xi_s^{(n)}(x) u^{(n)}(s, x) ds \mid \mathcal{F}_t \rangle \end{aligned}$$

and then one rewrites the integral between t and $t+h$ as a sum of integrals over intervals of the form $[k2^{-n}, (k+1)2^{-n}]$. These integrals are then rewritten as:

$$\begin{aligned} u^{(n)}(k2^{-n}, x) \int_{k2^{-n}}^{(k+1)2^{-n}} \xi_s^{(n)}(x) ds \\ + \int_{k2^{-n}}^{(k+1)2^{-n}} \xi_s^{(n)}(x) [u^{(n)}(s, x) - u^{(n)}(k2^{-n}, x)] ds \\ = u^{(n)}(k2^{-n}, x) \int_{k2^{-n}}^{(k+1)2^{-n}} \xi_s^{(n)}(x) ds \\ + \int_{k2^{-n}}^{(k+1)2^{-n}} \xi_s^{(n)}(x) \int_{k2^{-n}}^s \Delta u^{(n)}(\alpha, x) d\alpha ds \\ + \int_{k2^{-n}}^{(k+1)2^{-n}} \xi_s^{(n)}(x) \int_{k2^{-n}}^s \xi_\alpha^{(n)}(x) u^{(n)}(\alpha, x) d\alpha ds. \end{aligned}$$

The properties of the last integral imply the appearance, in the limit $n \rightarrow \infty$, of the Stratonovich correction $\int_t^{t+h} \Gamma_1 u(s, \cdot)(x) ds$. The details are rather cumbersome. We omit them. ■

II.3 Existence and Equations for the Moments

This last section is devoted to the proof of the existence of the moments of all orders of the solutions $u(t, x)$. The existence proof requires only a soft analysis argument and it applies consequently to both the white noise case and the finite time length correlation.

Let us assume that $x_1, \dots, x_p$ are fixed points of the lattice $\mathbb{Z}^d$ and let us also fix $t > 0$. Under these conditions one has:

$$\begin{aligned} m(t, x_1, \dots, x_p) &= \langle u(t, x_1) \cdots u(t, x_p) \rangle \\ &= \langle \mathbb{E}_{x_1} \{ e^{\int_0^t d\zeta_s(X_{t-s}^{(1)})} \} \cdots \mathbb{E}_{x_p} \{ e^{\int_0^t d\zeta_s(X_{t-s}^{(p)})} \} \rangle \\ &= \mathbb{E}_{x_1} \otimes \cdots \otimes \mathbb{E}_{x_p} \{ \{ e^{\int_0^t d\zeta_s(X_{t-s}^{(1)})} + \cdots + \int_0^t d\zeta_s(X_{t-s}^{(p)}) \} \} \end{aligned}$$

where $\{X_t^{(1)}; t \geq 0\}, \dots, \{X_t^{(p)}; t \geq 0\}$ denote p independent copies of the continuous time random walk on the lattice $\mathbb{Z}^d$ with diffusion coefficient κ . We shall use the following notation. $\mathbf{x} = (x_1, \dots, x_p)$ will denote a generic point of the lattice $\mathbb{Z}^{pd}$, while $\{\mathbf{X}_t = (X_t^{(1)}, \dots, X_t^{(p)}); t \geq 0\}$ denotes the standard continuous time random walk on the lattice $\mathbb{Z}^{pd}$ with diffusion coefficient κ and $\mathbb{E}_{(x_1, \dots, x_p)}\{\cdot\}$ denotes the corresponding expectation $\mathbb{E}_{x_1} \otimes \cdots \otimes \mathbb{E}_{x_p}\{\cdot\}$. Consequently we have:

$$m(t, x_1, \dots, x_p) = \mathbb{E}_{\mathbf{x}}\{e^A\}$$

with:

$$A = \int_0^t d\zeta_s(X_{t-s}^{(1)}) + \cdots + \int_0^t d\zeta_s(X_{t-s}^{(p)}).$$

Notice that, for each fixed path of the random walk, the random variable A is Gaussian with mean 0 and variance:

$$\langle A^2 \rangle = \sum_{i,j=1}^p \langle \int_0^t d\zeta_{t_1}(X_{t-t_1}^{(i)}) \int_0^t d\zeta_{t_2}(X_{t-t_2}^{(j)}) \rangle$$

$$\begin{aligned} &= \sum_{i,j=1}^p \sum_{h,k \geq 0} \langle [\zeta_{t-T_{h+1}^{(i)} \wedge t}(X_h^{(i)}) - \zeta_{t-T_h^{(i)} \wedge t}(X_h^{(i)})] \\ &\quad [\zeta_{t-T_{k+1}^{(j)} \wedge t}(X_k^{(j)}) - \zeta_{t-T_k^{(j)} \wedge t}(X_k^{(j)})] \rangle \\ &= \sum_{i,j=1}^p \sum_{h,k \geq 0} \mu([t - T_{h+1}^{(i)} \wedge t, t - T_h^{(i)} \wedge t] \times [t - T_{k+1}^{(j)} \wedge t, t - T_k^{(j)} \wedge t]) \\ &\quad \Gamma_1(X_h^{(i)} - X_k^{(j)}) \\ &= \int_0^t \int_0^t \sum_{i,j=1}^p \Gamma_1(X_{t-t_1}^{(i)} - X_{t-t_2}^{(j)}) d\mu(dt_1, dt_2). \end{aligned}$$

Consequently:

$$\langle A^2 \rangle \leq p^2 \Gamma_1(0) |\mu|([0, t] \times [0, t])$$

which is finite for every $t \geq 0$. We summarize the result of the above computation in the following:

Proposition II.3.1 *Let us assume that the mean zero Gaussian field $\{\zeta_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ has a covariance $\Gamma = \Gamma_0 \otimes \Gamma_1$ of the tensor product type with a temporal part given by a measure μ on the first quadrant $[0, \infty) \times [0, \infty)$. Then the random field $u(t, x)$ given by the Feynman-Kac formula (II.6) has moments of all orders. They are given by the formula:*

$$m(t, x_1, \dots, x_p) = \mathbb{E}_{(x_1, \dots, x_p)} \{ e^{\frac{1}{2} \int_0^t \int_0^t \sum_{i,j=1}^p \Gamma_1(X_{t-t_1}^{(i)} - X_{t-t_2}^{(j)}) \mu(dt_1, dt_2)} \}, \quad (\text{II.41})$$

whenever $t \geq 0$ and $x_1, \dots, x_p \in \mathbb{Z}^d$.

In the particular case $\zeta_t(x) = \int_0^t \xi_s(x) ds$ formula (II.41) becomes:

$$m(t, x_1, \dots, x_p) = \mathbb{E}_{(x_1, \dots, x_p)} \{ e^{\frac{1}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \}. \quad (\text{II.42})$$

We now concentrate on the limiting case of a white noise in time and we derive a set of closed equations for the moments of the solution. So we restrict ourselves, from now on, to the particular case:

$$\Gamma_0(s - t) = \delta_0(s - t). \quad (\text{II.43})$$

and we prove:

Theorem II.3.2 For each integer $p \geq 1$, each $t \geq 0$ and each $\mathbf{x} = (x_1, \dots, x_p) \in \mathbb{Z}^{pd}$ let us set:

$$m_p(t, \mathbf{x}) = \langle u(t, x_1) \cdots u(t, x_p) \rangle \quad (\text{II.44})$$

where $u(t, x) = u^{(I)}(t, x)$ is the solution of the Ito's equation (II.36). Then these moments satisfy the following parabolic equation:

$$\frac{\partial m_p}{\partial t} = \kappa(\Delta_{x_1} + \cdots + \Delta_{x_p})m_p + \sum_{1 \leq j < k \leq p} \Gamma_1(x_j - x_k)m_p \quad (\text{II.45})$$

with the initial condition $m_p(0, \cdot) \equiv 1$.

It is convenient to write $\Delta_{\mathbf{x}} = \Delta_{x_1} + \cdots + \Delta_{x_p}$ and:

$$V_p(\mathbf{x}) = \sum_{1 \leq j < k \leq p} \Gamma_1(x_j - x_k)$$

for $p \geq 2$ and $V_1(\mathbf{x}) \equiv 0$. With these notations, equation (II.45) becomes:

$$\frac{\partial m_p}{\partial t} = H_p(\kappa)m_p$$

where the operator $H_p(\kappa) = \kappa\Delta_{\mathbf{x}} + V(\mathbf{x})$ is a p -particle Schrödinger operator on the lattice $\mathbb{Z}^{pd}$.

Proof:

The proof we give does not use the explicit form of the moments as given by the Feynman-Kac formula. See the remark below. The proof is slightly longer than the proof discussed in the remark below but it is extremely useful in situations where the Feynman-Kac formula does not hold in a nice form. See for example [1]. It uses Ito's stochastic calculus and the fact that $u(t, x_1), \dots, u(t, x_p)$ are semimartingales. Consequently, if φ is a smooth function of p real variables, Ito's formula gives:

$$\begin{aligned} \varphi(u(t, x_1), \dots, u(t, x_p)) &= \varphi(u(0, x_1), \dots, u(0, x_p)) \\ &+ \sum_{j=1}^p \int_0^t \frac{\partial \varphi}{\partial x_j}(u(s, x_1), \dots, u(s, x_p)) du(s, x_j) \\ &+ \frac{1}{2} \sum_{j,k=1}^p \int_0^t \frac{\partial^2 \varphi}{\partial x_j \partial x_k}(u(s, x_1), \dots, u(s, x_p)) d[u(s, x_j), u(s, x_k)] \end{aligned}$$

and if we choose $\varphi(u_1, \dots, u_p) = u_1 \cdots u_p$ we get:

$$\begin{aligned} u(t, x_1) \cdots u(t, x_p) &= u(0, x_1) \cdots u(0, x_p) \\ &+ \sum_{j=1}^p \int_0^t \kappa u(s, x_1) \cdots u(s, x_{j-1}) \Delta u(s, x_j) u(s, x_{j+1}) \cdots u(s, x_p) ds \\ &+ \sum_{j=1}^p \int_0^t u(s, x_1) \cdots u(s, x_{j-1}) u(s, x_j) u(s, x_{j+1}) \cdots u(s, x_p) d\zeta_s(x) \\ &+ \sum_{1 \leq j < k \leq p} \Gamma_1(x_j - x_k) \int_0^t u(s, x_1) \cdots u(s, x_j) \cdots u(s, x_k) \cdots u(s, x_p) ds \end{aligned}$$

which gives the desired result. ■

Remark:

There is another way to derive the moment equation. It is appropriate for the solution $u(t, x) = u^{(S)}(t, x)$ when the equation is understood in the sense of Stratonovich. Indeed, the solution is then given by the Feynman-Kac formula. In this case, plugging (II.43) into (II.42) gives:

$$m_p(t, x_1, \dots, x_p) = \mathbb{E}_{(x_1, \dots, x_p)} \left\{ e^{\frac{1}{2} \int_0^t \sum_{i,j=1}^p \Gamma_1(X_s^{(i)} - X_s^{(j)}) ds} \right\}.$$

and it is easy from there that the moments $m_p(t, x_1, \dots, x_p)$ of the Stratonovich equation are still solution of the deterministic parabolic equation:

$$\frac{\partial m_p}{\partial t} = \kappa \Delta_{\mathbf{x}} m_p + V^{(S)}(\mathbf{x}) m_p$$

for the potential:

$$V^{(S)}(\mathbf{x}) = \frac{1}{2} \sum_{1 \leq j, k \leq p} \Gamma_1(x_j - x_k).$$

The difference between the Ito's moment equation and the Stratonovich's moment equation is of course consistent with formula (II.38) linking the two solutions.

We close this chapter with a remark concerning the moments of the fundamental equation (I.1) after the ϵ -scaling. In other words, let us set:

$$m^{(\epsilon)}(t, x_1, \dots, x_p) = \langle u^{(\epsilon)}(t, x_1) \cdots u^{(\epsilon)}(t, x_p) \rangle \quad (\text{II.46})$$

for the moments of the solution of the scaled equation. In this case we have:

$$m^{(\epsilon)}(t, x_1, \dots, x_p) = \mathbb{E}_{(x_1, \dots, x_p)} \left\{ e^{\int_0^t \int_0^t \Gamma_0^{(\epsilon)}(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \right\}, \quad (\text{II.47})$$

for each fixed ϵ . It is easy to see that:

$$\lim_{\epsilon \searrow 0} m^{(\epsilon)}(t, x_1, \dots, x_p) = m(t, x_1, \dots, x_p) \quad (\text{II.48})$$

and this is one more reason to study the easier case $\epsilon = 0$ as an approximation of the more realistic but more difficult case $\epsilon > 0$ and small!

Chapter III

MOMENT LYAPUNOV EXPONENTS AND INTERMITTENCY

This chapter is devoted to the discussion of the asymptotic properties of the moments of the solutions of the fundamental equation (I.1). The asymptotic behavior of these moments is exponential. It is determined by a limiting quantity called the average Lyapunov exponent. In the particular case of a white noise in time, we characterize them as the supremum of the spectrum of some N -body Schrödinger operators (on the lattice $\mathbb{Z}^d$). This characterization is of crucial importance because it makes it possible to use perturbation theory and the techniques developed in the spectral theory of many body Schrödinger operators to study the dependence of these Lyapunov exponents upon the diffusion coefficient κ and/or the coupling constant σ .

We explained in the introduction that the notion of intermittency of the solution field $u(t, x)$ is related to the properties of these exponents. We show existence of the Lyapunov exponents in full generality and we study the intermittency of the fields first in the white noise case, and second in some more general models by perturbation techniques.

We assume throughout this chapter that $\{\zeta_t(x); t \geq 0, x \in \mathbb{Z}^d\}$ is a mean zero Gaussian field of the tensor product type in the sense that its covariance function is of the form:

$$\Gamma((s, x), (t, y)) = \tilde{\Gamma}_0(s, t) \Gamma_1(x - y)$$

where $\Gamma_1(x - y)$ is the covariance of a stationary Gaussian field on $\mathbb{Z}^d$ and where $\tilde{\Gamma}_0(s, t)$ the covariance of a Gaussian process with homogeneous increments. In other words, we assume that $\tilde{\Gamma}_0$ is the distribution function of a symmetric μ measure on the first quadrant in the sense that:

$$\tilde{\Gamma}_0(s, t) = \mu([0, s] \times [0, t]).$$

The potential $\xi_t(x)$ appearing in the fundamental equation (I.1) is to be thought of as the time-derivative of this field $\zeta_t(x)$. When $\zeta_t(x)$ is actually differentiable with respect to t , its time derivative $\xi_t(x)$ is also a homogeneous Gaussian field with a covariance of the tensor product type. In this case, the measure μ is of the form:

$$\mu(ds, dt) = \Gamma_0(s - t) ds dt$$

where Γ_0 is the time component of the covariance of $\xi_t(x)$. In the particular case of a time white noise, the measure μ is a multiple of the Lebesgue measure on the first diagonal. In other words one has:

$$\Gamma_0(s - t) = \delta_0(s - t).$$

These are the two most important cases which we shall consider in this chapter.

III.1 The White Noise Case

Throughout this section we assume that:

$$\Gamma_0(s - t) = \delta_0(s - t),$$

and we denote by $u(t, x)$ the solution of the Ito's stochastic equation (II.36).

III.1.1 Existence of the Moment Lyapunov Exponents

The purpose of this subsection is to prove the existence, for each integer $p \geq 1$, of the limit:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) \quad (\text{III.1})$$

for all $p > 0$. This limit is called the p -th moment Lyapunov exponent of the solution $u(t, x)$. Later we study its dependence upon the diffusion parameter κ . We shall see

that it does not depend on the point $x = (x_1, \dots, x_p)$. We use the notation $\gamma_p(\kappa)$ for the limit (III.1).

We first identify this Lyapunov exponent with the spectral radius of the operator $H_p(\kappa)$ occurring in the parabolic equation satisfied by the moment $m_p(t, x)$.

Lemma III.1.1 *Let $n \geq 1$ be an integer, $V(x)$ a real valued bounded function on the lattice $\mathbb{Z}^n$ and let us define the operator $H(\kappa) = \kappa \Delta + V$ as a bounded self-adjoint operator on the Hilbert space $\mathcal{H} = \ell^2(\mathbb{Z}^n)$. We denote by $r^+(\kappa) = r^+(H(\kappa))$ the supremum of the spectrum of the operator $H(\kappa)$. As a function of κ , $r^+(\kappa)$ is nonincreasing continuous and convex and $r^+(0) = \|V\|_\infty$. Moreover, if $\varphi(x)$ is a real valued bounded nonnegative function on the lattice $\mathbb{Z}^n$ which is not identically zero, and if we let $m(t, x)$ be the solution of the parabolic equation:*

$$\frac{\partial m}{\partial t} = H(\kappa)m$$

with the initial condition $m(0, x) = \varphi(x)$, then the limit:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log m(t, x)$$

exists for every $x \in \mathbb{Z}^n$ and is equal to the number $r^+(\kappa)$.

This is a known result in the theory of Schrödinger operators. See for example [21]. We include a proof for the sake of completeness. We use the specifics of the lattice case to this end.

Proof:

We are obviously in the conditions of existence and uniqueness for the parabolic problem and the solution is given by the Feynman-Kac formula:

$$m(t, x) = \mathbb{E}_x \{ \varphi(X_t) e^{\int_0^t V(X_s) ds} \}$$

As before, $\{X_t; t \geq 0\}$ denotes the nearest neighbors continuous time random walk on $\mathbb{Z}^n$ with generator $\kappa \Delta$. We shall also denote by $N(t)$ the number of jumps before time t . If $a > 0$ is a constant (to be chosen later), we have:

$$\begin{aligned} m(t, x) &= \mathbb{E}_x \{ \varphi(X_t) e^{\int_0^t V(X_s) ds}; N(t) \leq at \} \\ &\quad + \mathbb{E}_x \{ \varphi(X_t) e^{\int_0^t V(X_s) ds}; N(t) > at \} \\ &= (i) + (ii) \end{aligned} \quad (\text{III.2})$$



If the constant a is chosen so that $a > 2n\kappa$ then a simple large deviation estimation gives:

$$\mathbb{P}\{N(t) \geq at\} \leq e^{-2n\kappa h(a/2n\kappa)}$$

where the function h is defined by:

$$h(t) = t \log t + 1 - t, \quad t \geq 1.$$

Notice that $h(t) \geq 0$ and that $h(t) \nearrow \infty$ as $t \nearrow \infty$. This estimate implies that:

$$(ii) \leq \|\varphi\|_\infty e^{t\|V\|_\infty} e^{-2n\kappa h(a/2n\kappa)} \quad (III.3)$$

where we used the notation $\|\cdot\|_\infty$ for the supremum of a function on the lattice $\mathbb{Z}^n$. On the other hand:

$$(i) \leq \mathbb{E}_x\{\varphi(X_t) e^{\int_0^t V(X_s) ds}; t < T\}$$

where T denotes the first exit time of the ball $B_t(x) = \{y \in \mathbb{Z}^n; |y - x| \leq at\}$. The above right hand side can be rewritten as $[e^{-tH(\kappa)|_{B_t(x)}} \varphi](x)$ if we denote by $H(\kappa)|_{B_t(x)}$ the restriction of the operator $H(\kappa)$ to the ball $B_t(x)$ (also called the operator $H(\kappa)$ in the ball $B_t(x)$ with Dirichlet boundary condition outside the ball). In any case, this remark gives:

$$(i) \leq \|\varphi\|_{L^2(B_t(x))} e^{r^+(\kappa, B_t(x))t}$$

where $r^+(\kappa, B_t(x))$ denotes the spectral radius of the operator $H(\kappa)|_{B_t(x)}$. Notice first that $r^+(\kappa, B_t(x)) \leq r^+(\kappa)$. Second that:

$$\|\varphi\|_{L^2(B_t(x))} \leq \|\varphi\|_\infty (at)^{n/2}$$

and consequently we have:

$$(i) \leq \|\varphi\|_\infty (at)^{n/2} e^{r^+(\kappa)t} \quad (III.4)$$

and putting together the estimates (III.3) and (III.4) one gets:

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log m(t, x) \leq r^+(\kappa) \quad (III.5)$$

if one recalls (III.2). In order to prove the lower bound:

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log m(t, x) \geq r^+(\kappa) \quad (III.6)$$

one proves that:

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log m(t, x) \geq r^+(\kappa, D)$$

for any finite subset $D \subset \mathbb{Z}^n$. As before, we denote by $r^+(\kappa, D)$ the spectral radius of the restriction $H|_D$ of the operator H to the domain D . The desired lower bound (III.6) will follow from the fact that $r^+(\kappa, D)$ increases to $r^+(\kappa)$ when $D \nearrow \mathbb{Z}^n$. Let D be a fixed finite subset of $\mathbb{Z}^n$. As before for any $x \in D$ one gets:

$$\begin{aligned} m(t, x) &\geq \mathbb{E}_x\{\varphi(X_t) e^{\int_0^t V(X_s) ds}; t < \tau_D\} \\ &= [e^{tH_D} \varphi](x) \\ &= \sum_{j \geq 1} e^{\lambda_{D,j} t} \langle \varphi, \psi_{D,j} \rangle \psi_{D,j}(x) \end{aligned}$$

where we used the notation $\tau_D = T_{D^c}$ for the first exit time of the domain D and where $\lambda_{D,1} \geq \lambda_{D,2} \geq \lambda_{D,3} \geq \dots$ are the eigenvalues of $H|_D$ and $\psi_{D,1}, \psi_{D,2}, \psi_{D,3}, \dots$ the corresponding normalized eigenfunctions. Notice that they are in finite number since the domain D is finite and the restriction $H|_D$ can be viewed as a matrix. Perron-Frobenius theory says that $\lambda_{D,1} > \lambda_{D,2}$ and that $\psi_{D,1}(x) > 0$ for all $x \in D$. Since φ is not identically zero, this implies that $\langle \varphi, \psi_{D,j} \rangle \neq 0$ for D large enough and consequently:

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log m(t, x) \geq \lambda_{D,1}$$

which gives the desired result since $\lambda_{D,1} = r^+(\kappa, D)$. This completes the proof of the lemma.

The remainder of the proof is easier. indeed, the classical variational principle gives:

$$\begin{aligned} r^+(\kappa) &= \sup_{\varphi \in \ell^2(\mathbb{Z}^n), \|\varphi\|=1} \langle H\varphi, \varphi \rangle \\ &= \sup_{\varphi \in \ell^2(\mathbb{Z}^n), \|\varphi\|=1} -\kappa \sum_{x \in \mathbb{Z}^n} \sum_{j=1}^n [\varphi(x + e_j) - \varphi(x)]^2 + \sum_{x \in \mathbb{Z}^n} V(x) |\varphi(x)|^2 \end{aligned}$$

where we used the notation $e_j = (0, \dots, 0, 1, 0, \dots, 0)$ where the 1 is the j -th entry, for the canonical basis of $\mathbb{Z}^n$. It follows that $r^+(\kappa)$ is a concave function of κ which is continuous (since it is finite) and nonincreasing. The formula $r^+(0) = \|V\|_\infty$ is plain. ■

Remark

If the potential function V is summable, then the multiplication operator by the function V is a compact perturbation of $\kappa\Delta$ and Weyl's theorem on the perturbation of the essential spectrum (see for example [6] Chapter II) says that the operators $H(\kappa)$ and $\kappa\Delta$ have the same essential spectrum, i.e.

$$\Sigma_{ess}(H(\kappa)) = \Sigma_{ess}(\kappa\Delta) = [-8\kappa d, 0].$$

In other words, either $r^+(\kappa) = 0$ or $r^+(\kappa)$ is an isolated eigenvalue of finite multiplicity. This implies in particular that $r^+(\kappa)$ is an analytic function of κ on the open domain where it is positive. See for example [18].

Let us come back to the moment equations. We reintroduce the parameter p , and we apply the above results with $n = pd$. The above lemma and Theorem II.3.2 of the previous chapter imply the existence of all the moment Lyapunov exponents $\gamma_p(\kappa)$. Notice that:

$$\gamma_1(\kappa) = r^+(H_1(\kappa)) = 0 \quad (\text{III.7})$$

because $V_1(x) \equiv 0$. Moreover, the case $\kappa = 0$ can be solved completely by inspection. Indeed:

$$\gamma_p(0) = r^+(H_p(0)) = \|V_p\|_\infty = \frac{p(p-1)}{2} \Gamma_1(0), \quad p \geq 2. \quad (\text{III.8})$$

We used the fact that $\|\Gamma_1\|_\infty = \Gamma_1(0)$.

The following *convexity* properties of the Lyapunov exponents will be of crucial importance in the sequel.

$$\frac{\gamma_p(\kappa)}{p} \leq \frac{\gamma_{p+1}(\kappa)}{p+1}, \quad p \geq 0, \kappa > 0. \quad (\text{III.9})$$

This first fact follows immediately from the easy consequence of Hölder's inequality:

$$\langle u(t, x)^p \rangle^{1/p} \leq \langle u(t, x)^{p+1} \rangle^{1/(p+1)}. \quad \blacksquare$$

The second property is hardly more difficult to prove. It reads:

$$\gamma_p(\kappa) \leq \frac{\gamma_{p+h}(\kappa) + \gamma_{p-h}(\kappa)}{2}, \quad p, h, p-h \geq 0, \kappa > 0. \quad (\text{III.10})$$

Indeed, because once more of Hölder's inequality, the obvious formula:

$$u(t, x)^p = u(t, x)^{(p+h)/2} u(t, x)^{(p-h)/2}$$

implies that, :

$$\langle u(t, x)^p \rangle^2 \leq \langle u(t, x)^{p+h} \rangle \langle u(t, x)^{p-h} \rangle$$

and the rest is plain. $\blacksquare$

The following result is an easy consequence of the properties (III.9) and (III.10) which we just proved. It was derived in [10] from a more involved argument.

Theorem III.1.2 *If $\gamma_p(\kappa) > 0$ for some integer $p \geq 1$, then we have:*

$$\frac{\gamma_p(\kappa)}{p} < \frac{\gamma_{p+1}(\kappa)}{p+1} < \frac{\gamma_{p+2}(\kappa)}{p+2} < \dots \quad (\text{III.11})$$

Proof:

Notice that $\gamma_0(\kappa) \equiv 0$ because $\langle u(t, x)^0 \rangle \equiv 1$. Also, for every integer $p \geq 1$, $\gamma_p(\kappa)/p$ is the slope of the line through the origin and the point $(p, \gamma_p(\kappa)/p)$. Let p be the smallest integer for which $\gamma_p(\kappa) > 0$. If $\gamma_p(\kappa)/p = \gamma_{p+1}(\kappa)/(p+1)$ then the point $(p+1, \gamma_{p+1}(\kappa))$ is on the line through the origin and the point $(p, \gamma_p(\kappa)/p)$. But the convexity property (III.10) implies that all the points $(n, \gamma_n(\kappa))$ must be, if $n > p$, above the line through the points $(p-1, 0)$ and $(p, \gamma_p(\kappa))$ and a fortiori above the line through the origin and the point $(p, \gamma_p(\kappa)/p)$. This gives a contradiction. The proof is now complete. $\blacksquare$

The result above justifies the following definition.

Definition III.1.1 *For each $\kappa > 0$ we let p be the smallest integer for which $\gamma_p(\kappa) > 0$. As usual we use the convention that the infimum of the empty set is ∞ . When $p < \infty$ we say that the solution field $u(t, x)$ shows (asymptotic) intermittency of order p and full intermittency when $p = 2$ (remember that $p \geq 2$ because $\gamma_1(\kappa) \equiv 0$).*

The main concerns of this chapter is to determine in which regime (i.e. for which values of κ) the solution $u(t, x)$ exhibits an intermittent behavior in the sense of the above definition.

III.1.2 An Explicitly Solvable Model

Throughout this subsection we restrict ourselves to the particular case of a space-time white noise. In particular, we assume that the space component of the covariance is also given by a delta function, i.e.

$$\Gamma_1(x) = \delta_0(x), \quad x \in \mathbb{Z}^d \quad (\text{III.12})$$

A space-time white noise is usually difficult to handle in the continuous case of $\mathbb{R}^d$, but, because we are considering the lattice $\mathbb{Z}^d$, this assumption is a simplifying factor. It implies for example that the Brownian motion processes $\{\zeta_t(x); t \geq 0\}$ are independent for different $x \in \mathbb{Z}^d$. Moreover, as if the space-time assumption was not restrictive enough, we limit ourselves to the second moment corresponding to $p = 2$. In our defense we shall see later that, because of the inequalities which we proved above, it will be possible to deduce many results from the detailed knowledge gained in this subsection.

In fact, the contents of this subsection are of crucial importance for the understanding of the intermittency phenomenon in general.

We first say a few words about the probabilistic approach. Its starting point is the formula:

$$\gamma_2(\kappa) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_{(0,0)} \{ e^{\int_0^t \Gamma_1(X_s^{(1)} - X_s^{(2)}) ds} \}.$$

It is easy to see that the process $\tilde{X}_t = X_t^{(1)} - X_t^{(2)}$ is a continuous time random walk on the lattice $\mathbb{Z}^d$. It starts from the origin and its generator is $2\kappa\Delta$. Consequently, one shall use the formula:

$$\gamma_2(\kappa) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_0 \{ e^{\int_0^t \Gamma_1(\tilde{X}_s) ds} \}. \quad (\text{III.13})$$

for the investigation of the properties of $\gamma_2(\kappa)$. In other words, $\gamma_2(\kappa)$ is the supremum of the spectrum of the (deterministic) Schrödinger operator:

$$H_2 = 2\kappa\Delta + \Gamma_1.$$

It is possible to rewrite $\gamma_2(\kappa)$ as a limit of expectations over the paths of the standard random walk with generator Δ , say X_s . This is done via a simple scaling argument.

$$\gamma_2(\kappa) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_0 \{ e^{\frac{1}{2\kappa} \int_0^t \Gamma_1(X_s) ds} \}. \quad (\text{III.14})$$

Computations are possible in the particular case of a space white noise for which $\Gamma_1(x - y) = \delta_0(x - y)$. Indeed $\int_0^t \Gamma_1(X_s) ds$ is the time spent at the origin before time t and it can be written as the sum of a random number of independent random variables. It is then possible from there to derive some of the results which we prove below. They depend upon the dimension d . In the probabilistic approach the dependence upon the dimension comes from the transience-recurrence properties of the random walk. But the analytic approach (based on the spectral analysis of the operator H_2) is simpler and we shall stick to it from now on. In this approach $\gamma_2(\kappa)$ is viewed as the supremum of the spectrum of H_2 . Γ_1 is a compact perturbation of the operator $2\kappa\Delta$ and consequently, H_2 and $2\kappa\Delta$ have the same essential spectrum (see for example [18] or [6]). Since $\Sigma(2\kappa\Delta) = [-8d\kappa, 0]$, either $\gamma_2(\kappa) = 0$ or H_2 has positive eigenvalues (of finite multiplicity). We make explicit computations in the particular case of a space white noise for which $\Gamma_1(x - y) = \delta_0(x - y)$. In this case Γ_1 is a rank one perturbation of $2\kappa\Delta$ and there is at most one such positive eigenvalue. Such an eigenvalue λ exists if there is an element $f \in \ell(\mathbb{Z}^d)$ such that $H_2 f = \lambda f$. Because we are interested in the supremum of the spectrum, one can assume without any loss of generality that f is positive. For convenience we normalize f in such a way that $f(0) = 1$. In this case $\Gamma_1 f = \Gamma_1$ and $H_2 f = \lambda f$ can be rewritten in the form:

$$f = (-2\kappa\Delta + \lambda)^{-1} \Gamma_1$$

which implies:

$$1 = (\delta_0, (-2\kappa\Delta + \lambda)^{-1} \delta_0).$$

Using Plancherel formula one gets:

$$1 = \frac{1}{(2\pi)^d} \int_{S^d} \frac{d\varphi}{2\kappa\Phi(\varphi) + \lambda}$$

where $\Phi(\varphi) = 2 \sum_{i=1}^d (1 - \cos \varphi_i)$ is the symbol (in Fourier domain) of the (negative) discrete Laplacian $-\Delta$ and where $\varphi = (\varphi_1, \dots, \varphi_d)$ is a generic element of the d -dimensional torus S^d . Consequently, the supremum of the spectrum of H_2 is a nonnegative solution of the equation:

$$2\kappa = \frac{1}{(2\pi)^d} \int_{S^d} \frac{d\varphi}{\Phi(\varphi) + \lambda_0/(2\kappa)}. \quad (\text{III.15})$$

Theorem III.1.3 *The spectrum $\Sigma(H_2)$ of H_2 satisfies the following alternative:*

a) *Either $\Sigma(H_2) = [-8d\kappa, 0]$.*

b) *or $\Sigma(H_2) = [-8d\kappa, 0] \cup \{\lambda_0(x)\}$, where $\lambda_0(x) > 0$ is the unique positive solution of equation (III.15).*

Equation (III.15) has a positive solution for any $\kappa > 0$ if $d = 1, 2$ and for $\kappa < \kappa_{2,cr}$ if $d \geq 3$. The critical value $\kappa_{2,cr}$ is defined by:

$$\kappa_{2,cr} = \frac{1}{(2\pi)^d} \int_{S^d} \frac{d\varphi}{2\Phi(\varphi)}$$

In particular, alternative a) occurs only when $\kappa \geq \kappa_{2,cr}$ and $d \geq 3$.

Proof:

Let us define the function $a \mapsto I(a)$ by the formula:

$$I(a) = \frac{1}{(2\pi)^d} \int_{S^d} \frac{d\varphi}{a + \Phi(\varphi)}.$$

The variations of $I(a)$ are sketched in the following figure.

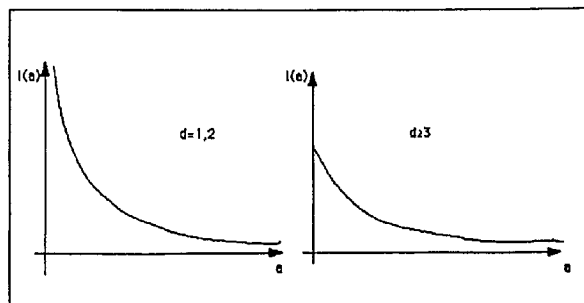


Figure III.1.1 Variations of the function $a \mapsto I(a)$.

Notice that:

$$I(a) \sim \frac{1}{a} \quad \text{as } a \rightarrow \infty$$

in all dimensions $d \geq 1$, that:

$$I(a) \sim \frac{c}{\sqrt{a}} \quad \text{as } a \searrow 0$$

in dimension $d = 1$, that:

$$I(a) \sim \log \frac{1}{a} \quad \text{as } a \searrow 0$$

in dimension $d = 2$ while:

$$\kappa_{cr} = \frac{1}{2} I(0) < \infty$$

in dimension $d \geq 3$. The proof is now complete. ■

Notice also that:

$$I(a) = I(0) - c\sqrt{a} + O(a) \quad \text{as } a \searrow 0$$

in dimension $d = 3$, that:

$$I(a) = I(0) - ca \log \frac{1}{a} + O(a) \quad \text{as } a \searrow 0$$

in dimension $d = 4$ and finally that:

$$I(a) = I(0) - ca + O(a^2) \quad \text{as } a \searrow 0$$

in dimension $d \geq 5$. This information can be translated into properties of the Lyapunov exponent $\gamma_2(\kappa)$. For example one gets:

$$\begin{aligned} \gamma_2(\kappa) &\sim \frac{c}{\kappa}, & \text{for } \kappa \rightarrow \infty \text{ and } d = 1 \\ \gamma_2(\kappa) &\sim 2\kappa e^{-1/(2\kappa)}, & \text{for } \kappa \rightarrow \infty \text{ and } d = 2 \\ \gamma_2(\kappa) &= 0, & \text{for } \kappa \geq \kappa_{cr} = I(0)/2 \text{ and } d \geq 3. \end{aligned}$$

It is not always possible to determine the behavior of $\gamma_p(\kappa)$ for $\kappa \nearrow \kappa_{p,cr}$. The particular case of the second moment for the space-time white noise model is one of the rare instances in which one can actually determine this behavior explicitly. For this reason we summarize the results we proved above on the approach to the critical value in a box:

$$\begin{aligned} \gamma_2(\kappa) &\sim (\kappa - \kappa_{cr})^2, & \text{for } \kappa \nearrow \kappa_{cr} \text{ and } d = 3 \\ \ln \gamma_2(\kappa) &\sim (\kappa_{cr} - \kappa) \ln \frac{1}{\kappa_{cr} - \kappa}, & \text{for } \kappa \nearrow \kappa_{cr} \text{ and } d = 4 \\ \gamma_2(\kappa) &\sim \kappa_{2,cr} - \kappa, & \text{for } \kappa \nearrow \kappa_{cr} \text{ and } d = 5. \end{aligned}$$

These results are summarized in the following figure.

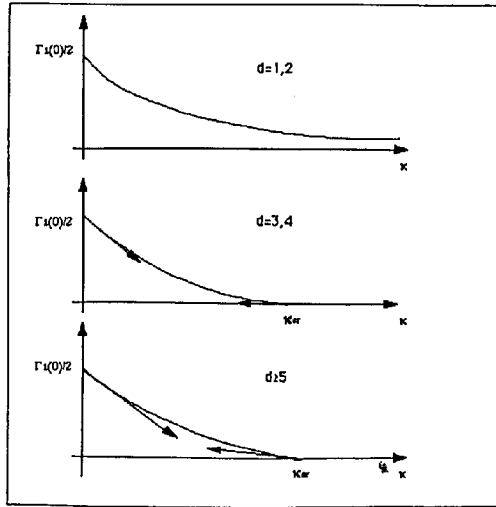


Figure III.1.2 Variations of the second moment Lyapunov exponent $\gamma_2(\kappa)$ as a function of κ .

III.1.3 First General Properties

Notice that a simple scaling argument gives:

$$\gamma_p(\kappa) = r^+(\kappa\Delta + V_p) = \kappa r^+(\Delta + \frac{1}{\kappa}V_p). \quad (\text{III.16})$$

We shall use this formula many times in the sequel. We first prove an elementary lower bound.

Proposition III.1.4 For any $d \geq 1$ and any $p \geq 2$ one has:

$$\frac{\gamma_p(\kappa)}{p} \geq \left(\frac{p-1}{2} \Gamma_1(0) - 2d\kappa \right)_+ \quad (\text{III.17})$$

Proof:

The proof of this lower bound is very easy, and as a consequence one might suspect that a finer estimate may be proved with more work. If one uses Proposition (II.3.1) expressing the moment $m(t, x)$ in terms of an expectation over the paths of the continuous time random walk on $\mathbb{Z}^{pd}$, one gets a lower bound by considering only the paths which did not leave the point x before time t . In this way one gets:

$$\begin{aligned} m(t, x) &= \mathbb{E}_x \left\{ e^{\int_0^t V_p(X_s) ds} \right\} \\ &\geq \mathbb{E}_x \left\{ e^{\int_0^t V_p(X_s) ds}; N(t) = 0 \right\} \\ &= e^{tV_p(0)} \mathbb{P}_0 \{ N(t) = 0 \} \\ &= e^{tV_p(0)} e^{-2d\kappa t} \end{aligned}$$

and this proves the lower bound. ■

We now prove a very useful upper bound:

Proposition III.1.5 If p is of the form $p = \sum_{i=1}^n \beta_i p_i$ for some $\beta_i \geq 1$ and $p_i \geq 2$ satisfying $p_i \neq p_j$ if $i \neq j$, then we have:

$$\gamma_p(\kappa) \leq (p-1) \sum_{i=1}^n \frac{\beta_i}{p_i-1} \gamma_{p_i} \left(\frac{p_i-1}{p-1} \kappa \right) \quad (\text{III.18})$$

and in particular one has:

$$\frac{\gamma_p(\kappa)}{p} \leq \frac{p-1}{2} \gamma_2 \left(\frac{\kappa}{p-1} \right). \quad (\text{III.19})$$

whenever $d \geq 1$ and p is even.

Proof:

Let us consider the partitions $\mathcal{I}$ of the set $\{1, \dots, p\}$ into β_1 groups containing p_1 elements, β_2 groups containing p_2 elements, ... and finally β_n elements with p_n elements each. We shall use the notation:

$$\{1, \dots, p\} = \bigcup_{i=1}^n \bigcup_{j=1}^{\beta_i} I_{i,j}$$

for such a partition $\mathcal{I}$. The β_i disjoint sets $I_{i,j}$ have exactly p_i elements. Notice that there are exactly:

$$P = p! \text{ over } (p_1!)^{\beta_1} (p_2!)^{\beta_2} \dots (p_n!)^{\beta_n} \beta_1! \beta_2! \dots \beta_n!$$

partitions of this type. The factors $\beta_1!, \beta_2!, \dots, \beta_n!$ appear in the denominator because we do not consider the order in which the groups occur. Each integer pair $i < j$ is contained in exactly:

$$P' = P'_1 + \dots + P'_n$$

such partitions where:

$$\begin{aligned} P'_i &= \binom{p-2}{p_i-2} \frac{(p-p_i)!}{(p_1!)^{\beta_1} (p_2!)^{\beta_2} \dots (p_i!)^{k_i-1} \dots (p_n!)^{\beta_n} \beta_1! \beta_2! \dots (\beta_i-1)! \dots \beta_n!} \\ &= \frac{p_i(p_i-1)\beta_i}{p(p-1)} \end{aligned}$$

Finally we set:

$$\lambda_i = \frac{p-1}{P(p_i-1)} \quad (\text{III.20})$$

so that:

$$\sum_{i=1}^n \lambda_i P'_i = 1.$$

With all the above notations we have:

$$\sum_{1 \leq i < j \leq p} \int_0^t \Gamma_1(X_s^{(i)} - X_s^{(j)}) ds = \sum_{\mathcal{I}} \sum_{i=1}^n \lambda_i \sum_{j=1}^{\beta_i} \sum_{i' < j', i', j' \in I_{i,j}} \int_0^t \Gamma_1(X_s^{(i')} - X_s^{(j')}) ds.$$

Consequently:

$$\begin{aligned} m_p(t, \mathbf{x}) &= \mathbb{E}_{\mathbf{x}} \left\{ \prod_{\mathcal{I}} \prod_{i=1}^n \exp \left[\lambda_i \sum_{j=1}^{\beta_i} \sum_{i' < j', i', j' \in I_{i,j}} \int_0^t \Gamma_1(X_s^{(i')} - X_s^{(j')}) ds \right] \right\} \\ &\leq \prod_{\mathcal{I}} \mathbb{E}_{\mathbf{x}} \left\{ \prod_{i=1}^n \exp \left[P \lambda_i \sum_{j=1}^{\beta_i} \sum_{i' < j', i', j' \in I_{i,j}} \int_0^t \Gamma_1(X_s^{(i')} - X_s^{(j')}) ds \right] \right\}^{1/P} \\ &\leq \prod_{\mathcal{I}} \prod_{i=1}^n \mathbb{E}_{\mathbf{x}} \left\{ \exp \left[P \lambda_i \sum_{j=1}^{\beta_i} \sum_{i' < j', i', j' \in I_{i,j}} \int_0^t \Gamma_1(X_s^{(i')} - X_s^{(j')}) ds \right] \right\}^{1/P} \\ &\leq \prod_{i=1}^n \mathbb{E}_{\mathbf{x}} \left\{ \exp \left[P \lambda_i \sum_{j=1}^{\beta_i} \sum_{i' < j', i', j' \in I_{i,j}} \int_0^t \Gamma_1(X_s^{(i')} - X_s^{(j')}) ds \right] \right\} \end{aligned}$$

where we first used Hölder's inequality and second the fact that the factors labeled by i are independent because of the choice of the partitions and finally the fact that the expectations do not depend on the particular partition in $\mathcal{I}$. Let us denote by $R_t(\mathbf{x})$ the quantity in the above right hand side. One can now easily see that:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log R_t(\mathbf{x}) = \lambda_1 P \beta_1 \gamma_{p_1} \left(\frac{\kappa}{\lambda_1 P} \right) + \dots + \lambda_n P \beta_n \gamma_{p_n} \left(\frac{\kappa}{\lambda_n P} \right)$$

which completes the proof of the estimate (III.18) if one recalls the definition (III.20) of the λ_i 's. Finally, we notice that the estimate (III.19) follows immediately from (III.18) by choosing $n = 1$, $p_1 = 2$ and $\beta_1 = p/2$. ■

Remark

Choosing the constants λ_i 's differently it is possible to prove the estimate:

$$\gamma_p(\kappa) \leq (p-1) \sum_{i=1}^n \beta_i \frac{p_i}{p} \gamma_{p_i} \left(\frac{\kappa}{p_i-1} \right)$$

when as before $p = \sum_{i=1}^n \beta_i p_i$, $2 \leq p_1 < p_2 < \dots < p_n$ and $\beta_1, \dots, \beta_n \geq 1$. We shall not prove this estimate because we do not need it in the sequel.

III.1.4 Small κ Behavior of $\gamma_p(\kappa)$

We first consider the small κ behavior of the Lyapunov exponent. We use classical perturbation theory to get the small κ behavior of the largest eigenvalue $\gamma_p(\kappa) = r^+(H_p(\kappa))$ of the operator $H_p(\kappa)$.

Proposition III.1.6 *For every $d \geq 1$ and $p \geq 1$ one has:*

$$\frac{\gamma_p(\kappa)}{p} = \frac{p-1}{2} \Gamma_1(0) - 2d\kappa + O(\kappa^2) \quad (\text{III.21})$$

for $\kappa \rightarrow 0$.

Proof:

We already saw that $\gamma_p(0) = p(p-1)\Gamma_1(0)/2$. Moreover we noticed that the maximum $\|V_p\|$ of the function $V_p(x)$ is attained only for the value $x = 0$, and that this maximum is isolated from the other values of $V_p(x)$. Consequently, since the operator $H_p(0)$ is the operator of multiplication V_p by the function $V_p(x)$, its largest eigenvalue $r^+(H_p(0))$ is equal to the largest value of the function V_p and it has multiplicity 1. In order to control the largest eigenvalue of $H_p(\kappa)$ for small κ we regard the operator $H_p(\kappa) = \kappa\Delta + V_p$ as a small perturbation of the self-adjoint operator V_p . The family $H_p(\kappa)$ is analytic in the sense of Kato and its largest eigenvalue is given by an asymptotic series. See for example the Chapter XII of [19]). We have:

$$\begin{aligned} r^+(H_p(\kappa)) &= r^+(H_p(0)) + \kappa(\Delta\delta_0, \delta_0) + O(\kappa^2) \\ &= \frac{p(p-1)}{2} \Gamma_1(0) + 2pd\kappa + O(\kappa^2) \end{aligned}$$

which is the desired result. ■

Notice that the above result gives a simple alternative proof of the lower bound (III.17). Indeed, the latter follows immediately from the fact that the function $\kappa \mapsto \gamma_p(\kappa)/p$ is a nonnegative convex function and that, by the above proposition, its derivative for $\kappa = 0$ is equal to $-2d$.

III.1.5 Large κ Behavior of $\gamma_p(\kappa)$

We now consider the asymptotic properties of $\gamma_p(\kappa)/p$ for large values of κ . These properties are dimension dependent. We first consider the case $d \geq 3$.

Proposition III.1.7 *If $p \geq 2$ and $d \geq 3$ we have:*

$$\gamma_p(\kappa) = 0, \quad \text{for } \kappa \geq \kappa_{d,p,cr}$$

for some critical constant $\kappa_{d,p,cr}$.

Proof:

We use the formula (III.16) and we consider the properties of the spectral radius $r^+(\Delta + \kappa^{-1}V_p)$. We use a simple lemma which is now standard in the path integral approach to the analysis of Schrödinger operators. It is often called Khasminkii's lemma. See for example [21]. In the present context it can be stated in the following way: if V is a nonnegative function on $\mathbb{Z}^n$ and if X_t is any Markov process in $\mathbb{Z}^n$ and if:

$$\sup_{x \in \mathbb{Z}^n} \mathbb{E}_x \left\{ \int_0^\infty V(X_t) dt \right\} < \infty \quad (\text{III.22})$$

then:

$$\sup_{x \in \mathbb{Z}^n} \mathbb{E}_x \{ e^{\int_0^\infty V(X_t) dt} \} < \infty.$$

We apply this result with $n = pd$ and $V = \kappa^{-1}V_p$. Indeed, if X_t denotes the continuous time random walk in $\mathbb{Z}^{pd}$ with diffusion coefficient $\kappa = 1$, then we have:

$$\begin{aligned} \mathbb{E}_x \left\{ \int_0^\infty |V_p(X_t)| dt \right\} &\leq \sum_{1 \leq j < k \leq p} \mathbb{E}_{(x_1, \dots, x_p)} \left\{ \int_0^\infty |\Gamma_1(X_t^{(j)} - X_t^{(k)})| dt \right\} \\ &= \sum_{1 \leq j < k \leq p} \mathbb{E}_{(x_j - x_k)} \left\{ \int_0^\infty |\Gamma_1(Y_t)| dt \right\} \end{aligned}$$

where $Y_t = X_t^{(j)} - X_t^{(k)}$ is a continuous time random walk in $\mathbb{Z}^{pd}$ with diffusion coefficient $\kappa = 2$. Consequently:

$$\mathbb{E}_x \left\{ \int_0^\infty V_p(X_t) dt \right\} \leq \frac{p(p-1)}{2} \sup_{x \in \mathbb{Z}^d} [G|\Gamma_1|](x)$$

where $G(x, y)$ denotes the Green's kernel of the continuous time random walk in $\mathbb{Z}^d$ with diffusion coefficient $\kappa = 2$. Since Γ_1 is assumed to be integrable,

$$[G|\Gamma_1|](x) = \sum_{y \in \mathbb{Z}^d} G(x, y) \Gamma_1(y)$$

is a bounded function of $x \in \mathbb{Z}^d$ if $d \geq 3$. Consequently, if $d \geq 3$, the assumption (III.22) is satisfied for κ large enough, say for $\kappa \geq \kappa_{d,p,cr}$. Under these conditions, Khasminskii's lemma implies that:

$$\begin{aligned} \gamma_p(\kappa) &= \kappa r^+(\Delta + \kappa^{-1} V_p) \\ &= \kappa \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_x \left\{ e^{\kappa^{-1} \int_0^\infty |V_p(X_t)| dt} \right\} \\ &= 0 \end{aligned}$$

if $\kappa \geq \kappa_{d,p,cr}$. We choose $\kappa_{d,p,cr}$ to be the smallest value of κ for which $\gamma_p(\kappa) = 0$ and, by convexity, we have automatically $\gamma_p(\kappa) > 0$ whenever $\kappa < \kappa_{d,p,cr}$. ■

Hence, when $d \geq 3$, the moment Lyapunov exponents $\gamma_p(\kappa)$ of order $p \geq 2$ vanish for large κ and the strict inequalities (III.11) giving the full intermittency are not satisfied. In other words, intermittency holds only for small values of κ .

We now consider the cases $d = 1$ and $d = 2$. The goal is to prove that the solutions of the parabolic Anderson problem are asymptotically intermittent for all the values of the diffusion constant κ . This will be a consequence of the following result.

Proposition III.1.8 *If $p \geq 2$ and $\kappa \rightarrow \infty$ we have:*

$$\gamma_p(\kappa) \asymp \frac{c_p}{\kappa}, \quad \text{for } d = 1$$

and:

$$\ln \gamma_p(\kappa) \asymp \frac{d_p}{\kappa}, \quad \text{for } d = 2$$

for some constants c_p and d_p .

We use the notation $a_p \asymp b_p$ to mean that $0 < c_1 < a_p/b_p < c_2 < \infty$ for some constants c_1 and c_2 when p is large enough.

Because of the convexity of $\gamma_1(\kappa) = r^+(\kappa)$ as a function of κ , we need only to consider large values of κ . In particular, full intermittency does follow from the proof that $r^+(\kappa) > 0$ for $\kappa \gg 1$. Equivalently, we consider the eigenvalue problem:

$$\Delta \psi = \epsilon \Gamma_1(x) \psi = E \psi \quad (\text{III.23})$$

and we try to prove that it has a solution $E = E(\epsilon) > 0$ for some strictly positive $\psi \in \ell^2(\mathbb{Z}^d)$. We set $\epsilon = 1/\kappa$ and $E = \lambda/\epsilon$ where we use the notation λ for the energy of the original eigenvalue problem for the operator $H = H(\kappa)$. The goal is thus to prove that $E(\epsilon) > 0$ for $\epsilon \ll 1$. We first assume that the covariance Γ_1 has finite support, i.e. $\Gamma_1(x) = 0$ if $|x| > R$ for some positive number R . If we compute the Fourier transform of both sides of equation (III.23) we get:

$$-\Phi(\varphi) \hat{\psi}(\varphi) + \epsilon \sum_{|x| \leq R} \Gamma_1(x) e^{ix\varphi} \psi(x) = E \hat{\psi}(\varphi)$$

or in other words:

$$\hat{\psi}(\varphi) = \epsilon \sum_{|x| \leq R} \frac{\Gamma_1(x) e^{ix\varphi}}{E + \Phi(\varphi)} \psi(x)$$

where, as before we used the notation:

$$\Phi(\varphi) = 2 \sum_{i=1}^d (1 - \cos \varphi_i)$$

for $\varphi = (\varphi_1, \dots, \varphi_d) \in S^d$ and $d = 1$ or $d = 2$. Consequently, taking inverse Fourier transform we get:

$$\psi(x) = \frac{\epsilon}{(2\pi)^d} \sum_{|y| \leq R} \Gamma_1(y) \left(\int_{S^d} \frac{e^{i(y-x)\varphi}}{E + \Phi(\varphi)} d\varphi \right) \psi(y), \quad |x| \leq R,$$

and for the eigenvalue $E = E(\epsilon)$ we have the dispersion relation:

$$\det([\delta_{x,y} - \epsilon I(E, x - y) \Gamma_1(y)]_{x,y}) = 0$$

where:

$$I(E, z) = \frac{1}{(2\pi)^d} \int_{S^d} \frac{e^{iz\varphi}}{E + \Phi(\varphi)} d\varphi.$$

Let us remark that:

$$\lim_{E \rightarrow \infty} I(E, z) = 0$$

and that we have in fact:

$$I(E, z) \sim \frac{c}{\sqrt{E}} \quad \text{for } d = 1$$

and:

$$I(E, z) \sim c \log \frac{1}{E} \quad \text{for } d = 2.$$

These facts follow from the expansion:

$$I(E, z) = I(E, 0) - \frac{1}{(2\pi)^d} \int_{S^d} \frac{1 - \cos(z\varphi)}{E + \Phi(\varphi)} d\varphi = I(E, 0) - \tilde{I}(E, z)$$

where:

$$\tilde{I}(E, z) \sim \tilde{I}(0, z) = \frac{1}{(2\pi)^d} \int_{S^d} \frac{1 - \cos(z\varphi)}{E\Phi(\varphi)} d\varphi \quad \text{as } E \searrow 0.$$

Let us denote by $x_0, \dots, x_n$ the points x of the lattice satisfying $|x| \leq R$. Then the determinant Δ can be rewritten in the form:

$$\Delta = \det \left([\delta_{i,j} - (\epsilon I(E, 0) - \epsilon \tilde{I}(E, x_i - x_j)) \Gamma_1(x_i)]_{i,j} \right).$$

Let us add all the rows with numbers $0, \dots, n-1$ to the n -th row. We get:

$$\begin{aligned} \Delta_n &= \left(1 - \epsilon I(E, 0) \sum_{k=0}^n \Gamma_1(x_k) + \epsilon \sum_{k=0}^n \tilde{I}(E, x_k - x_0) \Gamma_1(x_k) + \dots \right) \\ &= 1 - \epsilon I(E, 0) \sum_{k=0}^n \Gamma_1(x_k) + O(\epsilon). \end{aligned}$$

and the determinant can be shown to satisfy:

$$\Delta = 1 - \epsilon I(E, 0) \sum_{k=0}^n \Gamma_1(x_k) + O(\epsilon).$$

Consequently, if:

$$2\pi \hat{\Gamma}_1(0) = \sum_{k=0}^n \Gamma_1(x_k) > 0,$$

then the equation $\Delta = 0$, (at least for $d = 1$ and $d = 2$) takes the asymptotic form:

$$1 - c \hat{\Gamma}_1(0) \frac{\epsilon}{\sqrt{E}} + O\left(\frac{\epsilon}{\sqrt{E}}\right) + O(\epsilon) = 0$$

which gives:

$$E = E(\epsilon) \sim c \hat{\Gamma}_1(0)^2 \epsilon^2$$

for $d = 1$ and:

$$1 - c \epsilon \hat{\Gamma}_1(0) \log \frac{1}{E} + O(\epsilon \log \frac{1}{E}) = 0$$

which gives:

$$\log \frac{1}{E} \sim \frac{c}{\epsilon \hat{\Gamma}_1(0)}$$

and consequently:

$$E = E(\epsilon) \sim e^{-c/(\Gamma_1(0)\epsilon)}$$

for $d = 2$ provided this last equivalent sign is understood in terms of *logarithmic* equivalents. Notice that here and throughout this work we use the same letter c for a generic constant the value of which can change from line to line, but which does not depend upon the relevant parameters of the asymptotic in question.

We now consider the more complicated case:

$$2\pi \hat{\Gamma}_1(0) = \sum_{k=0}^n \Gamma_1(x_k) = 0.$$

In this case, the last row of the determinant has the form:

$$\Delta_n = \left(1 - \epsilon \sum_{k=0}^n \tilde{I}(E, x_k - x_n) \Gamma_1(x_k) \right).$$

Now, for $k = 0, \dots, n-1$ we multiply this last row Δ_n by $\epsilon I(E, 0)\Gamma_1(x_k)$ and we subtract the result from $\Delta_0, \dots, \Delta_{n-1}$ respectively, then we get:

$$\Delta = \det([\delta_{i,j} - \epsilon \tilde{I}(E, x_i - x_j)\Gamma_1(x_i) + \epsilon^2 H_j(E)I(E, 0)\Gamma_1(x_i)]_{i,j})$$

provided we set:

$$H_j(E) = \sum_{k=0}^n \tilde{I}(E, x_k - x_j).$$

It follows easily that equation $\Delta = 0$ can only be satisfied if $I(E, 0) = O(1/\epsilon^2)$ and in this case one has:

$$\Delta = \det([\delta_{i,j} + \epsilon^2 H_j(E)I(E, 0)\Gamma_1(x_i)]_{i,j}) + O(\epsilon).$$

Notice that:

$$\begin{aligned} H_j(E) &= \frac{1}{(2\pi)^d} \sum_{k=0}^n \Gamma_1(x_k) \int_{S^d} \frac{1 - \cos(x_k - x_j)\varphi}{E + \Phi(\varphi)} d\varphi \\ &= \frac{1}{(2\pi)^d} \int_{S^d} \sum_{k=0}^n \frac{\Gamma_1(x_k) e^{i(x_k - x_j)\varphi}}{E + \Phi(\varphi)} d\varphi \\ &= \frac{1}{(2\pi)^d} \int_{S^d} \frac{\hat{\Gamma}_1(\varphi) e^{i(x_k - x_j)\varphi}}{E + \Phi(\varphi)} d\varphi \end{aligned}$$

and if we use the fact that:

$$\det([\delta_{i,j} + a_i b_j]_{i,j}) = 1 + \sum_{k=0}^n a_k b_k$$

then we get:

$$\begin{aligned} \Delta &= 1 - c\epsilon^2 I(E, 0) \frac{1}{(2\pi)^d} \sum_{k=0}^n \int_{S^d} \frac{\hat{\Gamma}_1(\varphi) e^{ix_k \varphi}}{E + \Phi(\varphi)} \Gamma_1(x_k) d\varphi + O(\epsilon) \\ &= 1 - c\epsilon^2 I(E, 0) \frac{1}{(2\pi)^d} \int_{S^d} \frac{\hat{\Gamma}_1(\varphi)^2}{E + \Phi(\varphi)} d\varphi + O(\epsilon). \end{aligned}$$

Notice that:

$$\lim_{E \searrow 0} \int_{S^d} \frac{\hat{\Gamma}_1(\varphi)^2}{E + \Phi(\varphi)} d\varphi = \int_{S^d} \frac{\hat{\Gamma}_1(\varphi)^2}{\Phi(\varphi)} \Gamma_1(x_k) d\varphi = \alpha(\Gamma - 1)$$

and consequently the equation $\Delta = 0$ has (asymptotically) the form;

$$1 = c\alpha(\Gamma_1) \frac{\epsilon^2}{\sqrt{E}}$$

which gives:

$$E = E(\epsilon) \sim c\epsilon^4 \alpha(\Gamma_1)^2$$

for $d = 1$ and:

$$1 = c\alpha(\Gamma_1) \epsilon^2 \log \frac{1}{E}$$

which gives:

$$E = E(\epsilon) \sim e^{-c/(\alpha(\Gamma_1)\epsilon^2)}$$

for $d = 2$ where (as before) this last equivalent has to be understood in the sense of the logarithmic equivalents.

Remarks:

The above proof concerns the asymptotic $\epsilon \searrow 0$ when the size of the support, i.e. R is fixed. On the other hand, the fact that the function Γ_1 is a correlation function was not used. We only used the fact that the function Γ_1 has finite support and that $\hat{\Gamma}_1(0) \geq 0$.

We now consider the general case:

$$\sum_{x \in \mathbb{Z}^d} |\Gamma_1(x)| < \infty. \quad (\text{III.24})$$

We shall still assume that:

$$\sum_{x \in \mathbb{Z}^d} \Gamma_1(x) \geq 0. \quad (\text{III.25})$$

Lemma III.1.9 Under condition (IV.18) one has:

$$E(\epsilon) \leq c\epsilon^2$$

for ϵ small enough as long as $d = 1$ or $d = 2$.

Proof:

Using Hölder's inequality we get:

$$\begin{aligned} m(t) &= \mathbb{E}_0 \{ e^{\epsilon \int_0^t |\Gamma_1(X_s)| ds} \} \\ &= \mathbb{E}_0 \{ e^{\epsilon \sum_{x \in \mathbb{Z}^d} |\Gamma_1(x)| \int_0^t \delta_x(X_s) ds} \} \\ &\leq \prod_{x \in \mathbb{Z}^d} \mathbb{E}_0 \{ e^{\epsilon |S| \int_0^t \delta_x(X_s) ds} \} |\Gamma_1(x)| / S \end{aligned}$$

provided we set $S = \sum_{x \in \mathbb{Z}^d} |\Gamma_1(x)|$. If $d = 1$ we have:

$$\mathbb{E}_0 \{ e^{\epsilon |S| \int_0^t \delta_x(X_s) ds} \} \leq e^{c\epsilon^2 S^2}$$

and this implies that:

$$E(\epsilon) \leq c\epsilon^2 \left(\sum_{x \in \mathbb{Z}^d} |\Gamma_1(x)| \right)^2.$$

If $d = 2$ a similar computation gives:

$$E(\epsilon) \leq e^{-c/((\sum_{x \in \mathbb{Z}^d} |\Gamma_1(x)|)^2 \epsilon)}.$$

We summarize the above results in the following:

The solution of the Anderson's parabolic problem is asymptotically intermittent if:

$$d = 1, 2 \quad \text{or} \quad (d \geq 3 \quad \text{and} \quad \kappa < \kappa_{2,cr}).$$

III.1.6 Asymptotic Behavior of the Critical Diffusion Constant

It is possible to use formulas (III.17) and (III.19) to get upper and lower bounds on the critical value $\kappa_{p,cr}$ of the diffusion constant in dimension $d \geq 3$. One easily gets:

$$\frac{p-1}{4d} \leq \kappa_{p,cr} \leq (2[\frac{p}{2}] - 1) \kappa_{2,cr}.$$

In other words:

$$\kappa_{p,cr} \asymp p, \quad p \rightarrow \infty.$$

In fact, it is possible to prove a sharper result, namely that $\kappa_{p,cr}/p$ converges to a finite limit. First we notice that Proposition III.1.5 can be used to get more precise estimations of the critical values of the diffusion constant. Indeed, as a straightforward consequence one gets:

Proposition III.1.10 If p is of the form $p = \sum_{i=1}^n \beta_i p_i$ for some $\beta_i \geq 1$ and $p_i \geq 2$ satisfying $p_i \neq p_j$ if $i \neq j$, then we have:

$$\frac{1}{p-1} \kappa_{p,cr} \leq \max_i \frac{1}{p_i-1} \kappa_{p_i,cr}. \quad (\text{III.26})$$

It follows from (III.18) and (III.26) that (for $d \geq 3$)

$$\kappa_{2,cr} < \kappa_{3,cr} < \dots < \kappa_{p,cr} < \dots$$

But we also have:

Proposition III.1.11 Let us define the number $\bar{\kappa}$ by the formula:

$$\bar{\kappa} = \liminf_{p \rightarrow \infty} \frac{1}{p-1} \kappa_{p,cr}.$$

Then one has:

$$\lim_{p \rightarrow \infty} \frac{1}{p} \kappa_{p,cr} = \bar{\kappa}.$$

Proof:

Let $\epsilon > 0$ be arbitrarily small and let us choose p_0 such that:

$$\frac{\kappa_{p_0,cr}}{p_0 - 1} \leq \bar{\kappa} + \epsilon.$$

Notice that:

$$\gamma_{p_0-1}(\kappa) \leq \gamma_{p_0}(\kappa)$$

implies that:

$$\frac{\gamma_{p_0-1}(\kappa)}{p_0 - 1} \leq \frac{\gamma_{p_0}(\kappa)}{p_0 = 1}$$

and

$$\frac{\gamma_{p_0-1}(\kappa)}{p_0 - 2} - \frac{\gamma_{p_0-1}(\kappa)}{(p_0 - 2)(p_0 - 1)} \leq \frac{\gamma_{p_0}(\kappa)}{p_0 = 1}.$$

Consequently, if one chooses $\alpha > 0$ so that $\alpha < p_0\epsilon$, then one has:

$$\frac{\gamma_{p_0-1}(\kappa)}{p_0 - 2} \leq \bar{\kappa} + \epsilon, \quad \frac{\gamma_{p_0-1}(\kappa)}{p_0 - 1} \leq \bar{\kappa} + \epsilon.$$

We now choose p_1 large enough, and since any integer $p > p_1$ can be written in the form:

$$p = \beta_1 p_1 + \beta_2 (p_0 - 1)$$

for some $\beta_1, \beta_2 \geq 1$, then one has:

$$\limsup_{p \rightarrow \infty} \frac{\kappa_{cr}}{p - 1} = \limsup_{p \rightarrow \infty} \frac{\kappa_{cr}}{p} \leq \bar{\kappa}.$$

which completes the proof. ■

In other words, the critical values $\kappa_{p,cr}$ increase linearly (at least in the asymptotic regime $p \rightarrow \infty$) as a function of the order p of the moment. Notice that this p is also the number of particles in the deterministic Schrödinger operator whose supremum of the spectrum is the moment Lyapunov exponent. This leads to the following interpretation of the constant $1/\bar{\kappa}$: it is the minimal coupling energy (for large clusters) which guarantees the stability of the cluster under heat (diffusive) perturbations.

III.1.7 Summary

The results derived above are summarized in the figures 1 and 2 below.

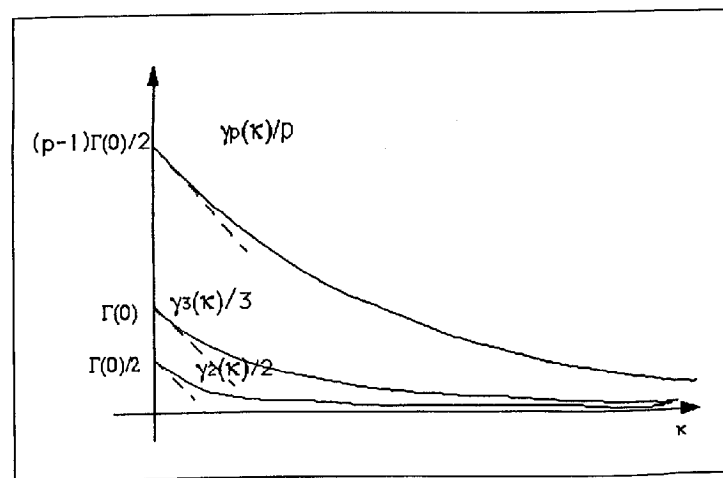


Figure III.1.3 Plot of $\gamma_p(\kappa)/p$ as a function of κ in dimensions $d = 1$ and $d = 2$.

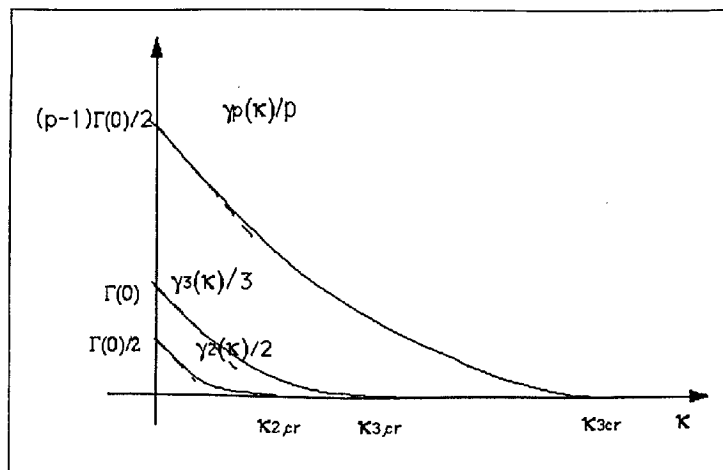


Figure III.1.4 Plot of $\gamma_p(\kappa)/p$ as a function of κ in dimensions $d \geq 3$.

Consequently we have:

- ◊ if $d = 1$ or $d = 2$, full intermittency
- ◊ if $d \geq 3$
 - full intermittency for κ in $[0, \kappa_{2,cr})$
 - intermittency for the moments of order $p \geq 2$ for κ in $[\kappa_{2,cr}, \kappa_{3,cr})$,
 -
 - moments of order greater than or equal to p for κ in $[\kappa_{p,cr}, \kappa_{p+1,cr})$.

The dichotomy between the case $d = 1, 2$ and $d \geq 3$ and small κ on one side and the case $d \geq 3$ and large κ on the other one, is typical of the predictions of Anderson's localization theory in disordered media. See [6] for example.

Notice that, in the case of Stratonovich equation, all the graphs appearing in Figure 1 and Figure 2 have to be shifted in the vertical directions by $\Gamma_1(0)/2$. In particular $\gamma_1(\kappa) \equiv \Gamma_1(0)/2$.

III.2 The Case of Finite Correlation Length

We now consider the case where the field $\zeta_t(x)$ is differentiable in time. Our interest concentrate on the time derivative $\xi_t(x)$ which is a mean zero homogeneous Gaussian field the covariance of which is a tensor product:

$$\Gamma((s, x), (t, y)) = \Gamma_0(s - t) \Gamma_1(x - y), \quad s, t \geq 0, \quad x, y \in \mathbb{Z}^d.$$

One usually assumes that the temporal part of the covariance is an integrable function. Throughout this section we shall need the following stronger condition:

$$\int_{-\infty}^{+\infty} |\tau| \Gamma_1(\tau) d\tau < \infty. \quad (\text{III.27})$$

The existence of the moment Lyapunov exponents is more difficult than in the case of the potential of the white noise type. This existence can be proved in the case of the problem (I.5) as well as in the case of the problem (I.20). Of course, the functional form of the Lyapunov exponent will depend on the particular form of the equation, but the existence will not.

We choose to work with the equation (I.20) for the sake of definiteness. The Feynman-Kac representation of the moments of the solution takes the form:

$$m_p^{(\sigma)}(t, x_1, \dots, x_p) = \mathbb{E}_{(x_1, \dots, x_p)} \left\{ e^{\sigma \int_0^t \int \Gamma_0^{(i)}(t_1 - t_2) \sum_{i \leq j} \Gamma_1(X_{t_1}^i - X_{t_2}^j) dt_1 dt_2} \right\}, \quad (\text{III.28})$$

where the X_t^i are for $i = 1, 2, \dots, p$ independent standard continuous time random walks on $\mathbb{Z}^d$ with generator Δ .

Recall that, when the random potential is a white noise in time, a duality argument can be used to get:

$$\gamma_p(\sigma) = \frac{1}{\sigma^2} \bar{\gamma}_p\left(\frac{1}{\sigma}\right).$$

Under these conditions, one can get the variations of $\bar{\gamma}_p(\sigma)$ from the variations of $\gamma_p(\kappa)$. The following picture gives these variations:

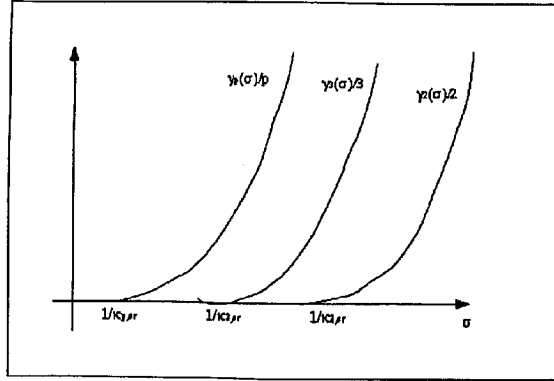


Figure III.2.1 Variations of $\bar{\gamma}_p(\sigma)/p$ in dimensions $d \geq 3$ when the potential is a white noise in time.

The main result of this section is the following.

Theorem III.2.1 For any integer $p \geq 1$ and for any $\sigma > 0$ the limit:

$$\bar{\gamma}_p(\sigma) = \lim_{t \rightarrow \infty} \frac{1}{t} \ln m_p(t, x_1, \dots, x_p)$$

exists and defines a nonnegative increasing continuous convex function $\bar{\gamma}_p(\sigma)$. Moreover, $\bar{\gamma}_p(0) = 0$ and:

$$\lim_{\sigma \rightarrow \infty} \frac{\bar{\gamma}_p(\sigma)}{\sigma} = \widehat{\Gamma}_0(0) \Gamma_1(0) \frac{p^2}{2}. \quad (\text{III.29})$$

Notice that $\bar{\gamma}_p(\sigma)/p \asymp p/2$ and not $(p-1)/2$ as one should expect from the asymptotic of $\gamma_p(\kappa)$ for $\kappa \rightarrow 0$ which we gave in the previous section. What could be considered as a contradiction is due to the fact that we worked with Ito's form of the stochastic integral equations and that in the Stratonovich case we have an additional shift $1/2$ and $\frac{p-1}{2} + \frac{1}{2} = \frac{p}{2}$!!!

III.2.1 Existence of $\bar{\gamma}_p(\sigma)$

Let us fix a large constant $T > 0$ and let us set $t = kT$ for some large integer $k \geq 1$. It is easy to see that:

$$\begin{aligned} & \frac{\sigma}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2 \\ &= \frac{\sigma}{2} \sum_{h=1}^k \int_{(h-1)T}^{hT} \int_{(h-1)T}^{hT} \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2 \\ & \quad + R_p(T, k) \end{aligned}$$

where the remainder term $R_p(T, k)$ can be easily controlled by:

$$\begin{aligned} |R_p(T, k)| &\leq 2k \frac{\sigma}{2} \int_0^\infty \int_0^\infty \Gamma_0(t_1 - t_2) p^2 \Gamma_1(0) dt_1 dt_2 \\ &\leq \Gamma_1(0) k \sigma p^2 \int_{-\infty}^{+\infty} |\tau| |\Gamma_0(\tau)| d\tau \\ &\leq k\theta \end{aligned}$$

if we set $\theta = \Gamma_1(0) \sigma p^2 \int |\tau| |\Gamma_0(\tau)| d\tau$. Consequently, one has:

$$\begin{aligned} & e^{-k\theta} \mathbb{E} \left\{ e^{\frac{\sigma}{2} \sum_{h=1}^k \int_{(h-1)T}^{hT} \int_{(h-1)T}^{hT} \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \right. \\ & \leq m_p(t, x_1, \dots, x_p) \\ & \left. \leq e^{+k\theta} \mathbb{E} \left\{ e^{\frac{\sigma}{2} \sum_{h=1}^k \int_{(h-1)T}^{hT} \int_{(h-1)T}^{hT} \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \right\} \right. \quad (\text{III.30}) \end{aligned}$$

In order to estimate the above expectation we introduce the following symmetric kernel. For each $\mathbf{x} = (x_1, \dots, x_p) \in \mathbb{Z}^{pd}$ and $\mathbf{y} = (y_1, \dots, y_p) \in \mathbb{Z}^{pd}$ we set:

$$q_T(\mathbf{x}, \mathbf{y}) = \mathbb{E}_{\mathbf{x}} \left\{ \delta_{\mathbf{y}}(\mathbf{X}_T) e^{\frac{\sigma}{2} \int_0^T \int_0^T \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \right\}$$

Notice that:

$$\begin{aligned} 0 < q_T(\mathbf{x}, \mathbf{y}) &\leq e^\theta \mathbb{P}_{\mathbf{x}} \{N(T) \geq |\mathbf{x} - \mathbf{y}|\} \\ &\leq e^\theta e^{-\frac{1}{2}|\mathbf{x} - \mathbf{y}| \log |\mathbf{x} - \mathbf{y}|} \end{aligned}$$

provided $|x - y| > T^2$. Indeed:

$$\begin{aligned} \mathbb{P}\{N(T) \geq n\} &\leq \inf_{\alpha} e^{-\alpha n} \langle e^{\alpha N(T)} \rangle \\ &= \inf_{\alpha} e^{-\alpha n} e^{T(e^{\alpha} - 1)} \\ &= e^{-n \log n + n \log T - T} \\ &\leq e^{-\frac{1}{2} n \log n} \end{aligned}$$

provided $n \geq T^2$. This implies that:

$$\sum_{y \in \mathbb{Z}^{pd}} q_T(x, y) < \infty$$

and consequently that the kernel q_T defines an operator Q_T which is bounded on the space $\ell^\infty(\mathbb{Z}^{pd})$. Using the Markov property, the bound (III.30) can now be rewritten in the form:

$$e^{-k\theta} [Q_T^k 1](x) \leq m_p(t, x) \leq e^{k\theta} [Q_T^k 1](x). \quad (\text{III.31})$$

The following result is a straightforward generalization to infinite dimensional spaces of a classical result of the so-called Perron-Frobenius theory of matrices.

Lemma III.2.2 *If a nonnegative self-adjoint operator, say Q , on $\ell^\infty(\mathbb{Z}^m)$ is given by a positive kernel $q(x, y)$, then for any non negative and nonidentically zero function $\psi(x)$ on $\mathbb{Z}^m$ and any $x \in \mathbb{Z}^m$ one has:*

$$\lim_{k \rightarrow \infty} \frac{1}{k} \log [Q^k \psi](x) = r^+(Q)$$

where we use as before the notation $r^+(Q)$ for the spectral radius of the operator Q .

With this in mind we get on one hand:

$$\frac{r^+(Q_T) - \theta}{T} \leq \liminf_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) \leq \limsup_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) \leq \frac{r^+(Q_T) + \theta}{T} \quad (\text{III.32})$$

but on the other hand we also have:

$$e^{-\frac{\sigma}{2} p^2 \Gamma_1(0) \int_0^t \int_0^t |\Gamma_0(t_1 - t_2)| dt_1 dt_2} \leq m_p(t, x) \leq e^{\frac{\sigma}{2} p^2 \Gamma_1(0) \int_0^t \int_0^t |\Gamma_0(t_1 - t_2)| dt_1 dt_2}$$

which in turn implies that:

$$\begin{aligned} & -\frac{\sigma}{2} p^2 \Gamma_1(0) \int |\tau| |\Gamma_0(\tau)| d\tau \\ & \leq \liminf_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) \\ & \leq \limsup_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) \\ & \leq +\frac{\sigma}{2} p^2 \Gamma_1(0) \int |\tau| |\Gamma_0(\tau)| d\tau. \end{aligned}$$

This implies that the limsup and liminf appearing in (III.32) are finite. Since their difference is smaller than θ/T one gets that the limits:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log m_p(t, x) = \lim_{T \rightarrow \infty} \frac{r^+(Q_T)}{T}$$

exist and are equal to a nonnegative number which we shall denote by $\gamma_p(\sigma)$. Moreover the second estimate implies that $\gamma_p(\sigma) = O(\sigma)$ as $\sigma \rightarrow \infty$.

Let us now prove the last claim. Notice that:

$$\log m_p(t, x) = \log \mathbb{E}_x \{e^{\sigma A_t}\}$$

if we set:

$$A_t = \frac{1}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2.$$

This implies that $\log m_p(t, x)$ is a convex function of σ . Indeed:

$$\frac{d}{d\sigma} \log \mathbb{E}_x \{e^{\sigma A_t}\} = \frac{\mathbb{E}_x \{A_t e^{\sigma A_t}\}}{\mathbb{E}_x \{e^{\sigma A_t}\}}$$

and:

$$\frac{d^2}{d\sigma^2} \log \mathbb{E}_x \{e^{\sigma A_t}\} = \frac{\mathbb{E}_x \{e^{\sigma A_t}\} \mathbb{E}_x \{A_t^2 e^{\sigma A_t}\} - \mathbb{E}_x \{A_t e^{\sigma A_t}\}^2}{\mathbb{E}_x \{e^{\sigma A_t}\}^2} \geq 0$$

by Schwarz inequality. Since the limit of a locally uniformly bounded family of convex functions is a continuous convex function we conclude that the limit $\gamma_p(\sigma)$ is also a continuous convex function of σ . Moreover, the main estimate of the existence proof implies that:

$$\bar{\gamma}_p(\sigma) \asymp c_p \sigma$$

for some positive constant c_p because we saw that $\bar{\gamma}_p(\sigma) = O(\sigma)$. Finally we remark that:

$$\bar{\gamma}_p(0) = 0.$$

III.2.2 Estimation of $\bar{\gamma}_p(\sigma)/p$

The purpose of this subsection is to prove the following estimate:

Proposition III.2.3 *There exist positive constants $c^\pm$ such that for each integer $p \geq 1$ and all $\sigma > 0$ one has:*

$$c^- \sigma \leq \frac{\bar{\gamma}_p(\sigma)}{p} \leq c^+ \sigma p.$$

The most important consequence of this estimation (compare with Figure 3) is the positivity of $\bar{\gamma}_p(\sigma)$ for all $\sigma > 0$ and $p \geq 1$ in any dimension $d \geq 1$.

Proof:

The upper bound is trivial. Indeed, the obvious bound:

$$m_p(t, \mathbf{x}) \leq e^{\frac{\sigma}{2} t p^2 \Gamma_1(0)} \int |\Gamma_0(t)| dt$$

implies immediately that:

$$\frac{\bar{\gamma}_p(\sigma)}{p} \leq \frac{1}{2} \Gamma_1(0) \int |\Gamma_0(t)| dt p \sigma. \quad (\text{III.33})$$

The derivation of the lower bound is more involved. Jensen's inequality gives:

$$m_p(t, \mathbf{x}) \geq \exp\left[\frac{\sigma}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) \sum_{i,j=1}^p \mathbb{E}_x\{\Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)})\} dt_1 dt_2\right]$$

and the problem reduces to the estimation of the expectation:

$$\mathbb{E}_x\{\Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)})\}.$$

Notice that for every element $\varphi = (\varphi_1, \dots, \varphi_d)$ of the d -dimensional torus $S_d = [-\pi, +\pi]^d$ one has:

$$\mathbb{E}\{e^{i\varphi \cdot X_t}\} = e^{-2t\Phi(\varphi)}$$

provided we set:

$$\Phi(\varphi) = \sum_{j=1}^d (1 - \cos \varphi_j).$$

Consequently, for $i \neq j$ we have:

$$\begin{aligned} \mathbb{E}_x\{\Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)})\} &= \mathbb{E}_x\left\{\int_{S_d} e^{i\varphi \cdot (X_{t_1}^{(i)} - X_{t_2}^{(j)})} d\nu_1(\varphi)\right\} \\ &= \int_{S_d} e^{-2(t_1+t_2)\Phi(\varphi)} d\nu_1(\varphi) \end{aligned}$$

which is independent of i and j . $\nu_1(d\varphi)$ denotes the spectral measure of the spatial part of the potential field (i.e. $\nu_1(d\varphi)$ is the Fourier transform of the nonnegative definite function Γ_1 on the lattice $\mathbb{Z}^d$). Now if we assume that this spectral measure is continuous and if we set:

$$H_1(t) = \int_{S_d} e^{-2t\Phi(\varphi)} d\nu_1(\varphi) \quad (\text{III.34})$$

then $H_1(t) \rightarrow 0$ as $t \rightarrow \infty$ and:

$$\int_0^t \int_0^t \Gamma_0(t_1 - t_2) H_1(t_1 + t_2) dt_1 dt_2$$

which is bounded in t as $t \rightarrow \infty$. We now consider the case $i = j$. A similar calculation gives:

$$\begin{aligned} \mathbb{E}_x\{\Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(i)})\} &= \int_{S_d} e^{-2|t_1-t_2|\Phi(\varphi)} d\nu_1(\varphi) \\ &= H_1(|t_1 - t_2|) \end{aligned}$$

which is independent of i . Moreover, as $t \rightarrow \infty$ one has:

$$\begin{aligned} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) H_1(|t_1 - t_2|) dt_1 dt_2 &\sim \frac{\sigma}{2} t \int_{-\infty}^{+\infty} \Gamma_0(\tau) H_1(|\tau|) d\tau \\ &= \frac{\sigma t}{4\pi} \int_{-\infty}^{+\infty} \hat{\Gamma}_0(\lambda) \hat{H}_1(\lambda) d\lambda \end{aligned}$$

by Parseval identity. The definition (III.34) of $H_1(t)$ implies that:

$$\hat{H}_1(\lambda) = \frac{2}{\pi} \int_{S^d} \frac{\Phi(\varphi)}{4\Phi(\varphi)^2 + \lambda^2} d\varphi$$

which gives:

$$\int_0^t \int_0^t \Gamma_0(t_1 - t_2) H_1(|t_1 - t_2|) dt_1 dt_2 \sim c \frac{\sigma t}{2\pi^2}$$

where the constant c is given by:

$$c = \int_{-\infty}^{+\infty} \int_{S^d} \frac{\hat{\Gamma}_0(\lambda) \Phi(\varphi)}{4\Phi(\varphi)^2 + \lambda^2} d\lambda d\varphi > 0.$$

Consequently, as $t \rightarrow \infty$ one has:

$$\log m_p(t, x) \geq \frac{\sigma}{2} p(p-1) O(t) + pc \frac{\sigma t}{2\pi^2}$$

which concludes the proof. ■

The above proof, together with the existence proof, contains more information than the statements of the results given above. Indeed, if one sets:

$$\bar{\gamma}_p^{(T)} = \frac{1}{T} \log \mathbb{E}_0 \left\{ e^{\sigma \int_0^T \int_0^T \Gamma_0(t_1 - t_2) \Gamma_1(X_{t_1} - X_{t_2}) dt_1 dt_2} \right\}$$

then the above proofs actually show that:

Theorem III.2.4 For any integer $p \geq 1$ it holds that:

$$\bar{\gamma}_p(\sigma) = \lim_{T \rightarrow \infty} \bar{\gamma}_p^{(T)}(\sigma)$$

and moreover:

$$|\bar{\gamma}_p(\sigma) - \bar{\gamma}_p^{(T)}(\sigma)| \leq C \frac{p^2}{T}$$

for some positive constant C independent of p .

Important Remark

There is an essential difference between the white noise case and the finite correlation case, especially in dimensions $d \geq 3$. Indeed, $\bar{\gamma}_p(\sigma) \equiv 0$ for $0 \leq \sigma \leq 1/\kappa_{p,cr}$ (recall

Figure 3) in the white noise case. But if in the finite correlation time case, then $\bar{\gamma}_p(\sigma) > 0$ for all $\sigma > 0$ and all $p \geq 1$.

It is possible to obtain upper and lower estimates for $\bar{\gamma}_p(\sigma)$ in the case $\epsilon > 0$.

Proposition III.2.5 If $d = 1, 2$ then we have full intermittency for all $\sigma > 0$ since:

$$0 < \bar{\gamma}_1(\sigma) < \frac{\bar{\gamma}_2(\sigma)}{2} < \dots$$

If $d \geq 3$ we have full intermittency for all $\sigma > \sigma_{2,cr}$ for some critical value $\sigma_{2,cr} > 0$ of the coupling constant and:

$$\bar{\gamma}_1(\sigma) \equiv \frac{\bar{\gamma}_2(\sigma)}{2}, \quad 0 \leq \sigma \leq \sigma_{2,cr}.$$

The proof of this result is much more difficult than in the δ -correlated case.

We close this section with a natural conjecture:

there should exist a decreasing sequence $\sigma_{2,cr} > \sigma_{3,cr} > \dots > \sigma_{p,cr} > \dots$ which converges to 0, i.e. such that $\sigma_{p,cr} \rightarrow 0$ when $p \rightarrow \infty$, and such that we have the behavior given in Figure 4

which is a natural generalization of the phenomenon contained in Figure 3 for the δ -correlated case).

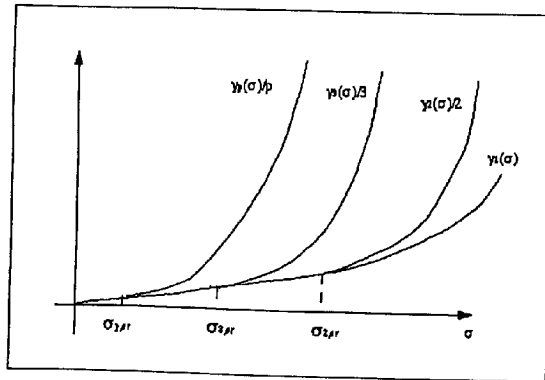


Figure III.2.2 Variations of $\bar{\gamma}_p(\sigma)/p$ in dimensions $d \geq 3$ in the case of finite time correlation.

III.2.3 Continuity Results

We now come back to the dependence upon epsilon of the properties of the moment Lyapunov and we try to prove continuity results for them. So we assume that:

$$\Gamma^{(\epsilon)}(s, t, x, y) = \frac{1}{\epsilon} \Gamma_0\left(\frac{s-t}{\epsilon}\right) \Gamma_1(x, y)$$

for some small parameter ϵ which we let go to 0. It is possible to give asymptotic formulas for $\bar{\gamma}_p(\sigma)$ in the regime $\epsilon \ll 1$. Generalizing a notation already introduced we define the operator $Q_p^{(T, \epsilon)}$ on $\ell^2(\mathbb{Z}^d)$ by its kernel:

$$q_p^{(T, \epsilon)}(x, y) = \mathbb{E}_0 \left\{ e^{\sigma \int_0^T \int_0^T \Gamma_0(t_1 - t_2) \Gamma_1(X_{t_1} - X_{t_2}) dt_1 dt_2} \mid X_t = y \right\} p_T(x, y)$$

where we used the notation $p_t(x, y)$ for the transition density of the standard continuous time random walk on $\mathbb{Z}^d$. We denote by $\lambda_p^{(T, \epsilon)}$ the largest eigenvalue of the operator $Q_p^{(T, \epsilon)}$. The latter is positive (and hence isolated from the rest of the spectrum) for all $\sigma > 0$ in dimensions $d = 1$ and $d = 2$ and for $\sigma > 1/\kappa_{p, cr}$ if $d \geq 3$. Moreover:

$$\lambda_p^{(T, \epsilon)} = e^{T \bar{\gamma}_p^{(T, \epsilon)}(\sigma)}.$$

Consequently, one can use classical perturbation theory of the discrete spectrum to get:

Theorem III.2.6 *One has:*

$$\lim_{\epsilon \searrow 0} \bar{\gamma}_p^{(\epsilon)}(\sigma) = \bar{\gamma}_p(\sigma, 0)$$

for all $\sigma > 0$ if $d = 1, 2$ and for $\sigma > 1/\kappa_{p, cr}$ if $d \geq 3$. We use the notation $\epsilon = 0$ for the case of a random potential which is a white noise in time. Moreover, one can write an asymptotic expansion:

$$\bar{\gamma}_p^{(\epsilon)}(\sigma) = \bar{\gamma}_p^{(\epsilon)}(\sigma, 0) + \epsilon c_1(p, \sigma) + \epsilon^2 c_2(p, \sigma) + \dots \quad (\text{III.35})$$

We do not give the details of the proof. They are very similar to those of the proofs of Theorem III.2.1 and Theorem III.2.4. As we saw earlier, the coefficients $c_i(p, \sigma)$ of the above expansion are given in terms of the eigenvalues and the eigenfunctions of the corresponding p -particle Schrödinger operator with pairwise potential $\Gamma_1(x_1 - x_2)$. It is interesting to notice that these eigenvalues are on degenerated for $\sigma \geq 1/\kappa_{p, cr}$! The expansion (III.35) is easier in the case $p = 1$. Indeed, in this case, the coefficients are given in terms of the moments of spectral measure of Γ_0 .

III.2.4 Lyapunov Exponents as Functions of κ

As we already explained, there is no simple formula linking the Lyapunov exponents $\gamma_p(\kappa)$ and $\hat{\gamma}_p(\sigma)$ of the equations:

$$\frac{\partial u}{\partial t} = \kappa \Delta u + \xi_t(x)u \quad \text{and} \quad \frac{\partial u}{\partial t} = \Delta u + \sqrt{\sigma} \xi_t(x)u.$$

Separate proofs are necessary to derive the existence and the properties of $\gamma_p(\kappa)$ when the time covariance is not a delta function. Recall that in this case one has:

$$m_p(t, 0) = \mathbb{E}_0 \left\{ e^{\frac{1}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) \left(\sum_{i,j=1}^d \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) \right) dt_1 dt_2} \right\} \quad (\text{III.36})$$

where the $X_s^{(i)}$ are independent nearest neighbor continuous time random walks with generators $\kappa \Delta$. The existence of the limit:

$$\gamma_p(\kappa) = \lim_{t \rightarrow \infty} \frac{1}{t} \log m_p(0)$$

is proved basically in the same way as for $\gamma_p(\sigma)$. We do not repeat the argument. The upper bound is also similar. Indeed:

$$m_p(t, 0) \leq e^{\frac{t}{2} p^2 \Gamma_1(0) \int_{-\infty}^{+\infty} |\Gamma_0(\tau)| d\tau}$$

which implies that:

$$\frac{\gamma_p(\kappa)}{p} \leq c + \frac{p}{2} \quad (\text{III.37})$$

for some positive constant c^+ depending only on the covariances Γ_0 and Γ_1 . Moreover, in the same way as before we prove that:

$$\begin{aligned} \frac{\gamma_p(\kappa)}{p} &\geq \frac{1}{2} \int_{-\infty}^{+\infty} |\Gamma_0(\tau)| d\tau \int_{S^d} e^{-2\kappa|\tau|\Phi(\varphi)} \nu_1(d\varphi) \\ &= \frac{1}{2\pi^2} \int_{-\infty}^{+\infty} \int_{S^d} \Gamma_0(\lambda) \frac{2\kappa\Phi(\varphi)}{\lambda^2 + 4\kappa^2\Phi(\varphi)^2} \nu_1(d\varphi) \end{aligned}$$

which is a positive constant. This proves the lower bound. This implies, among other things that:

$$\frac{\gamma_p(\kappa)}{p} \geq \frac{c}{\kappa} \quad \text{when} \quad \kappa \rightarrow \infty.$$

Finally we argue the convexity of the function $\gamma_p(\kappa)/\kappa$. Using the properties of the continuous time random walk one gets:

$$\begin{aligned} m_p(t, 0) &= \sum_{k=0}^{\infty} e^{-2d\kappa p t} \frac{(2d\kappa p t)^k}{k!} \mathbb{E}_0 \left\{ e^{\frac{1}{2} \int_0^t \int_0^t \Gamma_0(t_1 - t_2) (\sum_{i,j=1}^d \Gamma_1(X_{t_1}^{(i)} - X_{t_2}^{(j)}) dt_1 dt_2} \mid N(t) = k \right\} \\ &= e^{-2d\kappa p t} \sum_{k=0}^{\infty} \alpha_k(t) \kappa^k \end{aligned}$$

where the positive numbers $\alpha_k(t)$ do not depend upon κ . Indeed, once the number of jumps is known to be equal to k , the instants of jumps are independent and uniformly distributed over the interval $[0, t]$, and since the distribution of the sites actually visited

by the random walk are independent of κ , the above conditional expectation does not depend upon κ . Consequently:

$$\frac{1}{t} \log m_p(t, 0) = -2dp\kappa + \frac{1}{t} G_t(\theta)$$

if one sets $\kappa = e^\theta$ and:

$$G_t(\theta) = \log \left(\sum_{k=0}^{\infty} \alpha_k(t) e^{\theta k} \right).$$

Notice that, for each fixed $t > 0$, $G_t(\theta)$ is a monotone convex function of θ . Indeed:

$$\frac{d}{d\theta} G_t(\theta) = \frac{\sum_{k=0}^{\infty} k \alpha_k(t) e^{\theta k}}{\sum_{k=0}^{\infty} \alpha_k(t) e^{\theta k}}$$

and consequently:

$$\frac{d^2}{d\theta^2} G_t(\theta) = \frac{\left(\sum_{k=0}^{\infty} \alpha_k(t) e^{\theta k} \right) \left(\sum_{k=0}^{\infty} k^2 \alpha_k(t) e^{\theta k} \right) - \left(\sum_{k=0}^{\infty} k \alpha_k(t) e^{\theta k} \right)^2}{\left(\sum_{k=0}^{\infty} \alpha_k(t) e^{\theta k} \right)^2}$$

which is nonnegative by Schwarz inequality. Moreover equality holds if and only if $\alpha_k(t) = 0$ for all $k \geq 1$ which is not possible in the present situation. Consequently we proved that:

$$\frac{d^2}{d\theta^2} G_t(\theta) = \frac{d^2}{d\theta^2} \left(\frac{1}{t} \log m_p(t, 0) \right) > 0$$

and that:

$$\frac{d}{d\theta} \left(\frac{1}{t} \log m_p(t, 0) \right) = -2dp e^\theta + \frac{d}{d\theta} G_t(\theta)$$

where the second term in the right hand side is positive. Hence we proved that:

$$\gamma_p(\kappa) = \lim_{t \rightarrow \infty} \log m_p(t, 0)$$

is a continuous function of κ which can be written in the form:

$$\frac{\gamma_p(\kappa)}{p} = -2d\kappa + h(\log \kappa)$$

where the first and the second derivatives of the function h are positive.

The study of the highest Lyapunov exponent is of particular interest. See [16] for some applications to oceanography.

III.2.5 Another Explicitly Solvable Model

This subsection is devoted to the complete analysis of an explicitly solvable model of a random potential function which is not a white noise in time. The time dependence is simple: the potential is a piecewise constant function of time. More precisely we assume that there exist a sequence $\{\xi_n; n \geq 0\}$ of identically distributed mean zero ergodic Gaussian fields and a positive number τ for which:

$$\xi_t(x) = \frac{1}{\sqrt{\tau}} \xi_n(x) \quad \text{if} \quad n\tau \leq t < (n+1)\tau. \quad (\text{III.38})$$

The factor $1/\sqrt{\tau}$ is chosen for convenience. As usual we denote by Γ_1 the spatial covariance of the potential, i.e.

$$\langle \xi_n(x) \xi_m(y) \rangle = \delta_{m,n} \Gamma_1(x-y), \quad n, m \geq 0, \quad x, y \in \mathbb{Z}^d.$$

Throughout this subsection we shall assume that the fields ξ_n are delta correlated in space, namely that:

$$\Gamma_1(x-y) = \delta(x-y).$$

Obviously, τ can be interpreted as the time correlation length of the model. We shall prove below that, as announced in the introduction, the relevant parameter is the product $\kappa\tau$.

Let us denote by $q(0, \tau, x, y)$ the fundamental solution of the parabolic equation in the interval $[0, \tau]$. As usual, the Feynman-Kac formula can be used to express it in terms of the continuous time random walk X_s with generator $\kappa\Delta$:

$$q(0, \tau, x, y) = \mathbb{E}_x \{ \delta_\tau(X_\tau) e^{\int_0^\tau \xi_0(X_s) ds} \}.$$

We shall make use of the following fundamental remark: the fundamental solutions $q(n\tau, (n+1)\tau, \cdot, \cdot)$ are independent (over the probability space of the random potential) and identically distributed. Notice that the random potential is not stationary in time since it is piecewise constant. It is nevertheless homogeneous with respect to the time-shifts by integer multiples of τ . This last fact, together with the constancy of

the potential over the intervals $[n\tau, (n+1)\tau]$ implies that the first moment $m_1(t, x)$ is independent of x , i.e.:

$$m_1(t, x) = \langle u(t, x) \rangle \equiv m_1(t), \quad t > 0$$

and the limit:

$$\gamma_1^{(\tau)}(\kappa) = \lim_{t \rightarrow \infty} \frac{1}{t} \log m_1(t)$$

exists and is equal to:

$$\begin{aligned} \gamma_1^{(\tau)}(\kappa) &= \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log m_1(n\tau) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log m_1(\tau)^n \\ &= \frac{1}{\tau} \log m_1(\tau). \end{aligned}$$

Moreover:

$$\begin{aligned} m_1(\tau) &= \langle \mathbb{E}_0 \{ e^{\frac{1}{\sqrt{\tau}} \int_0^\tau \xi_0(X_s) ds} \} \rangle \\ &= \langle \mathbb{E}_0 \{ \exp \left[\frac{1}{\sqrt{\tau}} \sum_{x \in \mathbb{Z}^d} \xi_0(x) \ell_\tau(x) \right] \} \rangle \end{aligned}$$

where:

$$\ell_t(x) = \int_0^t \delta_x(X_s) ds$$

denote the total time spent by the random walk at the site x before time t . Using Fubini's theorem to interchange the two expectations and using the formula for the Laplace transform of a Gaussian random variable we get:

$$\begin{aligned} m_1(\tau) &= \mathbb{E}_0 \{ \exp \left[\frac{1}{2\tau} \left(\sum_{x \in \mathbb{Z}^d} \xi_0(x) \ell_\tau(x) \right)^2 \right] \} \\ &= \mathbb{E}_0 \{ \exp \left[\frac{1}{2\tau} \sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 \right] \} \end{aligned}$$

because we assume that the fields ξ_n are delta-correlated in space as well as in time. Obviously one has:

$$\sum_{x \in \mathbb{Z}^d} \ell_\tau(x) = \tau$$

and consequently:

$$\sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 = \tau^2$$

and:

$$\frac{1}{2\tau} \sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 = \frac{1}{\tau}.$$

We shall use the following elementary result, the proof of which we omit.

Lemma III.2.8 *If ξ is a random variable satisfying $0 \leq \xi \leq \epsilon$, then:*

$$\mathbb{E}\{\xi\} = 1 + (1 + O(\epsilon))\mathbb{E}\{\xi\}.$$

This implies that, for $\tau \searrow 0$ one has:

$$\begin{aligned} \gamma_1^{(\tau)}(\kappa) &= \frac{1}{\tau} \log \mathbb{E}_0 \left\{ \exp \left[\frac{1}{2\tau} \sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 \right] \right\} \\ &= \frac{1 + O(\tau)}{\tau} \mathbb{E}_0 \left\{ \sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 \right\} \end{aligned}$$

But:

$$\begin{aligned} \mathbb{E}_0 \{ \ell_\tau(x)^2 \} &= \mathbb{E}_0 \left\{ \int_0^\tau \int_0^\tau \delta_x(X_s) \delta_x(X_t) ds dt \right\} \\ &= 2 \int \int_{0 < s < t < \tau} \mathbb{E}_0 \{ \delta_x(X_s) \delta_x(X_t) \} ds dt \\ &= 2 \int \int_{0 < s < t < \tau} p_s(0, x) p_{t-s}(x, x) ds dt \\ &= 2 \int \int_{0 < s < t < \tau} p_s(0, x) p_{t-s}(0, 0) ds dt \end{aligned}$$

where we used the notation $p_t^{(\kappa)}(x, y)$ for the transition probability of the continuous time random walk with generator $\kappa \Delta$. After summation we get:

$$E E_0 \left\{ \sum_{x \in \mathbb{Z}^d} \ell_\tau(x)^2 \right\} = 2 \int_0^\tau (\tau - s) p_s(0, 0) ds.$$

Consequently, we proved that, uniformly in the diffusion parameter κ , one has:

$$\gamma_1^{(\tau)}(\kappa) = \frac{1}{\tau^2} \int_0^\tau (\tau - s) p_s(0, 0) ds (1 + O(\tau)).$$

Recall that:

$$p_s(0, 0) = \frac{1}{(2\pi)^d} \int_{S^d} e^{-\kappa \Phi(\varphi)} d\varphi$$

where:

$$\Phi(\varphi) = 2 \sum_{i=1}^d (1 - \cos \varphi_i) = \sum_{i=1}^d \sin^2 \frac{\varphi_i}{2} \quad (\text{III.39})$$

It follows from (III.39) that:

$$p_s(0, 0) = 1 - 2d\kappa s + O(\kappa^2 s^2)$$

for $\kappa \ll 1$, and consequently, in any dimension $d \geq 1$:

$$\gamma_1^{(\tau)}(\kappa) = \frac{1}{2} - \frac{2d\kappa}{\tau^2} \int_0^\tau s(\tau - s) ds + O(\tau^2 s^2)$$

and hence:

$$\gamma_1^{(\tau)}(\kappa) = \frac{1}{2} - \frac{d}{3} \kappa \tau + O(\tau^2 s^2) \quad (\text{III.40})$$

So, for $\tau \searrow 0$, not only the continuity property:

$$\gamma_1^{(\tau)}(\kappa) \rightarrow \gamma_1^{(0)}(\kappa)$$

for κ restricted to any fixed interval $[0, \kappa_0]$, but we also have the same convergence for κ restricted to intervals of the form $[0, O(\tau^{-1})]$. In any case, one should have a different

limiting regime when $\kappa\tau \rightarrow \infty$ and $\kappa \gg 1/\tau$. In this case, the asymptotic behavior of the first moment Lyapunov exponent depends on the dimension d as we are about to show. Using the Laplace method one to estimate integrals of exponentials (recall equation (III.39)) one sees that:

$$p_s(0,0) \sim \frac{1}{(2\pi\kappa s)^{d/2}} \quad (\text{III.41})$$

for $\kappa s \gg 1$. In this case one has:

$$\begin{aligned} \gamma_1^{(\tau)}(\kappa) &\sim 1\tau^2 \int_0^\tau \tau p_s(0,0) ds \\ &= \frac{1}{\tau\kappa} \int_0^{\tau\kappa} p_s^{(1)}(0,0) ds. \end{aligned}$$

Consequently:

In the regime $\kappa\tau \gg 1$ one has:

$$\begin{aligned} \gamma_1^{(\tau)}(\kappa) &\sim \frac{c_1}{\sqrt{\kappa\tau}}, & \text{for } d=1, \\ \gamma_1^{(\tau)}(\kappa) &\sim \frac{c_2 \log(\kappa\tau)}{\sqrt{\kappa\tau}}, & \text{for } d=2, \\ \gamma_1^{(\tau)}(\kappa) &\sim \frac{c_d}{(\kappa\tau)^{d-2}}, & \text{for } d \geq 3, \end{aligned}$$

In either case, the graph of the first moment Lyapunov exponent has the form given in Figure III.2.5 below.

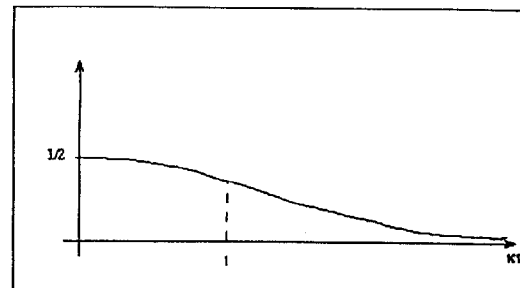


Figure III.2.5 Sketch of the variations of $\gamma_1^{(\tau)}(\kappa)$.

We close with the following important remark. When the random potential $\xi_t(x)$ is delta-correlated in time one has $\gamma_1(\kappa) \equiv 1/2$ when the fundamental equation is understood in the Stratonovich sense (and the space correlation is also given by the delta function). This corresponds to the case $\tau = 0$. But we just proved that, when $\tau > 0$, then:

$$\begin{aligned} \lim_{\kappa\tau \searrow 0} \gamma_1^{(\tau)}(\kappa) &= 1/2 \\ \lim_{\kappa\tau \nearrow \infty} \gamma_1^{(\tau)}(\kappa) &= 0 \end{aligned}$$

in all dimension $d \geq 1$!!!

Chapter IV

ALMOST SURE LYAPUNOV EXPONENTS

This chapter is devoted to the analysis of the almost sure asymptotic behavior of the solutions of the fundamental equation (1.1). We first prove the existence of the almost sure Lyapunov exponent when the initial condition $u_0(x)$ is localized. This Lyapunov exponent is shown to be independent of the initial condition and to be bounded from above by the moment Lyapunov exponents. Most of our efforts are devoted to the proof of a sharp asymptotic estimate when the diffusion coefficient κ tends to 0.

We assume throughout this chapter that the random potential $\{\xi_t(x)\}$ is delta correlated in time, or in other words, that the antiderivatives $\{\zeta_t(x); t \geq 0\}$ are Brownian motions.

IV.1 Existence

We now consider the more difficult problem of the existence and the analysis of the almost sure Lyapunov exponents. As before $u(t, x)$ denotes the unique solution of the problem (1.5). Unfortunately, some of the proofs of this chapter require that the initial condition $u_0(t)$ has compact (i.e. finite) support. We believe that all the results still hold for the initial condition $u_0(t) \equiv 1$ but a complete proof is still escaping us. For each $0 < s < t$ and $x, y \in \mathbb{Z}^d$ we consider the quantity:

$$q(s, x, t, y) = \mathbb{E}\{e^{\int_s^t d\zeta_\alpha(X_{t+s-\alpha})d\alpha} | X_s = x, X_t = y\} p_{t-s}(x, y). \quad (\text{IV.1})$$

The above expectation is computed for each (fixed) sample realization of the potential random field $\zeta_t(x)$. We shall sometime use the special notation:

$$q(s, x, u, z) = q^{(\zeta)}(s, x, u, z)$$

to emphasize the dependence of q upon the sample realization of the random field ζ . The definition of q involves an expectation over the sample paths of the continuous time random walk on $\mathbb{Z}^d$ when the paths are conditioned to be at x at time s and at y at time t . Recall that the notation $p_t(x, y)$ is used for the transition probability of this random walk. Notice that, by definition of the conditional expectation, we have:

$$\sum_{y \in \mathbb{Z}^d} u_s(y) q(s, x, t, y) = \mathbb{E}\{u_s(X_t) e^{\int_s^t d\zeta_\alpha(X_{t+s-\alpha})d\alpha} | X_s = x\}$$

which the solution $u(t, x)$ at time t when this solution is equal to $u_s(\cdot)$ at time s . Consequently, $q(s, x, t, y)$ so-defined can be regarded as the fundamental solution of our initial value parabolic problem. Indeed, when the potential field is actually the derivative of a differentiable function of t , i.e. when $\xi_t(x)dt = d\zeta_t(x)$, then it is easy to check that the function $q(s, x, t, y)$ is the solution, for each fixed $s > 0$ and $x \in \mathbb{Z}^d$ of the equation:

$$\frac{\partial q(s, x, t, y)}{\partial t} = \kappa \Delta_y q(s, x, t, y) + \xi_t(y) q(s, x, t, y)$$

with initial condition $q(s, x, s, y) = \delta_x(y)$. Notice that this fundamental solution is random. We are concerned with its almost sure properties. The following "semigroup" property follows easily from the Markov property of the random walk:

$$q^{(\zeta)}(s, x, u, z) = \sum_{y \in \mathbb{Z}^d} q^{(\zeta \cdot + (u-s))}(s, x, t, y) q^{(\zeta \cdot - (t-s))}(t, y, u, z)$$

where we used the notation $\zeta \cdot + \beta$ for the function $(\zeta \cdot + \beta)_t(x)$ defined by:

$$(\zeta \cdot + \beta)_t(x) = \zeta_{t+\beta}(x).$$

In any case, if x is fixed, one has:

$$q^{(\zeta)}(s, x, u, x) \geq q^{(\zeta \cdot + (u-s))}(s, x, t, x) q^{(\zeta \cdot - (t-s))}(t, x, u, x).$$

and:

$$q^{(\zeta)}(s, x, u, x) \geq q^{(\zeta)}(s, x, t, z) q^{(\zeta)}(t, z, u, y). \quad (\text{IV.2})$$

whenever $s < t < u$. Consequently, it is possible to use the stationarity in time of the random field ζ and Kingman's subadditive ergodic theorem to conclude that almost surely over the sample path of ζ , for all $x \in \mathbb{Z}^d$ and $s > 0$ the limit:

$$\lim_{t \rightarrow \infty} \frac{\ln q(s, x, t, x)}{t} = \tilde{\gamma}(\kappa)$$

exists. We call this limit the almost sure Lyapunov exponent of the problem. Using the super additivity (IV.2) in several forms, it is easy to check that the limit:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \ln q(s, x, t, y)$$

exists for all $s > 0$ and all $x, y \in \mathbb{Z}^d$ and that this limit is still equal to $\tilde{\gamma}(\kappa)$. In other words, this limit does not depend upon the choices of s, x and y . The Lyapunov exponent $\tilde{\gamma}(\kappa)$ as defined corresponds to an initial condition with a support concentrated on a point. It is easy to check that, if the support of the initial condition $u_0(t)$ is finite, the limit:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log u(t, x)$$

still exists and is equal to $\tilde{\gamma}(\kappa)$. We suspect that this is also true for the initial condition $u_0(x) \equiv 1$.

Unfortunately, it is extremely difficult to obtain any analytic expression for $\tilde{\gamma}(\kappa)$. Obviously, $\tilde{\gamma}(0) = 0$ and:

$$\tilde{\gamma}(\kappa) \leq \gamma_1(\kappa) \leq \gamma_2(\kappa)/2 \leq \dots$$

In our analysis of the moment Lyapunov exponents for finitely time correlation Gaussian fields, we rescaled the time variable and we considered the Lyapunov exponents as functions of both the coupling constant σ and of the diffusion constant κ . Two separate analyses were needed because there is no simple relationship between the corresponding exponents. We are now restricting ourselves to the delta correlated case. We show now that it is possible to take advantage of the duality $\kappa \leftrightarrow \sigma$ available in this case to analyze the Lyapunov exponents.

The differential equation:

$$\frac{\partial u}{\partial t} = \kappa \Delta u + \xi_t(x)u$$

can be rewritten in the form:

$$\frac{\partial u}{\partial(\kappa t)} = \Delta u + \frac{1}{\sqrt{\kappa}} \left(\frac{1}{\sqrt{\kappa}} \xi_{\kappa t/\kappa}(x) \right) u$$

or equivalently:

$$\frac{\partial u}{\partial s} = \Delta u + \frac{1}{\sqrt{\kappa}} \tilde{\xi}_s(x)u \quad (\text{IV.3})$$

provided we set $s = \kappa t$ and we define the new potential process $\tilde{\xi}$ by:

$$\tilde{\xi}_s(x) = \frac{1}{\sqrt{\kappa}} \xi_{s/\kappa}(x). \quad (\text{IV.4})$$

This potential field has the same distribution as the original one. As we already pointed out, this is one of the crucial differences between the finite time correlation and the delta correlated case considered in this chapter. The above derivation is only formal but it can be made completely rigorous by using the antiderivative $\zeta_t(x)$ and integral equations instead of differential equations.

We study the asymptotic behavior (as $t \rightarrow \infty$) of the solution $u(t, x)$ which we write by means of the Feynman-Kac formula:

$$u(t, x) = \mathbb{E}_x \{ u_0(X_t) e^{\sigma \int_0^t d\zeta_s(X_t-s)} \} \quad (\text{IV.5})$$

where we used the notation $\sigma = 1/\sqrt{\kappa}$ and where the expectation is now over the paths of the standard (i.e. $\kappa = 1$) continuous time random walk on the lattice $\mathbb{Z}^d$. Notice that, according to the results of Chapter II, the fact that we are working with this Feynman-Kac formula implies that we are considering the solution $u(t, x) = u^{(S)}(t, x)$ of the Stratonovich equation. The proof of the existence of the almost sure Lyapunov exponent which we gave at the beginning of this section can be easily adapted to the present situation. It gives the existence of the limit:

$$\tilde{\gamma}(\sigma) = \lim_{t \rightarrow \infty} \frac{1}{t} \log u(t, x)$$

and its independence from the initial condition u_0 and the point x . The Feynman-Kac formula (IV.5) shows that the solution $u(t, x)$ is a nice function of σ . Indeed, it is easy to compute the derivatives of $u(t, x)$ with respect to σ . One sees that the second

derivative is nonnegative. This shows that $u(t, x)$ and hence its limit $\tilde{\gamma}(\sigma)$ are convex functions of the coupling constant σ .

Remark

It is important to emphasize that the convexity argument which we gave above applies only to the solutions and the Lyapunov exponents in the Stratonovich sense. Indeed, the solutions in the Ito and Stratonovich sense are related by the formula:

$$u^{(I)}(t, x) = e^{-\sigma^2 \Gamma_1(0)t/2} u^{(S)}(t, x)$$

and consequently one has:

$$\tilde{\gamma}^{(I)}(\sigma) = -\frac{\Gamma_1(0)}{2} \sigma^2 + \tilde{\gamma}^{(S)}(\sigma)$$

and we cannot conclude that $\tilde{\gamma}^{(I)}(\sigma)$ is a convex function of σ .

It is in fact possible to show that:

$$\frac{d\tilde{\gamma}(\sigma)}{d\sigma} \geq 0$$

and this implies that $\tilde{\gamma}(\sigma)$ is a monotone increasing convex function of σ .

The results of the analysis of the almost sure Lyapunov exponent $\tilde{\gamma}(\sigma)$ as a function of the coupling constant σ can be immediately translated into properties of the almost sure Lyapunov exponent $\tilde{\gamma}(\kappa)$ as a function of κ because of the duality relation:

$$\tilde{\gamma}(\kappa) = \kappa \tilde{\gamma}\left(\frac{1}{\sqrt{\kappa}}\right)$$

which follows immediately from (IV.3) and which we already encountered in the discussion of the moment Lyapunov exponents.

The following theorem gives the small κ behavior of the (Stratonovich) almost sure Lyapunov exponent. It is the main technical result of this chapter.

Theorem IV.1.1 *With probability 1 there exist positive constants c_1 , c_2 and $\bar{\kappa}$ such that:*

$$\frac{c_1}{\log(1/\kappa)} \leq \tilde{\gamma}(\kappa) \leq c_2 \frac{\log \log(1/\kappa)}{\log(1/\kappa)}, \quad 0 < \kappa < \bar{\kappa}.$$

The calculations of the proof use the general idea of [26] but it is important to remark that the result of [26] is only true for discrete time. Indeed, the transition from discrete time to continuous time is not as easy as it could be thought of at first. In particular the small κ estimates have a different form as we shall see below. The estimate:

$$\tilde{\gamma}(\kappa) \geq c\sqrt{\kappa} \quad \text{as } \kappa \rightarrow 0$$

from [25] takes the form:

$$\tilde{\gamma}(\kappa) \geq \frac{c}{\log 1/\kappa} \quad \text{as } \kappa \rightarrow 0$$

in the present situation of continuous time fields. The proof is rather lengthy and technical. The lower estimate is proved in the next section and the upper estimate is proved in the following one.

In particular, $\tilde{\gamma}(\kappa)$ is a continuous function of κ at $\kappa = 0$. Obviously;

$$\tilde{\gamma}(\kappa) \leq \Gamma_1(0)/2 \quad \kappa \geq 0$$

because $\langle u(t, x) \rangle \equiv \Gamma_1(0)/2$. In fact, it is easy to prove that:

$$\tilde{\gamma}(\kappa) = \Gamma_1(0)/2 \quad d \geq 3, \quad \kappa \geq \kappa_{2,cr}.$$

Indeed, if $\kappa \geq \kappa_{2,cr}$ is fixed, then we saw that the second moment of the solution $u^{(I)}(t, x)$ is bounded while $\langle u^{(I)}(t, x) \rangle = m_1^{(I)}(t, x) \equiv 1$. This implies that one can find, for each $\epsilon > 0$ a $\delta > 0$ for which $\mathbb{P}\{u^{(I)}(t, x) > \delta\} > \epsilon$ for all $t > 0$. This implies that $\tilde{\gamma}^{(I)}(\kappa) \geq 0$. But since $\tilde{\gamma}^{(I)}(\kappa) \leq 0$ we can conclude that $\tilde{\gamma}^{(I)}(\kappa) = 0$ or equivalently that $\tilde{\gamma}(\kappa) = \Gamma_1(0)/2$.

Rather than stating a lengthy proposition, we summarize the results which we proved above (in the case of random potentials $\xi_i(x)$ which are white noise in time) about the variations of $\gamma_p(\kappa)/p$ and of $\tilde{\gamma}(\kappa)$ as functions of κ in a couple of figures. The first figure deals with dimensions $d = 1$ and $d = 2$. All the properties of $\tilde{\gamma}(\kappa)$ given in this figure have been proved except for the fact that $\tilde{\gamma}(\kappa) < 1/2$. Indeed this strict inequality has only been proved for small κ . We conjecture that, if $d = 1, 2$, then $\tilde{\gamma}(\kappa) < 1/2$ for all $\kappa > 0$ $\mathbb{P}$ -almost surely.

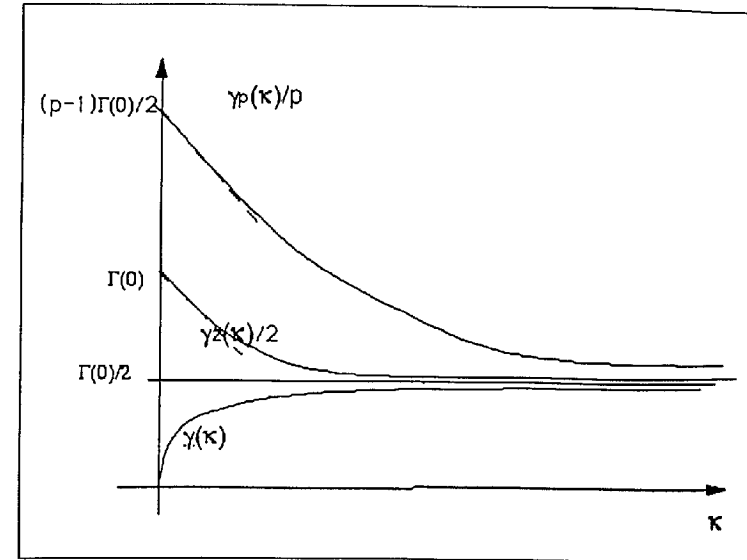


Figure IV.1.1 Plot of the almost sure Lyapunov exponent $\tilde{\gamma}_p(\kappa)/p$ as a function of κ in dimensions $d = 1$ and $d = 2$.

We also conjecture that, in dimensions $d \geq 3$, the variations of $\tilde{\gamma}(\kappa)$ are of the form given in the figure below. In contrast with the cases $d = 1$ and $d = 2$, all the statements illustrated by this figure are proved in this chapter.

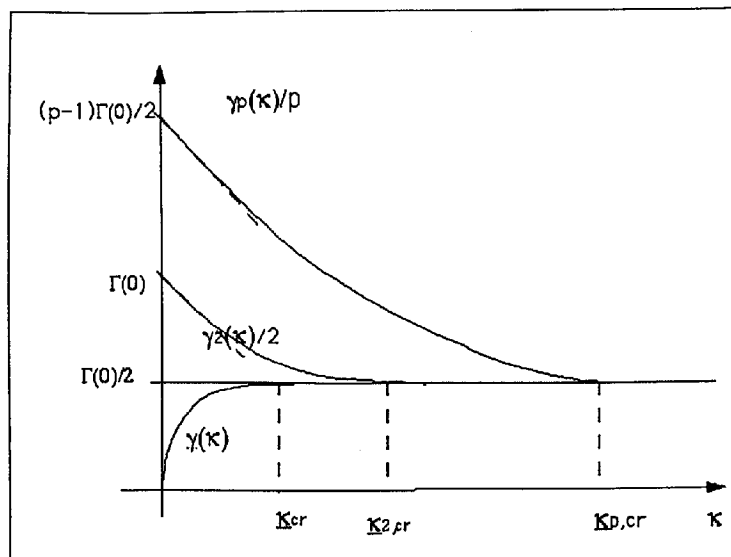


Figure IV.1.2 Plot of the almost sure Lyapunov exponent $\tilde{\gamma}_p(\kappa)/p$ as a function of κ in dimensions $d \geq 3$.

More precisely, we conjecture that $\tilde{\gamma}(\kappa) < 1/2$ for $\kappa < \tilde{\kappa}_{cr}$. As above, the thicker lines correspond to what we can actually prove.

IV.2 Proof of the Lower Bound

In order to get a lower bound on the almost sure Lyapunov exponent $\tilde{\gamma}(\kappa)$, we first prove a lower bound on the solution $u(t, 0)$ where we assume that this solution is computed with the initial condition $u_0 = \delta_0$. Our approach is very much in the spirit of the usual proofs of the lower bounds in the theory of large deviations. Let us fix $t > 0$ momentarily and let us pick an even integer $n = 2k$ to be chosen later as a function of t and κ . We get a lower bound on $u(t, 0)$ by averaging only over the paths of the continuous time random walk which have exactly n jumps before time t .

$$\begin{aligned} u(t, 0) &= \mathbb{E}_0\{\delta_0(X_t) e^{\int_0^t d\zeta_s(X_{t-s})}\} \\ &\geq \mathbb{E}_0\{\delta_0(X_t) e^{\int_0^t d\zeta_s(X_{t-s})}; N(t) = n\} \\ &= e^{-2d\kappa t} \frac{(2d\kappa t)^n}{n!} \mathbb{E}_0\{\delta_0(X_t) e^{\int_0^t d\zeta_s(X_{t-s})} | N(t) = n\}. \end{aligned}$$

Under the condition that the number of jumps before time t is equal to n , the values $T_1, \dots, T_n$ of the instants of jumps are the increasing rearrangements (order statistics) of n independent random variables uniformly distributed over the interval $[0, t]$. This gives:

$$u(t, 0) \geq e^{-2d\kappa t} (2d\kappa)^n \int \int_D \mathbb{E}_0\{\delta_0(X_t) e^{\sum_{j=0}^n [\zeta_{t-t_j}(X_j) - \zeta_{t-t_{j+1}}(X_j)]}\} dt_1 \dots dt_n$$

where $D = \{t_0 = 0 < t_1 < \dots < t_n < t_{n+1} = t\}$ and where the expectation $\mathbb{E}_0$ is now over the discrete time symmetric nearest neighbor random walk $\{X_j; j \geq 0\}$ on the lattice $\mathbb{Z}^d$. We now restrict the possible instants of jumps. In fact, we consider only the case where the actual jumps take place in n small intervals separated by $(n-1)$ larger intervals without jumps. More precisely, we choose positive numbers $\rho > 0$ and $r > 0$ in such a way that $t = (n-1)(\rho+r) + \rho$. One should think of ρ as being much smaller than r . Then we set:

$$I_j = [(j-1)(\rho+r), (j-1)(\rho+r) + \rho), \quad j = 1, \dots, n$$

and we demand that the j -th jump occurs in the interval I_j . We now have:

$$u(t, 0) \geq e^{-2d\kappa t} (2d\kappa)^n \int_{I_1} \dots \int_{I_n} \mathbb{E}_0\{\delta_0(X_t) e^{\sum_{j=0}^n [\zeta_{t-t_j}(X_j) - \zeta_{t-t_{j+1}}(X_j)]}\} dt_1 \dots dt_n$$

We rewrite the above exponent in the following form:

$$\begin{aligned} \sum_{j=0}^n [\zeta_{t-t_j}(X_j) - \zeta_{t-t_{j+1}}(X_j)] &= [\zeta_t(X_0) - \zeta_{t-t_1}(X_0)] \\ &+ [\zeta_{t-t_1}(X_1) - \zeta_{t-\rho}(X_1)] + [\zeta_{t-\rho}(X_1) - \zeta_{t-\rho-r}(X_1)] \\ &+ [\zeta_{t-\rho-r}(X_1) - \zeta_{t-t_2}(X_1)] + [\zeta_{t-t_2}(X_2) - \zeta_{t-2\rho-r}(X_2)] \\ &+ [\zeta_{t-2\rho-r}(X_2) - \zeta_{t-2\rho-2r}(X_2)] + [\zeta_{t-2\rho-2r}(X_2) - \zeta_{t-t_3}(X_2)] \end{aligned}$$

$$\begin{aligned}
& + \dots + [\zeta_{t-t_j}(X_j) - \zeta_{t-(j-1)(\rho+r)-\rho}(X_j)] \\
& + [\zeta_{t-(j-1)(\rho+r)-\rho}(X_j) - \zeta_{t-j(\rho+r)}(X_j)] + [\zeta_{t-j(\rho+r)}(X_j) - \zeta_{t-t_{j+1}}(X_j)] \\
& + \dots + [\zeta_{t-t_n}(X_n) - \zeta_0(X_0)]
\end{aligned}$$

so that:

$$e^{\sum_{j=0}^n [\zeta_{t-t_j}(X_j) - \zeta_{t-t_{j+1}}(X_j)]} = \prod_{j=1}^n e^{\tilde{\zeta}_j(X_j)} \prod_{j=1}^n e^{\eta_j(t_j, X_j)}$$

where:

$$\tilde{\zeta}_j(X_j) = \zeta_{t-(j-1)(\rho+r)-\rho}(X_j) - \zeta_{t-j(\rho+r)}(X_j), \quad j = 1, \dots, n$$

and:

$$\begin{aligned}
\eta_j(t_j, X_j) &= [\zeta_{t-(j-1)(\rho+r)}(X_j) - \zeta_{t-t_j}(X_j)] \\
&+ [\zeta_{t-t_j}(X_{j-1}) - \zeta_{t-(j-1)(\rho+r)-\rho}(X_j)] \quad j = 1, \dots, n.
\end{aligned}$$

Consequently

$$\begin{aligned}
u(t, 0) &\geq e^{-2d\kappa t} (2d\kappa\rho)^n \int_{I_1} \cdot \int_{I_n} \mathbb{E}_0\{\delta_0(X_t) \prod_{j=1}^n e^{\tilde{\zeta}_j(X_j)} \prod_{j=1}^n \left(\frac{1}{\rho} \int_{I_j} e^{\eta_j(t_j, X_j)} dt_j \right)\} \\
&\geq e^{-2d\kappa t} (2d\kappa\rho)^n \mathbb{E}_0\{\delta_0(X_t) \prod_{j=1}^n e^{\tilde{\zeta}_j(X_j)} \prod_{j=1}^n e^{\eta_j(X_j)}\}
\end{aligned}$$

if we use Jensen's inequality and if we set:

$$\eta_j(X_j) = \frac{1}{\rho} \int_{I_j} \eta_j(t_j, X_j) dt_j \quad j = 1, \dots, n.$$

This gives:

$$u(t, 0) \geq e^{-2d\kappa t} (2d\kappa\rho)^n \mathbb{E}_0\{\delta_0(X_t) e^{\sum_{j=1}^n [\tilde{\zeta}_j(X_j) + \eta_j(X_j)]}\}$$

Notice that, for each fixed path $X_0 X_1 \dots X_n \dots$ of the random walk, the $\zeta_j(X_j)$'s are i.i.d. normal random variable with mean zero and variance $\rho\Gamma_1(0)$, that the $\eta_j(X_j)$'s are also i.i.d. normal mean zero random variables and that the $\zeta_j(X_j)$'s and the

$\eta_{j'}(X_{j'})$'s are independent since they involve increments of ζ over disjoint intervals. The variance of $\eta_j(X_j)$'s is given by:

$$\begin{aligned}
\sigma_j^2 &= \frac{1}{\rho^2} < \left(\int_{I_j} [\zeta_{t-(j-1)(\rho+r)}(X_j) - \zeta_{t-t_j}(X_j)] \right. \\
&\quad \left. + [\zeta_{t-t_j}(X_{j-1}) - \zeta_{t-(j-1)(\rho+r)-\rho}(X_j)] dt_j \right)^2 > \\
&= \frac{1}{3} [2\Gamma_1(0) - \Gamma_1(X_j - X_{j-1})].
\end{aligned}$$

Finally, we choose a specific path for the (discrete time) random walk instead of integrating over all the possible paths which start from 0 and return to 0 at time t . Here is the path which we choose. Notice that this path depends upon the sample realization of the random environment ζ . We first assume that $N(t) = n$, i.e. the path jumps exactly n times before time t , and $X_j = x_j$ for $j = 1, \dots, n$ where the points $x_1, \dots, x_n$ are constructed inductively in the following way. We set $x_0 = 0$ and for $j = 1, \dots, k$ we choose x_j among the $2d$ neighbors of x_{j-1} as the site x which maximizes the increment:

$$\zeta_{t-(j-1)(\rho+r)-\rho}(x) - \zeta_{t-j(\rho+r)}(x).$$

We then set:

$$x_{k+1} = x_{k-1}, \quad x_{k+2} = x_{k-2}, \dots, x_{2k-1} = x_1, \quad x_{2k+1} = x_0 = 0$$

to force the path to be back at the origin at time t . Consequently we have:

$$u(t, 0) \geq e^{-2d\kappa t} (2d\kappa\rho)^n \left(\frac{1}{2d} \right)^n e^{\sum_{j=1}^k \zeta_j + \sum_{j=1}^k \eta_j}$$

where the ζ_j 's and the η_j 's are the above $\tilde{\zeta}_j(X_j)$'s and $\eta_j(X_j)$'s for the specific choices $X_j = x_j$ which we made.

Note that the probability space of the continuous time random walk has been disposed with: the remaining random variables ζ_j and η_j are defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ of the random field ζ . Even though the X_j 's were chosen as random variables on Ω , ζ_j and the $\eta_{j'}$ are still independent because of the independence of the increments of $\zeta_t(x)$ and of the particular definition of the X_j 's. Each ζ_j has the distribution of $\sqrt{\Gamma} \max\{\theta_1, \dots, \theta_{2d}\}$ where $(\theta_1, \dots, \theta_{2d})$ is a mean zero Gaussian in $\mathbb{R}^{2d}$ with covariance

determined in an obvious way by the values of $\Gamma_1(z)$ for $|z| = 1$. The ζ_j are independent and identically distributed.

On the other hand, the random variables η_j 's are independent but not identically distributed. Each η_j has the distribution $N(0, [2\Gamma_1(0) - \Gamma_1(x_j - x_{j-1})]/3)$ for the appropriate choice of the x_j 's which we described above.

Now, for any $\gamma > 0$ and $z > 0$ one has:

$$\begin{aligned} \mathbb{P}\{u(t, 0) \leq e^{\gamma t}\} &\leq \mathbb{P}\{e^{\sum_{j=1}^n \zeta_j + \sum_{j=1}^n \eta_j} \leq (\kappa\rho)^{-n} e^{(\gamma+2d\kappa)t}\} \\ &\leq \mathbb{P}\{e^{-z \sum_{j=1}^n \zeta_j - z \sum_{j=1}^n \eta_j} \geq (\kappa\rho)^{zn} e^{-z(\gamma+2d\kappa)t}\} \\ &\leq (\kappa\rho)^{-zn} e^{z(\gamma+2d\kappa)t} \mathbb{E}\{e^{-z \sum_{j=1}^n \zeta_j - z \sum_{j=1}^n \eta_j}\} \\ &\leq (\kappa\rho)^{-zn} e^{z(\gamma+2d\kappa)t} \mathbb{E}\{e^{-z\zeta_j}\}^n \prod_{j=1}^n \mathbb{E}\{e^{-z\eta_j}\} \quad (\text{IV.6}) \end{aligned}$$

The conclusion of the proof of the lower bound depends upon the following Cramer type estimate. This result is elementary and presumably well known. We include a proof for the sake of completeness.

Lemma IV.2.1 *Let us assume that the random variable X satisfies:*

$$\log \mathbb{P}\{X > t\} \asymp -t^2 \quad t \rightarrow \infty, \quad (\text{IV.7})$$

and:

$$\log \mathbb{P}\{X < t\} \asymp -t^2 \quad t \rightarrow -\infty. \quad (\text{IV.8})$$

Then there exist positive constants ρ_0 and ρ_1 such that the estimate:

$$\mu z + \rho_1 z^2 \leq \log \mathbb{E}\{e^{zX}\} \leq \mu z + \rho_2 z^2$$

holds for all the values of $z \in \mathbb{R}$ where $\mu = \mathbb{E}\{X\}$.

Proof:

The Laplace method and the assumptions (IV.7) and (IV.8) on the tails of the distribution of X imply that:

$$\log \mathbb{E}\{e^{zX}\} \asymp z^2 \quad |z| \rightarrow \infty.$$

Moreover, the convex function $\varphi(z) = \log \mathbb{E}\{e^{zX}\}$ satisfies $\varphi'(0) = \mu$ and $\varphi''(0) = \mathbb{E}\{X^2\} - \mathbb{E}\{X\}^2 > 0$ because of our assumption on the distribution of X . Consequently, the quantity:

$$\frac{\log \mathbb{E}\{e^{zX}\} - \mu z}{z^2}$$

is bounded near the infinities and near the origin. This completes the proof. ■

We can now return to the proof of the lower bound. Indeed, it is a simple exercise to prove that the distribution of the maximum of $2d$ Gaussian random variables satisfies the assumptions (IV.7) and (IV.8) of the above lemma. Since the common expectation of the ζ_j 's is a positive number, say $\alpha > 0$, as a consequence of the above lemma one concludes that:

$$\mathbb{E}\{e^{-z\zeta_j}\} \leq e^{-\alpha\sqrt{r}z + \beta rz^2}$$

for some positive constant β . Consequently the inequality (IV.6) gives:

$$\begin{aligned} \mathbb{P}\{u(t, 0) \leq e^{\gamma t}\} &\leq (\kappa\rho)^{-zn} e^{z(\gamma+2d\kappa)t - n\alpha\sqrt{r}z + n\beta rz^2} \prod_{j=1}^n e^{\rho_2 \sigma_j} \\ &\leq e^{-zt((1/r)\log(\kappa) - \gamma - 2d\kappa + \alpha/\sqrt{r} - \beta z - cz)} \end{aligned}$$

if one sets $\rho = 1$ and $n \sim t/r$ for some $r = r(\kappa) \rightarrow \infty$. Now, if one chooses $z = z(\kappa) \searrow 0$ faster than $r(\kappa)^{-1/2}$, then the right hand side will be summable if $\gamma = \gamma(\kappa)$ is chosen to be smaller than $\alpha/r(\kappa)^{1/2}$. One can now choose:

$$r = r(\kappa) = (\log(1/\kappa))^2$$

and apply the first Borel Cantelli lemma to finish the proof of the lower bound. ■

IV.3 Proof of the Upper Bound

The proof of the upper bound relies on a specific procedure of discretization of the time of the random walk. We divide the interval $[0, t]$ into k disjoint subintervals of equal lengths, say $I_1, I_2, \dots, I_k$, and we denote the common length by δ so that $t = k\delta$. For each sample path of the continuous time random walk, these intervals are

classified into two disjoint groups. An interval is said to be *white* if it does not contain an instant of jump of the continuous time random walk X_t and it is said to be *black* otherwise.

We say that Γ is a nontrivial increment sequence if $\Gamma = \{x_0, x_1, \dots, x_n\}$ is a finite sequence with $n \geq 1$ of elements $x_i \in \mathbb{Z}^d$ such that $|x_i| = 1$. $\Gamma = \{0\}$ will be called the trivial increment sequence. Notice that the same x can appear several times in such a sequence. If $\Gamma_1, \Gamma_2, \dots, \Gamma_k$ are increment sequences, i.e.

$$\Gamma_i = \{x_{i,0}, x_{i,1}, \dots, x_{i,n_i}\}$$

with $|x_{i,j}| = 1$ then we denote by $\{\Gamma_1, \Gamma_2, \dots, \Gamma_k\}$ the set of sample paths X , of the continuous time random walk which have exactly n_i jumps, say $T_{i,1}, T_{i,2}, \dots, T_{i,n_i}$, in the interval I_i and for which:

$$x_{i,0} = X_{T_{i,1}} - X_{(i-1)\delta}, x_{i,1} = X_{T_{i,2}} - X_{T_{i,1}}, \dots, x_{i,n_i} = X_{i\delta} - X_{T_{i,n_i}}.$$

$\{\Gamma_1, \Gamma_2, \dots, \Gamma_k\}$ will be called the common *skeleton* of this set of sample paths. Notice that, on $\{\Gamma_1, \Gamma_2, \dots, \Gamma_k\}$, only the values of the instants of the jumps (and not their number) are free to vary. With this new notation one has:

$$\begin{aligned} u(t, x) &= \sum_{\{\Gamma_1, \dots, \Gamma_k\}} \mathbb{E}_x \left\{ \exp \left[\sum_{i=1}^k \int_{I_i} d\zeta_{t-s}(X_s) \right] \mathbf{1}_{\{\Gamma_1, \dots, \Gamma_k\}} \right\} \\ &= \sum_{\{\Gamma_1, \dots, \Gamma_k\}} \mathbb{E}_x \left\{ \mathbf{1}_{\{\Gamma_1, \dots, \Gamma_k\}} \prod_{i=1}^k \mathbb{E}_x \left\{ \exp \left[\int_{I_i} d\zeta_{t-s}(X_s) \right] \mid \Gamma_i \right\} \right\} \end{aligned}$$

because the sequence of the instants of the jumps is independent of the sequence of the sites visited by the random walk, and because of the independence of the time increments. Now, once the interval I_i and the skeleton Γ_i in I_i are fixed, one has:

$$\begin{aligned} \int_{I_i} d\zeta_{t-s}(X_s) &= \zeta_{t-i\delta}(X_{(i-1)\delta} + x_{i,0} + x_{i,1} + \dots + x_{i,n_i-2} + x_{i,n_i-1}) \\ &\quad - \zeta_{t-T_{i,n_i}}((X_{(i-1)\delta} + x_{i,0} + x_{i,1} + \dots + x_{i,n_i-2} + x_{i,n_i-1}) \\ &\quad + \zeta_{t-T_{i,n_i}}(X_{T_{i,n_i-1}}) - \zeta_{t-T_{i,n_i-1}}(X_{T_{i,n_i-1}})) \\ &\quad + \dots \\ &\quad + \zeta_{t-T_{i,2}}(X_{T_{i,1}}) - \zeta_{t-T_{i,1}}(X_{T_{i,1}}) \end{aligned}$$

$$\begin{aligned} &+ \zeta_{t-T_{i,1}}(X_{(i-1)\delta}) - \zeta_{t-(i-1)\delta}(X_{(i-1)\delta}) \\ = &\zeta_{t-i\delta}(X_{T_{i,n_i}}) - \zeta_{t-T_{i,n_i}}(X_{T_{i,n_i}}) \\ &+ \zeta_{t-T_{i,n_i}}(X_{(i-1)\delta} + x_{i,0} + x_{i,1} + \dots + x_{i,n_i-2}) \\ &- \zeta_{t-T_{i,n_i-1}}(X_{(i-1)\delta} + x_{i,0} + x_{i,1} + \dots + x_{i,n_i-2}) \\ &+ \dots \\ &+ \zeta_{t-T_{i,2}}(X_{(i-1)\delta} + x_{i,0}) - \zeta_{t-T_{i,1}}(X_{(i-1)\delta} + x_{i,0}) \\ &+ \zeta_{t-T_{i,1}}(X_{(i-1)\delta}) - \zeta_{t-(i-1)\delta}(X_{(i-1)\delta}) \end{aligned}$$

Moreover, the number n_i of jumps in I_i is a Poisson random variable with parameter $2d\kappa\delta$ and, conditioned on this number n_i , the instants of jumps in I_i have the joint distribution of n_i points chosen independently and uniformly over the interval I_i . Consequently one can write:

$$u(t, x) = \sum_{\{\Gamma_1, \dots, \Gamma_k\}} \mathbb{E}_x \{ \mathbf{1}_{\{\Gamma_1, \dots, \Gamma_k\}} \exp \left[\sum_{i=1}^k \eta(\Gamma_i) \right] \}$$

where, for example:

$$\eta(\Gamma_i) = \zeta_{t-i\delta}(X_{(i-1)\delta}) - \zeta_{t-(i-1)\delta}(X_{(i-1)\delta})$$

when the interval $I_i = [(i-1)\delta, i\delta]$ is white. The expression of η_{Γ_i} is more complicated when the interval I_i is black. It depends on the number of jumps n_i and of the skeleton in the interval. It is the logarithm of the integral of the exponential of a multiparameter Gaussian process over the tetrahedron $\{0 < s_1 < s_2 < \dots < s_{n_i} < t\}$. For example, when $n_i = 2$ and $\Gamma_i = \{x_{i,0}, x_{i,1}, x_{i,2}\}$, one has:

$$\eta(\Gamma_i) = \log \left(\frac{2!}{\delta^2} \int \int_{0 < s_1 < s_2 < \delta} \exp \left[\left[\zeta_{t-i\delta}(X_{(i-1)\delta} + x_{i,0} + x_{i,1}) \right. \right. \right. \quad (\text{IV.9})$$

$$\begin{aligned} &\quad \left. - \zeta_{t-T_{i,2}}(X_{(i-1)\delta} + x_{i,0} + x_{i,1}) \right] \\ &\quad + \left[\zeta_{t-T_{i,2}}(X_{(i-1)\delta} + x_{i,0}) - \zeta_{t-T_{i,1}}(X_{(i-1)\delta} + x_{i,0}) \right] \\ &\quad \left. + \left[\zeta_{t-T_{i,1}}(X_{(i-1)\delta}) - \zeta_{t-(i-1)\delta}(X_{(i-1)\delta}) \right] ds_1 ds_2 \right). \quad (\text{IV.10}) \end{aligned}$$

Notice that $X_{i\delta} = X_{(i-1)\delta} + x_{i,0} + x_{i,1} + x_{i,2}$. Notice also that one has:

$$< e^{a\eta(\Gamma_i)} > = e^{a^2\delta/2} \quad (\text{IV.11})$$

for all $a \in \mathbb{R}$ whenever the interval is white. Indeed, if the skeleton $\{\Gamma_1, \Gamma_2, \dots, \Gamma_k\}$ is known, then $X_{(i-1)\delta}$ is also known and the random variable $\eta(\Gamma_i)$ is $N(0, \delta)$. Unfortunately, formula (IV.11) holds only for $a = 1$ when the interval is black. This last fact is obvious if one recall the definition of $\eta(\Gamma_i)$. See for example (IV.9).

Let us use the obvious notation $\{\Gamma_i\}$ for the set of sample paths of the continuous time random walk which have exactly n_i jumps in the interval I_i , the sizes of the jumps being given by the elements $x_{i,j}$ of Γ_i . Obviously, the $\{\Gamma_i\}$'s are independent. Moreover:

$$\mathbb{E}_x\{1_{\{\Gamma_i\}}\} = \frac{(\kappa\delta)^{n_i}}{n_i!} e^{-2d\kappa\delta}. \quad (\text{IV.12})$$

One can use Schwarz inequality to separate the contribution of the white intervals from the (smaller) contribution of the black intervals. Indeed:

$$\begin{aligned} u(t, 0) &= \mathbb{E}_0\{\exp[\sum_{i, n_i=0} \int_{I_i} d\zeta_{t-s}(X_s) + \sum_{i, n_i \neq 0} \int_{I_i} d\zeta_{t-s}(X_s)]\} \\ &\leq (u^{(w)}(t) u^{(b)}(t))^{1/2} \end{aligned}$$

provided we set:

$$u^{(w)}(t) = \mathbb{E}_0\{\exp[2 \sum_{i, n_i=0} \int_{I_i} d\zeta_{t-s}(X_s)]\} \quad (\text{IV.13})$$

and:

$$u^{(b)}(t) = \mathbb{E}_0\{\exp[2 \sum_{i, n_i \neq 0} \int_{I_i} d\zeta_{t-s}(X_s)]\}. \quad (\text{IV.14})$$

Obviously one has:

$$\tilde{\gamma}(\kappa) \leq \frac{1}{2} \left(\limsup_{t \rightarrow \infty} \frac{1}{t} \log u^{(w)}(t) + \limsup_{t \rightarrow \infty} \frac{1}{t} \log u^{(b)}(t) \right). \quad (\text{IV.15})$$

But using (IV.9) with $a = 1$ one has:

$$\langle u^{(b)}(t) \rangle = \langle \mathbb{E}_0\{\exp[2 \sum_{i, n_i \neq 0} \int_{I_i} d\zeta_{t-s}(X_s)]\} \rangle$$

$$\begin{aligned} &= \mathbb{E}_0\{\langle \exp[2 \sum_{i, n_i \neq 0} \int_{I_i} d\zeta_{t-s}(X_s)] \rangle\} \\ &= \mathbb{E}_0\{e^{2\delta N_b}\} \end{aligned}$$

where we used the independence of the increments of ζ_t and where N_b denotes the number of black intervals. The latter has a binomial distribution with parameters k and $p = 1 - e^{-2d\kappa\delta}$. As a consequence one gets:

$$\langle u^{(b)}(t) \rangle = [(1 - e^{-2d\kappa\delta})e^{2\delta} + e^{-2d\kappa\delta}]^k.$$

Chebychev's inequality implies:

$$\mathbb{P}\{u^{(b)}(k\delta) > k^2 \langle u^{(b)}(t) \rangle\} \leq \frac{1}{k^2}$$

and the first Borel-Cantelli lemma gives:

$$\limsup_{k \rightarrow \infty} \frac{1}{k\delta} \log u^{(b)}(k\delta) \leq \frac{1}{\delta} \log[(1 - e^{-2d\kappa\delta})e^{2\delta} + e^{-2d\kappa\delta}] \quad (\text{IV.16})$$

almost surely. For $\delta \rightarrow \infty$ one has:

$$\begin{aligned} \frac{1}{\delta} \log[(1 - e^{-2d\kappa\delta})e^{2\delta} + e^{-2d\kappa\delta}] &\leq \frac{c}{\delta} (1 - e^{-2d\kappa\delta})e^{2\delta} \\ &\leq 2cd\kappa e^{2\delta} \end{aligned}$$

for some constant $c > 1$. We shall choose (later) the free parameter δ giving the length of the intervals to be of the form $\delta = \delta' \log(1/\kappa)$ for some small parameter $\delta' > 0$ independent of κ . In this case the left hand side of (IV.16) is smaller than the upper bound which we shall derive for the contribution of the white intervals.

Consequently the proof of the upper bound reduces to the estimation of the quantity $u^{(w)}(t)$ for large t and small κ . Notice that we considered $t \rightarrow \infty$ restricted to the subsequence $k\delta$. This is enough because we already know that the almost sure Lyapunov exponent exists and hence its value can be computed from any deterministic subsequence.

Let us introduce the following notation. For a given skeleton sequence $\tilde{\Gamma} = \{\Gamma_1, \dots, \Gamma_k\}$ we denote by $k_0 = k_0(\tilde{\Gamma})$ the number of white intervals, i.e. the number of i 's for which $n_i = 0$. Here we use the notation $n_i = |\Gamma_i| - 1$. One should think of n_i as the number of jumps in the interval I_i . For each nontrivial increment sequence $\Gamma = \{x_0, x_1, \dots, x_n\}$

we define the integers $k(\tilde{\Gamma}, \Gamma)$ as the number of (black) intervals I_i 's for which $\Gamma_i = \Gamma$. Obviously one has:

$$k = k_0(\tilde{\Gamma}) + \sum_{|\Gamma|} k(\tilde{\Gamma}, \Gamma)$$

Notice also that for each fixed sequence of integers $k_0, k(\Gamma)$ for $|\Gamma| \neq 2$, the number of skeletons $\tilde{\Gamma} = (\Gamma_1, \dots, \Gamma_k)$ of length k for which $k(\tilde{\Gamma}, \Gamma)$ for all Γ is given by:

$$\frac{k!}{k_0! \prod_{|\Gamma| \neq 0} k(\Gamma)!}$$

Moreover, the statistical weight of such a skeleton sequence is given by the probability that the continuous time random walk has, for $i = 1, \dots, k$, exactly n_i jumps in the interval I_i , namely:

$$\prod_{i=1}^k \frac{(2d\kappa\delta)^{n_i}}{n_i!} e^{-2d\kappa\delta}$$

Let us choose a sequence $a_0, \dots, a_n \dots$ of positive constants and let us consider the event (defined on the probability space of the random potential $d\zeta_i(x)$):

$$A^{(k)}(\Gamma_1, \dots, \Gamma_k) = \left\{ \sum_{n_i=0} \int_{I_i} \zeta_{t-s}(x_{(i)}) \geq a_0 k_0 + \sum_{j \geq 1} a_j \sum_{n_i=j} k(\Gamma_i) \right\}.$$

where $x_{(i)}$ denotes the position at the beginning of the time interval I_i given by the skeleton, i.e.

$$x_{(i)} = \sum_{i'=1}^{i-1} \sum_{j=0}^{n_{i'}} x_{i',j}.$$

But because of the very definition of a white interval one has:

$$\sum_{n_i=0} \int_{I_i} \zeta_{t-s}(x_{(i)}) \sim N(0, k_0\delta).$$

Using the standard upper bound:

$$\mathbb{P}\{\zeta > \lambda\} \leq e^{-\lambda^2/2}$$

which holds for $\zeta \sim N(0, 1)$ and $\lambda > 0$, we get:

$$\begin{aligned} \mathbb{P}\{A^{(k)}(\Gamma_1, \dots, \Gamma_k)\} &\leq \exp\left[-\frac{1}{2k_0\delta}(a_0 k_0 + \sum_{|\Gamma| \geq 1} a_{|\Gamma|} k(\Gamma))^2\right] \\ &\leq \exp\left[-\frac{a_0^2 k_0}{2\delta} - \frac{a_0}{\delta} \sum_{|\Gamma| \geq 1} a_{|\Gamma|}\right] \end{aligned} \quad (\text{IV.17})$$

The strategy for the end of the proof is the following. We choose the sequence $a_0, a_1, \dots$ in such a way that:

$$\sum_{\{\Gamma_1, \dots, \Gamma_k\}} \mathbb{P}\{A^{(k)}(\Gamma_1, \dots, \Gamma_k)\} \leq q^k \quad (\text{IV.18})$$

for some positive real number $q < 1$. Then, the first Borel Cantelli lemma gives the almost sure existence of an integer $k' = k'(\omega)$ such that, whenever $\{\Gamma_1, \dots, \Gamma_k\}$ is a skeleton of length $k > k'$, then:

$$\sum_{1 \leq i \leq k; n_i=0} \int_{I_i} \zeta_{t-s}(x_{(i)}) \leq a_0 k_0 + \sum_{|\Gamma_i| \geq 2} a_{|\Gamma_i|} k(\Gamma_i).$$

Consequently, for such a $k > k'(\omega)$ one has:

$$\begin{aligned} u^{(w)}(t, 0) &\leq \sum_{\{\Gamma_1, \dots, \Gamma_k\}} e^{-2d\kappa k\delta} \prod_{i=1}^k \frac{(\kappa\delta)^{|\Gamma_i|}}{|\Gamma_i|!} \exp\left[2 \sum_{|\Gamma_i|=1} \int_{I_i} \zeta_{t-s}(x_{(i)})\right] \\ &\leq \sum_{\{\Gamma_1, \dots, \Gamma_k\}} e^{-2d\kappa k\delta} \prod_{i=1}^k \frac{(\kappa\delta)^{|\Gamma_i|}}{|\Gamma_i|!} \exp\left[2a_0 k_0 + 2 \sum_{|\Gamma_i| \geq 2} a_{|\Gamma_i|} k(\Gamma_i)\right] \\ &\leq e^{-2d\kappa k\delta} \sum_{\{\Gamma_1, \dots, \Gamma_k\}} \left(\frac{2d\kappa\delta}{2!}\right)^{2 \sum_{|\Gamma_i|=2} k(\Gamma_i)} \left(\frac{2d\kappa\delta}{3!}\right)^{2 \sum_{|\Gamma_i|=3} k(\Gamma_i)} \dots \\ &\quad \dots \frac{k!}{k_0! \prod_{|\Gamma_i| \geq 2} k(\Gamma_i)!} \exp\left[2a_0 k_0 + 2 \sum_{|\Gamma_i| \geq 2} a_{|\Gamma_i|} k(\Gamma_i)\right] \end{aligned} \quad (\text{IV.19})$$

if one remembers that the definition of $u^{(w)}(t)$ involves only white intervals and if one computes the expectation $\mathbb{E}_0\{\cdot\}$ appearing in the definition (IV.13) of $u^{(w)}(t)$ by first summing over all the possible skeletons $\{\Gamma_1, \dots, \Gamma_k\}$ of length k . The upper bound appearing in the right hand side depends on the skeleton only via the numbers

$|\Gamma_i|$ and the integers $k(\Gamma_i)$. Consequently one can sum over the various choices of these integers. One can for example sum first over all the possible choices of integers $k_0, k_1, \dots$ summing up to k . One gets:

$$u^{(w)}(t, 0) \leq e^{-2d\kappa\delta} \left(e^{2a_0} + \frac{(2d\kappa\delta)}{1!} e^{2a_1} + \dots + \frac{(2d\kappa\delta)^\ell}{\ell!} e^{2a_\ell} + \dots \right)^k \quad (\text{IV.20})$$

because there exist exactly $2d$ different Γ_i with $|\Gamma_i| = 1$, $(2d)^2$ Γ_i with $|\Gamma_i| = 2d$, ... The same sort of calculations (using again (IV.17)) show that, for each fixed k one also has:

$$\begin{aligned} \sum_{(\Gamma_1, \dots, \Gamma_k)} \mathbb{P}\{A^{(k)}(\Gamma_1, \dots, \Gamma_k)\} &\leq \left(e^{-a_0^2/(2\delta)} + \sum_{|\Gamma_i| \neq 0} e^{-a_0 a_{|\Gamma_i|}/\delta} \right)^k \\ &\leq \exp\left[-\frac{a_0^2 k_0}{2\delta} - \frac{a_0}{\delta} \sum_{|\Gamma_i| > 1} a_{|\Gamma_i|}\right] \\ &= \left(e^{-a_0^2/(2\delta)} + \sum_{\ell=2}^{\infty} (2d)^\ell e^{-a_0 a_\ell/\delta} \right)^k. \end{aligned}$$

Consequently we have:

$$\begin{aligned} \tilde{\gamma}^{(w)}(\kappa) &= \limsup_{t \rightarrow \infty} \frac{1}{t} \log u^{(w)}(t) \\ &\leq -2d\kappa + \frac{1}{\delta} \log \left(e^{2a_0} + \sum_{\ell=2}^{\infty} \frac{(2d\kappa\delta)^\ell}{\ell!} e^{2a_\ell} \right) \end{aligned}$$

provided the condition:

$$e^{-a_0^2/(2\delta)} + \sum_{\ell=2}^{\infty} (2d)^\ell e^{-a_0 a_\ell/\delta} < 1$$

is satisfied. Indeed, because of (IV.21) this last condition implies that (IV.18) is satisfied so that one can use the first Borel-Cantelli lemma. At this stage of the proof, the *right thing to do* would be to solve the optimization problem:

$$\inf_{(a_0, a_1, \dots)} e^{2a_0} + \sum_{\ell=2}^{\infty} \frac{(2d\kappa\delta)^\ell}{\ell!} e^{2a_\ell}$$

under the condition (constraint):

$$e^{-a_0^2/(2\delta)} + \sum_{\ell=2}^{\infty} (2d)^\ell e^{-a_0 a_\ell/\delta} = 1.$$

Indeed, the solution of this minimization problem would give the best upper bound on $\tilde{\gamma}^{(w)}(\kappa)$ in the present approach. We shall limit ourselves to the construction of an *approximate* solution. We choose:

$$a_0 = (1 + \delta_1) \log \log(1/\kappa) \quad \text{and} \quad a_\ell = \ell\delta, \quad \ell = 2, 3, \dots$$

and $\delta = \delta' \log(1/\kappa)$ for some small $\delta' > 0$ as in the estimation of $\tilde{\gamma}^{(b)}(\kappa)$. Under these conditions we have:

$$\begin{aligned} e^{-a_0^2/(2\delta)} + \sum_{\ell=2}^{\infty} (2d)^\ell e^{-a_0 a_\ell/\delta} &= e^{-a_0^2/(2\delta)} + \frac{(2d)^2 e^{-2a_0}}{1 - (2d)^2 e^{-2a_0}} \\ &\leq 1 - \frac{a_0^2}{2\delta} + \frac{a_0^4}{8\delta^2} + 8d^2 e^{-2a_0} \end{aligned}$$

provided κ is small enough so that $2(2d)^2 e^{-2a_0} < 1$. It is then easy to see that the above right hand side is strictly smaller than 1 with the choices of a_0 and δ which we made. Moreover, in the asymptotic regime we chose we have:

$$e^{2a_0} + \sum_{\ell=2}^{\infty} \frac{(2d\kappa\delta)^\ell}{\ell!} e^{2a_\ell} \sim e^{2a_0}$$

as $\kappa \rightarrow 0$ provided $\delta' < 1/2$. Consequently we proved:

$$\begin{aligned} \tilde{\gamma}^{(w)}(\kappa) &\leq \frac{2a_0}{\delta} - 2d\kappa \\ &\leq C \frac{\log \log(1/\kappa)}{\log(1/\kappa)} \end{aligned}$$

and this concludes the proof of the upper bound. ■

Chapter V

CONCLUDING REMARKS

We conclude the present memoir with a discussion of the above results and conjectures from the point view of the qualitative behavior of the solution $u(t, x)$ of the random parabolic equation (I.1) which we understand as a stochastic partial differential equation (in the sense of Stratonovich) when the time correlation length is zero.

When $\kappa \ll 1$, then the almost sure Lyapunov exponent $\tilde{\gamma}(\kappa) < \Gamma_1(0)/2$ and Ito's solution $u^{(I)}(t, x) = \exp[-\Gamma_1(0)t/2]u(t, x)$ tends to 0 exponentially fast. Moreover, all the statistical moments of order $p \geq 2$ tend to infinity. This means that all the particles (remember that $u(t, x)$ has the interpretation of a mean density of particles!) are concentrated, for large times t , in a system of rare (with respect to t) exponential picks, the space between these picks being essentially empty. This is the analog of the localization phenomenon so typical of the classical Anderson's model in quantum disordered systems. This model is concerned with the random self-adjoint Schrödinger operator $H = \kappa\Delta + \xi(x)$ with a time independent (stationary in our terminology) potential. In fact, in the classical Anderson model, the $\xi(x)$ are assumed to be i.i.d. random variables. Exponential localization is known to hold (see for example [6]) for all κ in dimension $d = 1$. It is expected to hold for all κ in dimension $d = 2$, and it is also proven in all dimensions $d \geq 1$ for sufficiently small κ . The conjectures we formulated in Chapter IV above concerning the almost sure Lyapunov exponent are mere translations of these facts to the nonstationary case.

If $\kappa \gg 1$ and $d \geq 3$ we saw that:

$$\tilde{\gamma}(\kappa) = \gamma_1(\kappa) = \frac{\gamma_2(\kappa)}{2} \equiv 0,$$

that the finite dimensional marginal distributions of the solution $u^{(I)}(t, x)$ of the Ito's problem are tight and that the second moment converges to a finite limit. In fact, we suspect (but we cannot prove) that this Ito's solution has a nontrivial limit in distribution when $t \rightarrow \infty$. In other words we conjecture that:

$$u^{(I)}(t, \cdot) \xrightarrow{\text{dist}} u^{(\infty)}(\infty, \cdot)$$

for some nontrivial limit. We have intermittency of the highest moments for all κ , i.e. we still have a system of high picks, but between these picks we now have a nondegenerate (bounded in probability) density of particles. This phenomenon can be considered as the analog in the nonstationary case of the existence of extended states and absolutely continuous spectrum in dimensions $d \geq 3$ in the case of the classical Anderson's model with $\kappa \gg 1$. Let us emphasize that this last result has not been proved rigorously even though it is believed to be true. The critical value $\tilde{\kappa}_{cr}$ plays the role in the non-stationary case, of the (hypothetical) Anderson's *mobility edge*.

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